

CHAPTER 5

Linear Mappings

5.1 Introduction

The main subject matter of linear algebra is the study of linear mappings and their representation by means of matrices. This chapter introduces us to these linear maps and [Chapter 6](#) shows how they can be represented by matrices. First, however, we begin with a study of mappings in general.

5.2 Mappings, Functions

Let A and B be arbitrary nonempty sets. Suppose to each element in $a \in A$ there is assigned a unique element of B ; called the *image* of a . The collection f of such assignments is called a *mapping* (or map) from A into B , and it is denoted by

$$f: A \rightarrow B$$

The set A is called the *domain* of the mapping, and B is called the *target set*. We write $f(a)$, read “ f of a ,” for the unique element of B that f assigns to $a \in A$.

One may also view a mapping $f: A \rightarrow B$ as a computer that, for each input value $a \in A$, produces a unique output $f(a) \in B$.

Remark: The term *function* is used synonymously with the word *mapping*, although some texts reserve the word “function” for a real-valued or complex-valued mapping.

Consider a mapping $f: A \rightarrow B$. If A' is any subset of A , then $f(A')$ denotes the set of images of elements of A' ; and if B' is any subset of B , then $f^{-1}(B')$ denotes the set of elements of A ; each of whose image lies in B' . That is,

$$f(A') = \{f(a) : a \in A'\} \quad \text{and} \quad f^{-1}(B') = \{a \in A : f(a) \in B'\}$$

We call $f(A')$ the *image* of A' and $f^{-1}(B')$ the *inverse image* or *preimage* of B' . In particular, the set of all images (i.e., $f(A)$) is called the image or *range* of f .

To each mapping $f: A \rightarrow B$ there corresponds the subset of $A \times B$ given by $\{(a, f(a)) : a \in A\}$. We call this set the *graph* of f . Two mappings $f: A \rightarrow B$ and $g: A \rightarrow B$ are defined to be *equal*, written $f = g$, if $f(a) = g(a)$ for every $a \in A$ — that is, if they have the same graph. Thus, we do not distinguish between a function and its graph. The negation of $f = g$ is written $f \neq g$ and is the statement:

There exists an $a \in A$ for which $f(a) \neq g(a)$.

Sometimes the “barred” arrow $\mapsto$ is used to denote the image of an arbitrary element $x \in A$ under a mapping $f: A \rightarrow B$ by writing

$$x \mapsto f(x)$$

This is illustrated in the following example.

EXAMPLE 5.1

- (a) Let $f: \mathbf{R} \rightarrow \mathbf{R}$ be the function that assigns to each real number x its square x^2 . We can denote this function by writing

$$f(x) = x^2 \quad \text{or} \quad x \mapsto x^2$$

Here the image of -3 is 9 , so we may write $f(-3) = 9$. However, $f^{-1}(9) = \{3, -3\}$. Also, $f(\mathbf{R}) = [0, \infty) = \{x : x \geq 0\}$ is the image of f .

- (b) Let $A = \{a, b, c, d\}$ and $B = \{x, y, z, t\}$. Then the following defines a mapping $f: A \rightarrow B$:

$$f(a) = y, f(b) = x, f(c) = z, f(d) = y \quad \text{or} \quad f = \{(a, y), (b, x), (c, z), (d, y)\}$$

The first defines the mapping explicitly, and the second defines the mapping by its graph. Here,

$$f(\{a, b, d\}) = \{f(a), f(b), f(d)\} = \{y, x, y\} = \{x, y\}$$

Furthermore, $f(A) = \{x, y, z\}$ is the image of f .

EXAMPLE 5.2 Let V be the vector space of polynomials over $\mathbf{R}$, and let $p(t) = 3t^2 - 5t + 2$.

- (a) The derivative defines a mapping $\mathbf{D}: V \rightarrow V$ where, for any polynomials $f(t)$, we have $\mathbf{D}(f) = df/dt$. Thus,

$$\mathbf{D}(p) = \mathbf{D}(3t^2 - 5t + 2) = 6t - 5$$

- (b) The integral, say from 0 to 1, defines a mapping $\mathbf{J}: V \rightarrow \mathbf{R}$. That is, for any polynomial $f(t)$,

$$\mathbf{J}(f) = \int_0^1 f(t) dt, \quad \text{and so} \quad \mathbf{J}(p) = \int_0^1 (3t^2 - 5t + 2) dt = \frac{1}{2}$$

Observe that the mapping in (b) is from the vector space V into the scalar field $\mathbf{R}$, whereas the mapping in (a) is from the vector space V into itself.

Matrix Mappings

Let A be any $m \times n$ matrix over K . Then A determines a mapping $F_A: K^n \rightarrow K^m$ by

$$F_A(u) = Au$$

where the vectors in K^n and K^m are written as columns. For example, suppose

$$A = \begin{bmatrix} 1 & -4 & 5 \\ 2 & 3 & -6 \end{bmatrix} \quad \text{and} \quad u = \begin{bmatrix} 1 \\ 3 \\ -5 \end{bmatrix}$$

then

$$F_A(u) = Au = \begin{bmatrix} 1 & -4 & 5 \\ 2 & 3 & -6 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ -5 \end{bmatrix} = \begin{bmatrix} -36 \\ 41 \end{bmatrix}$$

Remark: For notational convenience, we will frequently denote the mapping F_A by the letter A , the same symbol as used for the matrix.

Composition of Mappings

Consider two mappings $f: A \rightarrow B$ and $g: B \rightarrow C$, illustrated below:

$$A \xrightarrow{f} B \xrightarrow{g} C$$

The *composition* of f and g , denoted by $g \circ f$, is the mapping $g \circ f: A \rightarrow C$ defined by

$$(g \circ f)(a) \equiv g(f(a))$$

That is, first we apply f to $a \in A$, and then we apply g to $f(a) \in B$ to get $g(f(a)) \in C$. Viewing f and g as “computers,” the composition means we first input $a \in A$ to get the output $f(a) \in B$ using f , and then we input $f(a)$ to get the output $g(f(a)) \in C$ using g .

Our first theorem tells us that the composition of mappings satisfies the associative law.

THEOREM 5.1: Let $f: A \rightarrow B$, $g: B \rightarrow C$, $h: C \rightarrow D$. Then

$$h \circ (g \circ f) = (h \circ g) \circ f$$

We prove this theorem here. Let $a \in A$. Then

$$\begin{aligned}(h \circ (g \circ f))(a) &= h((g \circ f)(a)) = h(g(f(a))) \\ ((h \circ g) \circ f)(a) &= (h \circ g)(f(a)) = h(g(f(a)))\end{aligned}$$

Thus, $(h \circ (g \circ f))(a) = ((h \circ g) \circ f)(a)$ for every $a \in A$, and so $h \circ (g \circ f) = (h \circ g) \circ f$.

One-to-One and Onto Mappings

We formally introduce some special types of mappings.

DEFINITION: A mapping $f: A \rightarrow B$ is said to be *one-to-one* (or 1-1 or *injective*) if different elements of A have distinct images; that is,

$$\text{If } f(a) = f(a'), \text{ then } a = a'.$$

DEFINITION: A mapping $f: A \rightarrow B$ is said to be *onto* (or f maps A onto B or *surjective*) if every $b \in B$ is the image of at least one $a \in A$.

DEFINITION: A mapping $f: A \rightarrow B$ is said to be a *one-to-one correspondence* between A and B (or *bijjective*) if f is both one-to-one and onto.

EXAMPLE 5.3 Let $f: \mathbf{R} \rightarrow \mathbf{R}$, $g: \mathbf{R} \rightarrow \mathbf{R}$, $h: \mathbf{R} \rightarrow \mathbf{R}$ be defined by

$$f(x) = 2^x, \quad g(x) = x^3 - x, \quad h(x) = x^2$$

The graphs of these functions are shown in [Fig. 5-1](#). The function f is one-to-one. Geometrically, this means that each horizontal line does not contain more than one point of f . The function g is onto. Geometrically, this means that each horizontal line contains at least one point of g . The function h is neither one-to-one nor onto. For example, both 2 and -2 have the same image 4, and -16 has no preimage.

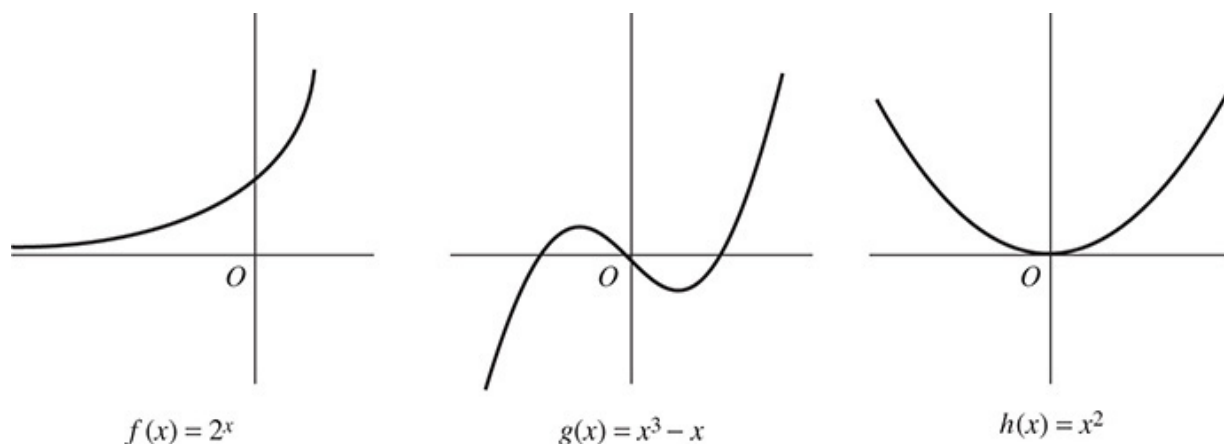


Figure 5-1

Identity and Inverse Mappings

Let A be any nonempty set. The mapping $f: A \rightarrow A$ defined by $f(a) = a$ —that is, the function that assigns to each element in A itself—is called *identity mapping*. It is usually denoted by $\mathbf{1}_A$ or $\mathbf{1}$ or I . Thus, for any $a \in A$, we have $\mathbf{1}_A(a) = a$.

Now let $f: A \rightarrow B$. We call $g: B \rightarrow A$ the inverse of f , written f^{-1} , if

$$f \circ g = \mathbf{1}_B \quad \text{and} \quad g \circ f = \mathbf{1}_A$$

We emphasize that f has an inverse if and only if f is a one-to-one correspondence between A and B ; that is, f is one-to-one and onto ([Problem 5.7](#)). Also, if $b \in B$, then $f^{-1}(b) = a$, where a is the unique element of A for which $f(a) = b$.

5.3 Linear Mappings (Linear Transformations)

We begin with a definition.

DEFINITION: Let V and U be vector spaces over the same field K . A mapping $F: V \rightarrow U$ is called a *linear mapping* or *linear transformation* if it satisfies the following two conditions:

- (1) For any vectors $u, w \in V$, $F(u + w) = F(u) + F(w)$.
- (2) For any scalar k and vector $u \in V$, $F(ku) = kF(u)$.

Namely, $F: V \rightarrow U$ is linear if it “preserves” the two basic operations of a vector space, that of vector addition and that of scalar multiplication.

Substituting $k = 0$ into condition (2), we obtain $F(0) = 0$. Thus, every linear mapping takes the zero vector into the zero vector.

Now for any scalars $a, b \in K$ and any vector $u, w \in V$, we obtain

$$F(av + bw) = F(av) + F(bw) = aF(v) + bF(w)$$

More generally, for any scalars $a_i \in K$ and any vectors $u_i \in V$, we obtain the following basic property of linear mappings:

$$F(a_1 v_1 + a_2 v_2 + \cdots + a_m v_m) = a_1 F(v_1) + a_2 F(v_2) + \cdots + a_m F(v_m)$$

Remark 1: A linear mapping $F: V \rightarrow U$ is completely characterized by the condition

$$F(av + bw) = aF(v) + bF(w)$$

and so this condition is sometimes used as its definition.

Remark 2: The term *linear transformation* rather than *linear mapping* is frequently used for linear mappings of the form $F: \mathbf{R}^n \rightarrow \mathbf{R}^m$.

EXAMPLE 5.4

- (a) Let $F: \mathbf{R}^3 \rightarrow \mathbf{R}^3$ be the “projection” mapping into the xy -plane; that is, F is the mapping defined by $F(x, y, z) = (x, y, 0)$. We show that F is linear. Let $v = (a, b, c)$ and $w = (a', b', c')$. Then

$$\begin{aligned} F(v + w) &= F(a + a', b + b', c + c') = (a + a', b + b', 0) \\ &= (a, b, 0) + (a', b', 0) = F(v) + F(w) \end{aligned}$$

and, for any scalar k ,

$$F(kv) = F(ka, kb, kc) = (ka, kb, 0) = k(a, b, 0) = kF(v)$$

Thus, F is linear.

- (b) Let $G: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ be the “translation” mapping defined by $G(x, y) = (x + 1, y + 2)$. [That is, G adds the vector $(1, 2)$ to any vector $v = (x, y)$ in $\mathbf{R}^2$.] Note that

$$G(0) = G(0, 0) = (1, 2) \neq 0$$

Thus, the zero vector is not mapped into the zero vector. Hence, G is not linear.

EXAMPLE 5.5 (Derivative and Integral Mappings) Consider the vector space $V = \mathbf{P}(t)$ of polynomials over the real field $\mathbf{R}$. Let $u(t)$ and $v(t)$ be any polynomials in V and let k be any scalar.

- (a) Let $\mathbf{D}: V \rightarrow V$ be the derivative mapping. One proves in calculus that

$$\frac{d(u + v)}{dt} = \frac{du}{dt} + \frac{dv}{dt} \quad \text{and} \quad \frac{d(ku)}{dt} = k \frac{du}{dt}$$

That is, $\mathbf{D}(u + v) = \mathbf{D}(u) + \mathbf{D}(v)$ and $\mathbf{D}(ku) = k\mathbf{D}(u)$. Thus, the derivative mapping is linear.

- (b) Let $\mathbf{J}: V \rightarrow \mathbf{R}$ be an integral mapping, say

$$\mathbf{J}(f(t)) = \int_0^1 f(t) dt$$

One also proves in calculus that,

$$\int_0^1 [u(t) + v(t)] dt = \int_0^1 u(t) dt + \int_0^1 v(t) dt$$

and

$$\int_0^1 ku(t) dt = k \int_0^1 u(t) dt$$

That is, $\mathbf{J}(u + v) = \mathbf{J}(u) + \mathbf{J}(v)$ and $\mathbf{J}(ku) = k\mathbf{J}(u)$. Thus, the integral mapping is linear.

EXAMPLE 5.6 (Zero and Identity Mappings)

(a) Let $F: V \rightarrow U$ be the mapping that assigns the zero vector $0 \in U$ to every vector $v \in V$. Then, for any vectors $v, w \in V$ and any scalar $k \in K$, we have

$$F(v + w) = 0 = 0 + 0 = F(v) + F(w) \quad \text{and} \quad F(kv) = 0 = k0 = kF(v)$$

Thus, F is linear. We call F the *zero mapping*, and we usually denote it by 0 .

(b) Consider the identity mapping $I: V \rightarrow V$, which maps each $v \in V$ into itself. Then, for any vectors $v, w \in V$ and any scalars $a, b \in K$, we have

$$I(av + bw) = av + bw = aI(v) + bI(w)$$

Thus, I is linear.

Our next theorem (proved in [Problem 5.13](#)) gives us an abundance of examples of linear mappings. In particular, it tells us that a linear mapping is completely determined by its values on the elements of a basis.

THEOREM 5.2: Let V and U be vector spaces over a field K . Let $\{v_1, v_2, \dots, v_n\}$ be a basis of V and let $u_1, u_2, \dots, u_n$ be any vectors in U . Then there exists a unique linear mapping $F: V \rightarrow U$ such that $F(v_1) = u_1, F(v_2) = u_2, \dots, F(v_n) = u_n$.

We emphasize that the vectors $u_1, u_2, \dots, u_n$ in [Theorem 5.2](#) are completely arbitrary; they may be linearly

dependent or they may even be equal to each other.

Matrices as Linear Mappings

Let A be any real $m \times n$ matrix. Recall that A determines a mapping $F_A : K^n \rightarrow K^m$ by $F_A(u) = Au$ (where the vectors in K^n and K^m are written as columns). We show F_A is linear. By matrix multiplication,

$$\begin{aligned}F_A(v + w) &= A(v + w) = Av + Aw = F_A(v) + F_A(w) \\F_A(kv) &= A(kv) = k(Av) = kF_A(v)\end{aligned}$$

In other words, using A to represent the mapping, we have

$$A(v + w) = Av + Aw \quad \text{and} \quad A(kv) = k(Av)$$

Thus, the matrix mapping A is linear.

Vector Space Isomorphism

The notion of two vector spaces being isomorphic was defined in [Chapter 4](#) when we investigated the coordinates of a vector relative to a basis. We now redefine this concept.

DEFINITION: Two vector spaces V and U over K are *isomorphic*, written $V \cong U$, if there exists a bijective (one-to-one and onto) linear mapping $F : V \rightarrow U$. The mapping F is then called an *isomorphism* between V and U .

Consider any vector space V of dimension n and let S be any basis of V . Then the mapping

$$x \mapsto (v)_S$$

which maps each vector $v \in V$ into its coordinate vector $[v]_S$, is an isomorphism between V and K^n .

5.4 Kernel and Image of a Linear Mapping

We begin by defining two concepts.

DEFINITION: Let $F : V \rightarrow U$ be a linear mapping. The *kernel* of F , written $\text{Ker } F$, is the set of elements in V that map into the zero vector 0 in U ; that is,

$$\text{Ker } F = \{v \in V : F(v) = 0\}$$

The *image* (or *range*) of F , written $\text{Im } F$, is the set of image points in U ; that is,

$$\text{Im } F = \{u \in U : \text{there exists } v \in V \text{ for which } F(v) = u\}$$

The following theorem is easily proved ([Problem 5.22](#)).

THEOREM 5.3: Let $F : V \rightarrow U$ be a linear mapping. Then the kernel of F is a subspace of V and the image of F is a subspace of U .

Now suppose that $v_1, v_2, \dots, v_m$ span a vector space V and that $F : V \rightarrow U$ is linear. We show that $F(v_1), F(v_2), \dots, F(v_m)$ span $\text{Im } F$. Let $u \in \text{Im } F$. Then there exists $v \in V$ such that $F(v) = u$. Because the v_i 's span V and $v \in V$, there exist scalars $a_1, a_2, \dots, a_m$ for which

$$v = a_1 v_1 + a_2 v_2 + \cdots + a_m v_m$$

Therefore,

$$u = F(v) = F(a_1 v_1 + a_2 v_2 + \cdots + a_m v_m) = a_1 F(v_1) + a_2 F(v_2) + \cdots + a_m F(v_m)$$

Thus, the vectors $F(v_1), F(v_2), \dots, F(v_m)$ span $\text{Im } F$.

We formally state the above result.

PROPOSITION 5.4: Suppose $v_1, v_2, \dots, v_m$ span a vector space V , and suppose $F: V \rightarrow U$ is linear. Then $F(v_1), F(v_2), \dots, F(v_m)$ span $\text{Im } F$.

EXAMPLE 5.7

(a) Let $F: \mathbf{R}^3 \rightarrow \mathbf{R}^3$ be the projection of a vector v into the xy -plane [as pictured in Fig. 5-2(a)]; that is,

$$F(x, y, z) = (x, y, 0)$$

Clearly the image of F is the entire xy -plane—that is, points of the form $(x, y, 0)$. Moreover, the kernel of F is the z -axis—that is, points of the form $(0, 0, c)$. That is,

$$\text{Im } F = \{(a, b, c) : c = 0\} = xy\text{-plane} \quad \text{and} \quad \text{Ker } F = \{(a, b, c) : a = 0, b = 0\} = z\text{-axis}$$

(b) Let $G: \mathbf{R}^3 \rightarrow \mathbf{R}^3$ be the linear mapping that rotates a vector v about the z -axis through an angle θ [as pictured in Fig. 5-2(b)]; that is,

$$G(x, y, z) = (x \cos \theta - y \sin \theta, x \sin \theta + y \cos \theta, z)$$

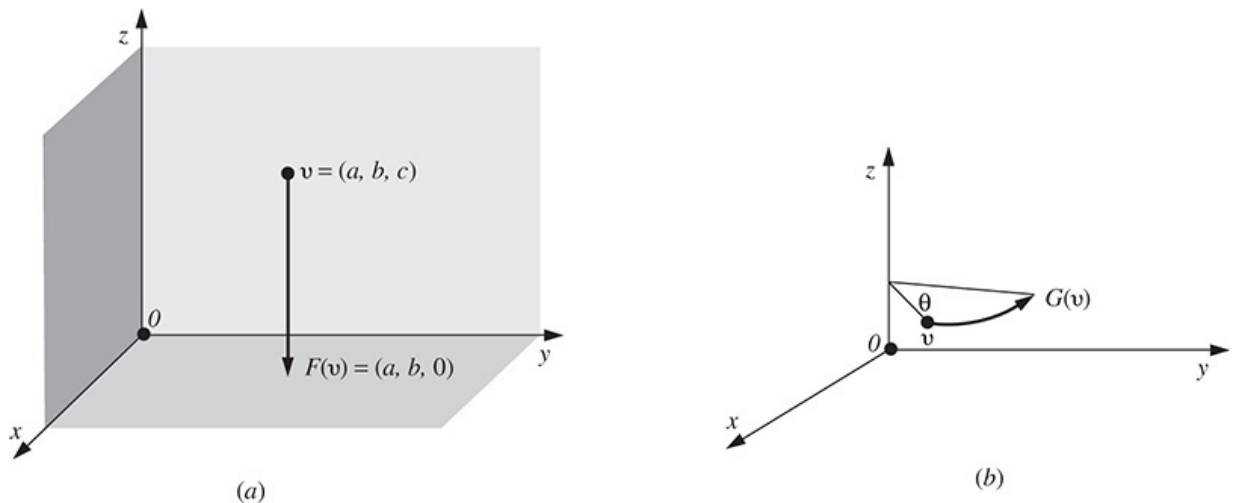


Figure 5-2

Observe that the distance of a vector v from the origin O does not change under the rotation, and so only the zero vector 0 is mapped into the zero vector 0 . Thus, $\text{Ker } G = \{0\}$. On the other hand, every vector

u in $\mathbf{R}^3$ is the image of a vector v in $\mathbf{R}^3$ that can be obtained by rotating u back by an angle of θ . Thus, $\text{Im } G = \mathbf{R}^3$, the entire space.

EXAMPLE 5.8 Consider the vector space $V = \mathbf{P}(t)$ of polynomials over the real field $\mathbf{R}$, and let $H: V \rightarrow V$ be the third-derivative operator; that is, $H[f(t)] = d^3 f/dt^3$. [Sometimes the notation $\mathbf{D}^3$ is used for H , where $\mathbf{D}$ is the derivative operator.] We claim that

$$\text{Ker } H = \{\text{polynomials of degree } \leq 2\} = \mathbf{P}_2(t) \quad \text{and} \quad \text{Im } H = V$$

The first comes from the fact that $H(at^2 + bt + c) = 0$ but $H(t^n) \neq 0$ for $n \geq 3$. The second comes from that fact that every polynomial $g(t)$ in V is the third derivative of some polynomial $f(t)$ (which can be obtained by taking the antiderivative of $g(t)$ three times).

Kernel and Image of Matrix Mappings

Consider, say, a 3×4 matrix A and the usual basis $\{e_1, e_2, e_3, e_4\}$ of K^4 (written as columns):

$$A = \begin{bmatrix} a_1 & a_2 & a_3 & a_4 \\ b_1 & b_2 & b_3 & b_4 \\ c_1 & c_2 & c_3 & c_4 \end{bmatrix}, \quad e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad e_4 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Recall that A may be viewed as a linear mapping $A: K^4 \rightarrow K^3$, where the vectors in K^4 and K^3 are viewed as column vectors. Now the usual basis vectors span K^4 , so their images Ae_1, Ae_2, Ae_3, Ae_4 span the image of A . But the vectors Ae_1, Ae_2, Ae_3, Ae_4 are precisely the columns of A :

$$Ae_1 = [a_1, b_1, c_1]^T, \quad Ae_2 = [a_2, b_2, c_2]^T, \quad Ae_3 = [a_3, b_3, c_3]^T, \quad Ae_4 = [a_4, b_4, c_4]^T$$

Thus, the image of A is precisely the column space of A .

On the other hand, the kernel of A consists of all vectors v for which $Av = 0$. This means that the kernel of A is the

solution space of the homogeneous system $AX = 0$, called the *null space* of A .

We state the above results formally.

PROPOSITION 5.5: Let A be any $m \times n$ matrix over a field K viewed as a linear map $A : K^n \rightarrow K^m$. Then

$$\text{Ker } A = \text{nullsp}(A) \quad \text{and} \quad \text{Im } A = \text{colsp}(A)$$

Here $\text{colsp}(A)$ denotes the column space of A , and $\text{nullsp}(A)$ denotes the null space of A .

Rank and Nullity of a Linear Mapping

Let $F : V \rightarrow U$ be a linear mapping. The *rank* of F is defined to be the dimension of its image, and the *nullity* of F is defined to be the dimension of its kernel; namely,

$$\text{rank}(F) = \dim(\text{Im } F) \quad \text{and} \quad \text{nullity}(F) = \dim(\text{Ker } F)$$

The following important theorem (proved in [Problem 5.23](#)) holds.

THEOREM 5.6 Let V be of finite dimension, and let $F : V \rightarrow U$ be linear. Then

$$\dim V = \dim(\text{Ker } F) + \dim(\text{Im } F) = \text{nullity}(F) + \text{rank}(F)$$

Recall that the rank of a matrix A was also defined to be the dimension of its column space and row space. If we now view A as a linear mapping, then both definitions correspond, because the image of A is precisely its column space.

EXAMPLE 5.9 Let $F : \mathbf{R}^4 \rightarrow \mathbf{R}^3$ be the linear mapping defined by

$$F(x, y, z, t) = (x - y + z + t, \quad 2x - 2y + 3z + 4t, \quad 3x - 3y + 4z + 5t)$$

(a) Find a basis and the dimension of the image of F .

First find the image of the usual basis vectors of $\mathbf{R}^4$,

$$F(1, 0, 0, 0) = (1, 2, 3), \quad F(0, 0, 1, 0) = (1, 3, 4)$$

$$F(0, 1, 0, 0) = (-1, -2, -3), \quad F(0, 0, 0, 1) = (1, 4, 5)$$

By Proposition 5.4, the image vectors span $\text{Im } F$. Hence, form the matrix M whose rows are these image vectors and row reduce to echelon form:

$$M = \begin{bmatrix} 1 & 2 & 3 \\ -1 & -2 & -3 \\ 1 & 3 & 4 \\ 1 & 4 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 2 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Thus, $(1, 2, 3)$ and $(0, 1, 1)$ form a basis of $\text{Im } F$. Hence, $\dim(\text{Im } F) = 2$ and $\text{rank}(F) = 2$.

(b) Find a basis and the dimension of the kernel of the map F .

Set $F(v) = 0$, where $v = (x, y, z, t)$,

$$F(x, y, z, t) = (x - y + z + t, \quad 2x - 2y + 3z + 4t, \quad 3x - 3y + 4z + 5t) = (0, 0, 0)$$

Set corresponding components equal to each other to form the following homogeneous system whose solution space is $\text{Ker } F$:

$$\begin{array}{lll} x - y + z + t = 0 & & x - y + z + t = 0 \\ 2x - 2y + 3z + 4t = 0 & \text{or} & z + 2t = 0 \\ 3x - 3y + 4z + 5t = 0 & & z + 2t = 0 \end{array} \quad \text{or} \quad \begin{array}{l} x - y + z + t = 0 \\ z + 2t = 0 \end{array}$$

The free variables are y and t . Hence, $\dim(\text{Ker } F) = 2$ or $\text{nullity}(F) = 2$.

(i) Set $y = 1, t = 0$ to obtain the solution $(1, 1, 0, 0)$,

(ii) Set $y = 0, t = 1$ to obtain the solution $(1, 0, -2, 1)$.

Thus, $(-1, 1, 0, 0)$ and $(1, 0, -2, 1)$ form a basis for $\text{Ker } F$.

As expected from [Theorem 5.6](#), $\dim(\text{Im } F) + \dim(\text{Ker } F) = 4 = \dim \mathbf{R}^4$.

Application to Systems of Linear Equations

Let $AX = B$ denote the matrix form of a system of m linear equations in n unknowns. Now the matrix A may be viewed

as a linear mapping

$$A: K^n \rightarrow K^m$$

Thus, the solution of the equation $AX = B$ may be viewed as the preimage of the vector $B \in K^m$ under the linear mapping A . Furthermore, the solution of the associated homogeneous system

$$AX = 0$$

may be viewed as the kernel of the linear mapping A . Applying [Theorem 5.6](#) to this homogeneous system yields

$$\dim(\text{Ker } A) = \dim K^n - \dim(\text{Im } A) = n - \text{rank } A$$

But n is exactly the number of unknowns in the homogeneous system $AX = 0$. Thus, we have proved the following theorem of [Chapter 4](#).

THEOREM 4.19: The dimension of the solution space W of a homogenous system $AX = 0$ of linear equations is $s = n - r$, where n is the number of unknowns and r is the rank of the coefficient matrix A .

Observe that r is also the number of pivot variables in an echelon form of $AX = 0$, so $s = n - r$ is also the number of free variables. Furthermore, the s solution vectors of $AX = 0$ described in [Theorem 3.14](#) are linearly independent ([Problem 4.52](#)). Accordingly, because $\dim W = s$, they form a basis for the solution space W . Thus, we have also proved [Theorem 3.14](#).

5.5 Singular and Nonsingular Linear Mappings, Isomorphisms

Let $F: V \rightarrow U$ be a linear mapping. Recall that $F(0) = 0$. F is said to be *singular* if the image of some nonzero vector v is 0 —that is, if there exists $v \neq 0$ such that $F(v) = 0$. Thus, $F: V \rightarrow U$ is *nonsingular* if the zero vector 0 is the only vector whose image under F is 0 or, in other words, if $\text{Ker } F = \{0\}$.

EXAMPLE 5.10 Consider the projection map $F: \mathbf{R}^3 \rightarrow \mathbf{R}^3$ and the rotation map $G: \mathbf{R}^3 \rightarrow \mathbf{R}^3$ appearing in Fig. 5-2. (See Example 5.7.) Because the kernel of F is the z -axis, F is singular. On the other hand, the kernel of G consists only of the zero vector 0 . Thus, G is nonsingular.

Nonsingular linear mappings may also be characterized as those mappings that carry independent sets into independent sets. Specifically, we prove (Problem 5.28) the following theorem.

THEOREM 5.7: Let $F: V \rightarrow U$ be a nonsingular linear mapping. Then the image of any linearly independent set is linearly independent.

Isomorphisms

Suppose a linear mapping $F: V \rightarrow U$ is one-to-one. Then only $0 \in V$ can map into $0 \in U$, and so F is nonsingular. The converse is also true. For suppose F is nonsingular and $F(v) = F(w)$, then $F(v - w) = F(v) - F(w) = 0$, and hence, $v - w = 0$ or $v = w$. Thus, $F(v) = F(w)$ implies $v = w$ —that is, F is one-to-one. We have proved the following proposition.

PROPOSITION 5.8: A linear mapping $F: V \rightarrow U$ is one-to-one if and only if F is nonsingular.

Recall that a mapping $F: V \rightarrow U$ is called an *isomorphism* if F is linear and if F is bijective (i.e., if F is one-to-one and

onto). Also, recall that a vector space V is said to be *isomorphic* to a vector space U , written $V \cong U$, if there is an isomorphism $F: V \rightarrow U$.

The following theorem (proved in [Problem 5.29](#)) applies.

THEOREM 5.9: Suppose V has finite dimension and $\dim V = \dim U$. Suppose $F: V \rightarrow U$ is linear. Then F is an isomorphism if and only if F is nonsingular.

5.6 Operations with Linear Mappings

We are able to combine linear mappings in various ways to obtain new linear mappings. These operations are very important and will be used throughout the text.

Let $F: V \rightarrow U$ and $G: V \rightarrow U$ be linear mappings over a field K . The sum $F + G$ and the scalar product kF , where $k \in K$, are defined to be the following mappings from V into U :

$$(F + G)(v) \equiv F(v) + G(v) \quad \text{and} \quad (kF)(v) \equiv kF(v)$$

We now show that if F and G are linear, then $F + G$ and kF are also linear. Specifically, for any vectors $v, w \in V$ and any scalars $a, b \in K$,

$$\begin{aligned} (F + G)(av + bw) &= F(av + bw) + G(av + bw) \\ &= aF(v) + bF(w) + aG(v) + bG(w) \\ &= a[F(v) + G(v)] + b[F(w) + G(w)] \\ &= a(F + G)(v) + b(F + G)(w) \end{aligned}$$

and

$$\begin{aligned}(kF)(av + bw) &= kF(av + bw) = k[aF(v) + bF(w)] \\ &= akF(v) + bkF(w) = a(kF)(v) + b(kF)(w)\end{aligned}$$

Thus, $F + G$ and kF are linear.

The following theorem holds.

THEOREM 5.10: Let V and U be vector spaces over a field K . Then the collection of all linear mappings from V into U with the above operations of addition and scalar multiplication forms a vector space over K .

The vector space of linear mappings in [Theorem 5.10](#) is usually denoted by

$$\text{Hom}(V, U)$$

Here Hom comes from the word “homomorphism.” We emphasize that the proof of [Theorem 5.10](#) reduces to showing that $\text{Hom}(V, U)$ does satisfy the eight axioms of a vector space. The zero element of $\text{Hom}(V, U)$ is the *zero mapping* from V into U , denoted by $\mathbf{0}$ and defined by

$$\mathbf{0}(v) = \mathbf{0}$$

for every vector $v \in V$.

Suppose V and U are of finite dimension. Then we have the following theorem.

THEOREM 5.11: Suppose $\dim V = m$ and $\dim U = n$. Then $\dim[\text{Hom}(V, U)] = mn$.

Composition of Linear Mappings

Now suppose V , U , and W are vector spaces over the same field K , and suppose $F: V \rightarrow U$ and $G: U \rightarrow W$ are linear mappings. We picture these mappings as follows:

$$V \xrightarrow{F} U \xrightarrow{G} W$$

Recall that the composition function $G \circ F$ is the mapping from V into W defined by $(G \circ F)(v) = G(F(v))$. We show that $G \circ F$ is linear whenever F and G are linear. Specifically, for any vectors $v, w \in V$ and any scalars $a, b \in K$, we have

$$\begin{aligned}(G \circ F)(av + bw) &= G(F(av + bw)) = G(aF(v) + bF(w)) \\ &= aG(F(v)) + bG(F(w)) = a(G \circ F)(v) + b(G \circ F)(w)\end{aligned}$$

Thus, $G \circ F$ is linear.

The composition of linear mappings and the operations of addition and scalar multiplication are related as follows.

THEOREM 5.12: Let V, U, W be vector spaces over K . Suppose the following mappings are linear:

$$F: V \rightarrow U, \quad F': V \rightarrow U \quad \text{and} \quad G: U \rightarrow W, \quad G': U \rightarrow W$$

Then, for any scalar $k \in K$:

- (i) $G \circ (F + F') = G \circ F + G \circ F'$.
- (ii) $(G + G') \circ F = G \circ F + G' \circ F$.
- (iii) $k(G \circ F) = (kG) \circ F = G \circ (kF)$.

5.7 Algebra $A(V)$ of Linear Operators

Let V be a vector space over a field K . This section considers the special case of linear mappings from the vector space V into itself—that is, linear mappings of the form $F: V \rightarrow V$.

They are also called *linear operators* or *linear transformations* on V . We will write $A(V)$, instead of $\text{Hom}(V, V)$, for the space of all such mappings.

Now $A(V)$ is a vector space over K (Theorem 5.8), and, if $\dim V = n$, then $\dim A(V) = n^2$. Moreover, for any mappings $F, G \in A(V)$, the composition $G \circ F$ exists and also belongs to $A(V)$. Thus, we have a “multiplication” defined in $A(V)$. [We sometimes write FG instead of $G \circ F$ in the space $A(V)$.]

Remark: An *algebra* A over a field K is a vector space over K in which an operation of multiplication is defined satisfying, for every $F, G, H \in A$ and every $k \in K$:

- (i) $F(G + H) = FG + FH$,
- (ii) $(G + H)F = GF + HF$,
- (iii) $k(GF) = (kG)F = G(kF)$.

The algebra is said to be *associative* if, in addition, $(FG)H = F(GH)$.

The above definition of an algebra and previous theorems give us the following result.

THEOREM 5.13: Let V be a vector space over K . Then $A(V)$ is an associative algebra over K with respect to composition of mappings. If $\dim V = n$, then $\dim A(V) = n^2$.

This is why $A(V)$ is called the *algebra of linear operators* on V .

Polynomials and Linear Operators

Observe that the identity mapping $I: V \rightarrow V$ belongs to $A(V)$. Also, for any linear operator F in $A(V)$, we have $FI = IF = F$. We can also form “powers” of F . Namely, we define

$$F^0 = I, \quad F^2 = F \circ F, \quad F^3 = F^2 \circ F = F \circ F \circ F, \quad F^4 = F^3 \circ F, \quad \dots$$

Furthermore, for any polynomial $p(t)$ over K , say,

$$p(t) = a_0 + a_1t + a_2t^2 + \cdots + a_st^s$$

we can form the linear operator $p(F)$ defined by

$$p(F) = a_0I + a_1F + a_2F^2 + \cdots + a_sF^s$$

(For any scalar k , the operator kI is sometimes denoted simply by k .) In particular, we say F is a *zero* of the polynomial $p(t)$ if $p(F) = 0$.

EXAMPLE 5.11 Let $F : K^3 \rightarrow K^3$ be defined by $F(x, y, z) = (0, x, y)$. For any $(a, b, c) \in K^3$,

$$(F + I)(a, b, c) = (0, a, b) + (a, b, c) = (a, a + b, b + c)$$

$$F^3(a, b, c) = F^2(0, a, b) = F(0, 0, a) = (0, 0, 0)$$

Thus, $F^3 = 0$, the zero mapping in $A(V)$. This means F is a zero of the polynomial $p(t) = t^3$.

Square Matrices as Linear Operators

Let $\mathbf{M} = \mathbf{M}_{n,n}$ be the vector space of all square $n \times n$ matrices over K . Then any matrix A in M defines a linear mapping $F_A : K^n \rightarrow K^n$ by $F_A(u) = Au$ (where the vectors in K^n are written as columns). Because the mapping is from K^n into itself, the square matrix A is a linear operator, not simply a linear mapping.

Suppose A and B are matrices in M . Then the matrix product AB is defined. Furthermore, for any (column) vector u in K^n ,

$$F_{AB}(u) = (AB)u = A(Bu) = A(F_B(u)) = F_A(F_B(u)) = (F_A \circ F_B)(u)$$

In other words, the matrix product AB corresponds to the composition of A and B as linear mappings. Similarly, the

matrix sum $A + B$ corresponds to the sum of A and B as linear mappings, and the scalar product kA corresponds to the scalar product of A as a linear mapping.

Invertible Operators in $A(V)$

Let $F: V \rightarrow V$ be a linear operator. F is said to be *invertible* if it has an inverse—that is, if there exists F^{-1} in $A(V)$ such that $FF^{-1} = F^{-1}F = I$. On the other hand, F is invertible as a mapping if F is both one-to-one and onto. In such a case, F^{-1} is also linear and F^{-1} is the inverse of F as a linear operator (proved in [Problem 5.15](#)).

Suppose F is invertible. Then only $0 \in V$ can map into itself, and so F is nonsingular. The converse is not true, as seen by the following example.

EXAMPLE 5.12 Let $V = \mathbf{P}(t)$, the vector space of polynomials over K . Let F be the mapping on V that increases by 1 the exponent of t in each term of a polynomial; that is,

$$F(a_0 + a_1t + a_2t^2 + \cdots + a_st^s) = a_0t + a_1t^2 + a_2t^3 + \cdots + a_st^{s+1}$$

Then F is a linear mapping and F is nonsingular. However, F is not onto, and so F is not invertible.

The vector space $V = \mathbf{P}(t)$ in the above example has infinite dimension. The situation changes significantly when V has finite dimension. Namely, the following theorem applies.

THEOREM 5.14: Let F be a linear operator on a finite-dimensional vector space V . Then the following four conditions are equivalent.

- | | |
|---|-------------------------------|
| (i) F is nonsingular: $\text{Ker } F = \{0\}$. | (iii) F is an onto mapping. |
| (ii) F is one-to-one. | (iv) F is invertible. |

The proof of the above theorem mainly follows from [Theorem 5.6](#), which tells us that

$$\dim V = \dim(\text{Ker } F) + \dim(\text{Im } F)$$

By Proposition 5.8, (i) and (ii) are equivalent. Note that (iv) is equivalent to (ii) and (iii). Thus, to prove the theorem, we need only show that (i) and (iii) are equivalent. This we do below.

- (a) Suppose (i) holds. Then $\dim(\text{Ker } F) = 0$, and so the above equation tells us that $\dim V = \dim(\text{Im } F)$. This means $V = \text{Im } F$ or, in other words, F is an onto mapping. Thus, (i) implies (iii).
- (b) Suppose (iii) holds. Then $V = \text{Im } F$, and so $\dim V = \dim(\text{Im } F)$. Therefore, the above equation tells us that $\dim(\text{Ker } F) = 0$, and so F is nonsingular. Therefore, (iii) implies (i).

Accordingly, all four conditions are equivalent.

Remark: Suppose A is a square $n \times n$ matrix over K . Then A may be viewed as a linear operator on K^n . Because K^n has finite dimension, [Theorem 5.14](#) holds for the square matrix A . This is why the terms “nonsingular” and “invertible” are used interchangeably when applied to square matrices.

EXAMPLE 5.13 Let F be the linear operator on $\mathbf{R}^2$ defined by $F(x, y) = (2x + y, 3x + 2y)$.

- (a) To show that F is invertible, we need only show that F is nonsingular. Set $F(x, y) = (0, 0)$ to obtain the homogeneous system $2x + y = 0$ and $3x + 2y = 0$

$$2x + y = 0 \quad \text{and} \quad 3x + 2y = 0$$

Solve for x and y to get $x = 0$, $y = 0$. Hence, F is nonsingular and so invertible.

(b) To find a formula for F^{-1} , we set $F(x, y) = (s, t)$ and so $F^{-1}(s, t) = (x, y)$. We have

$$(2x + y, 3x + 2y) = (s, t) \quad \text{or} \quad \begin{aligned} 2x + y &= s \\ 3x + 2y &= t \end{aligned}$$

Solve for x and y in terms of s and t to obtain $x = 2s - t$, $y = -3s + 2t$. Thus,

$$F^{-1}(s, t) = (2s - t, -3s + 2t) \quad \text{or} \quad F^{-1}(x, y) = (2x - y, -3x + 2y)$$

where we rewrite the formula for F^{-1} using x and y instead of s and t .

SOLVED PROBLEMS

Mappings

5.1. State whether each diagram in Fig. 5-3 defines a mapping from $A = \{a, b, c\}$ into $B = \{x, y, z\}$.

(a) No. There is nothing assigned to the element $b \in A$.

(b) No. Two elements, x and z , are assigned to $c \in A$.

(c) Yes.

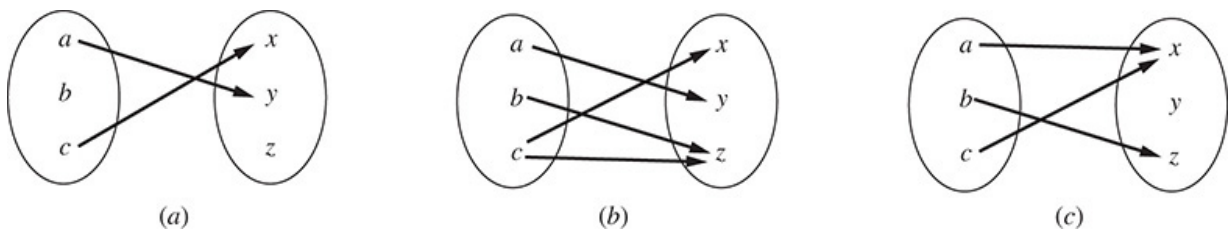


Figure 5-3

5.2. Let $f: A \rightarrow B$ and $g: B \rightarrow C$ be defined by Fig. 5-4.

(a) Find the composition mapping $(g \circ f): A \rightarrow C$.

(b) Find the images of the mappings f , g , $g \circ f$.

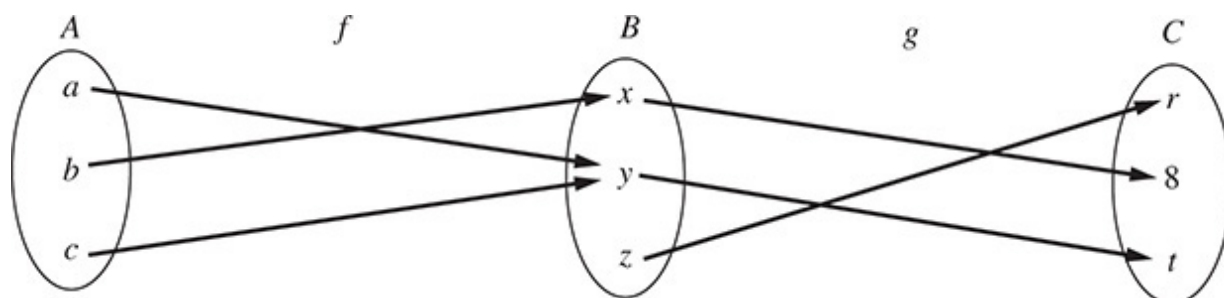


Figure 5-4

(a) Use the definition of the composition mapping to compute

$$\begin{aligned} (g \circ f)(a) &= g(f(a)) = g(y) = t, & (g \circ f)(b) &= g(f(b)) = g(x) = s \\ (g \circ f)(c) &= g(f(c)) = g(y) = t \end{aligned}$$

Observe that we arrive at the same answer if we “follow the arrows” in Fig. 5-4:

$$a \rightarrow y \rightarrow t, \quad b \rightarrow x \rightarrow s, \quad c \rightarrow y \rightarrow t$$

(b) By Fig. 5-4, the image values under the mapping f are x and y , and the image values under g are r , s , t .

Hence,

$$\text{Im } f = \{x, y\} \quad \text{and} \quad \text{Im } g = \{r, s, t\}$$

Also, by part (a), the image values under the composition mapping $g \circ f$ are t and s ; accordingly, $\text{Im } g \circ f = \{s, t\}$. Note that the images of g and $g \circ f$ are different.

5.3. Consider the mapping $F: \mathbf{R}^3 \rightarrow \mathbf{R}^2$ defined by $F(x, y, z) = (yz, x^2)$. Find

(a) $F(2, 3, 4)$; (b) $F(5, 2, 7)$; (c) $F^{-1}(0, 0)$, that is, all $u \in \mathbf{R}^3$ such that $F(u) = 0$.

(a) Substitute in the formula for F to get $F(2, 3, 4) = (3 \cdot 4, 2^2) = (12, 4)$.

(b) $F(5, -2, 7) = (-2 \cdot 7, 5^2) = (14, 25)$.

(c) Set $F(u) = 0$, where $u = (x, y, z)$, and then solve for x, y, z :

$$F(x, y, z) = (yz, x^2) = (0, 0) \quad \text{or} \quad yz = 0, x^2 = 0$$

Thus, $x = 0$ and either $y = 0$ or $z = 0$. In other words, $x = 0, y = 0$ or $x = 0, z = 0$ —that is, the z -axis and the y -axis.

5.4. Consider the mapping $F: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ defined by $F(x, y) = (3y, 2x)$. Let S be the unit circle in $\mathbf{R}^2$, that is, the solution set of $x^2 + y^2 = 1$. (a) Describe $F(S)$. (b) Find $F^{-1}(S)$.

(a) Let (a, b) be an element of $F(S)$. Then there exists $(x, y) \in S$ such that $F(x, y) = (a, b)$. Hence,

$$(3y, 2x) = (a, b) \quad \text{or} \quad 3y = a, 2x = b \quad \text{or} \quad y = \frac{a}{3}, x = \frac{b}{2}$$

Because $(x, y) \in S$ —that is, $x^2 + y^2 = 1$ —we have

$$\left(\frac{b}{2}\right)^2 + \left(\frac{a}{3}\right)^2 = 1 \quad \text{or} \quad \frac{a^2}{9} + \frac{b^2}{4} = 1$$

Thus, $F(S)$ is an ellipse.

(b) Let $F(x, y) = (a, b)$, where $(a, b) \in S$. Then $(3y, 2x) = (a, b)$ or $3y = a, 2x = b$. Because $(a, b) \in S$, we have $a^2 + b^2 = 1$. Thus, $(3y)^2 + (2x)^2 = 1$. Accordingly, $F^{-1}(S)$ is the ellipse $4x^2 + 9y^2 = 1$.

5.5. Let the mappings $f: A \rightarrow B, g: B \rightarrow C, h: C \rightarrow D$ be defined by Fig. 5-5. Determine whether or not each function is (a) one-to-one; (b) onto; (c) invertible (i.e., has an inverse).

(a) The mapping $f: A \rightarrow B$ is one-to-one, as each element of A has a different image. The mapping $g: B \rightarrow C$ is not one-to-one, because x and z both have the same image 4. The mapping $h: C \rightarrow D$ is one-to-one.

(b) The mapping $f: A \rightarrow B$ is not onto, because $z \in B$ is not the image of any element of A . The mapping $g: B \rightarrow C$ is onto, as each element of C is the image of some element of B . The mapping $h: C \rightarrow D$ is also onto.

(c) A mapping has an inverse if and only if it is one-to-one and onto. Hence, only h has an inverse.

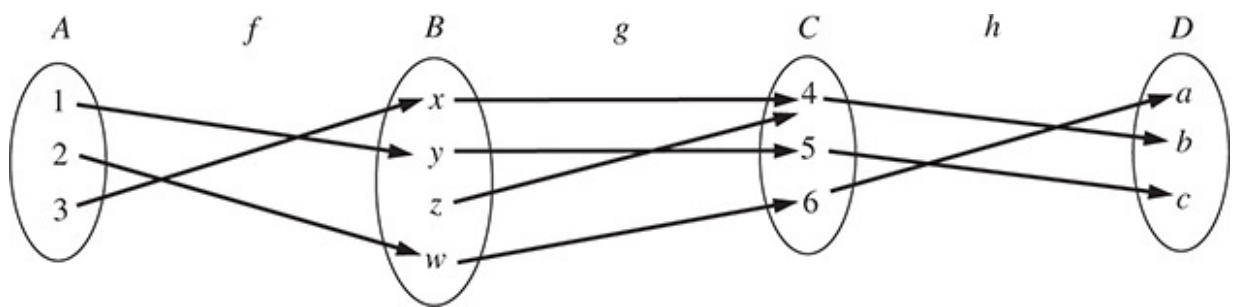


Figure 5-5

5.6. Suppose $f: A \rightarrow B$ and $g: B \rightarrow C$. Hence, $(g \circ f): A \rightarrow C$ exists. Prove

(a) If f and g are one-to-one, then $g \circ f$ is one-to-one.

- (b) If f and g are onto mappings, then $g \circ f$ is an onto mapping.
 - (c) If $g \circ f$ is one-to-one, then f is one-to-one.
 - (d) If $g \circ f$ is an onto mapping, then g is an onto mapping.
- (a) Suppose $(g \circ f)(x) = (g \circ f)(y)$. Then $g(f(x)) = g(f(y))$. Because g is one-to-one, $f(x) = f(y)$. Because f is one-to-one, $x = y$. We have proven that $(g \circ f)(x) = (g \circ f)(y)$ implies $x = y$; hence $g \circ f$ is one-to-one.
 - (b) Suppose $c \in C$. Because g is onto, there exists $b \in B$ for which $g(b) = c$. Because f is onto, there exists $a \in A$ for which $f(a) = b$. Thus, $(g \circ f)(a) = g(f(a)) = g(b) = c$. Hence, $g \circ f$ is onto.
 - (c) Suppose f is not one-to-one. Then there exist distinct elements $x, y \in A$ for which $f(x) = f(y)$. Thus, $(g \circ f)(x) = g(f(x)) = g(f(y)) = (g \circ f)(y)$. Hence, $g \circ f$ is not one-to-one. Therefore, if $g \circ f$ is one-to-one, then f must be one-to-one.
 - (d) If $a \in A$, then $(g \circ f)(a) = g(f(a)) \in g(B)$. Hence, $(g \circ f)(A) \subseteq g(B)$. Suppose g is not onto. Then $g(B)$ is properly contained in C and so $(g \circ f)(A)$ is properly contained in C ; thus, $g \circ f$ is not onto. Accordingly, if $g \circ f$ is onto, then g must be onto.

5.7. Prove that $f: A \rightarrow B$ has an inverse if and only if f is one-to-one and onto.

Suppose f has an inverse—that is, there exists a function $f^{-1}: B \rightarrow A$ for which $f^{-1} \circ f = \mathbf{1}_A$ and $f \circ f^{-1} = \mathbf{1}_B$. Because $\mathbf{1}_A$ is one-to-one, f is one-to-one by Problem 5.6(c), and because $\mathbf{1}_B$ is onto, f is onto by Problem 5.6(d); that is, f is both one-to-one and onto.

Now suppose f is both one-to-one and onto. Then each $b \in B$ is the image of a unique element in A , say b^* . Thus, if $f(a) = b$, then $a = b^*$; hence, $f(b^*) = b$. Now let g denote the mapping from B to A defined by $b \mapsto b^*$. We have

$$(i) (g \circ f)(a) = g(f(a)) = g(b) = b^* = a \text{ for every } a \in A; \text{ hence, } g \circ f = \mathbf{1}_A.$$

$$(ii) (f \circ g)(b) = f(g(b)) = f(b^*) = b \text{ for every } b \in B; \text{ hence, } f \circ g = \mathbf{1}_B.$$

Accordingly, f has an inverse. Its inverse is the mapping g .

5.8. Let $f: \mathbf{R} \rightarrow \mathbf{R}$ be defined by $f(x) = 2x - 3$. Now f is one-to-one and onto; hence, f has an inverse mapping f^{-1} . Find a formula for f^{-1} .

Let y be the image of x under the mapping f ; that is, $y = f(x) = 2x - 3$. Hence, x will be the image of y under the inverse mapping f^{-1} . Thus, solve for x in terms of y in the above equation to obtain $x = \frac{1}{2}(y + 3)$. Then the formula defining the inverse function is $f^{-1}(y) = \frac{1}{2}(y + 3)$, or, using x instead of y , $f^{-1}(x) = \frac{1}{2}(x + 3)$.

Linear Mappings

5.9. Suppose the mapping $F: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ is defined by $F(x, y) = (x + y, x)$. Show that F is linear.

We need to show that $F(u + w) = F(u) + F(w)$ and $F(kv) = kF(v)$, where u and v are any elements of $\mathbf{R}^2$ and k is any scalar. Let $u = (a, b)$ and $w = (a', b')$. Then

$$u + w = (a + a', b + b') \quad \text{and} \quad kv = (ka, kb)$$

We have $F(u) = (a + b, a)$ and $F(w) = (a' + b', a')$. Thus,

$$\begin{aligned} F(v+w) &= F(a+a', b+b') = (a+a'+b+b', a+a') \\ &= (a+b, a) + (a'+b', a') = F(v) + F(w) \end{aligned}$$

and

$$F(kv) = F(ka, kb) = (ka+kb, ka) = k(a+b, a) = kF(v)$$

Because v, w, k were arbitrary, F is linear.

5.10. Suppose $F: \mathbf{R}^3 \rightarrow \mathbf{R}^2$ is defined by $F(x, y, z) = (x + y + z, 2x - 3y + 4z)$. Show that F is linear.

We argue via matrices. Writing vectors as columns, the mapping F may be written in the form $F(v) = Av$, where $v = [x, y, z]^T$ and

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & -3 & 4 \end{bmatrix}$$

Then, using properties of matrices, we have

$$F(v+w) = A(v+w) = Av + Aw = F(v) + F(w)$$

$$F(kv) = A(kv) = k(Av) = kF(v)$$

and

Thus, F is linear.

5.11. Show that the following mappings are not linear:

(a) $F: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ defined by $F(x, y) = (xy, x)$

(b) $F: \mathbf{R}^2 \rightarrow \mathbf{R}^3$ defined by $F(x, y) = (x + 3, 2y, x + y)$

(c) $F: \mathbf{R}^3 \rightarrow \mathbf{R}^2$ defined by $F(x, y, z) = (|x|, y + z)$

(a) Let $v = (1, 2)$ and $w = (3, 4)$; then $v + w = (4, 6)$.

Also,

$$F(v) = (1(2), 1) = (2, 1) \quad \text{and} \quad F(w) = (3(4), 3) = (12, 3)$$

Hence,

$$F(v + w) = (4(6), 4) = (24, 6) \neq F(v) + F(w)$$

(b) Because $F(0, 0) = (3, 0, 0) \neq (0, 0, 0)$, F cannot be linear.

(c) Let $v = (1, 2, 3)$ and $k = -3$. Then $kv = (-3, -6, -9)$. We have

$$F(v) = (1, 5) \quad \text{and} \quad kF(v) = -3(1, 5) = (-3, -15).$$

Thus,

$$F(kv) = F(-3, -6, -9) = (3, -15) \neq kF(v)$$

Accordingly, F is not linear.

5.12. Let V be the vector space of n -square real matrices. Let M be an arbitrary but fixed matrix in V . Let $F: V \rightarrow V$ be defined by $F(A) = AM + MA$, where A is any matrix in V . Show that F is linear.

For any matrices A and B in V and any scalar k , we have

$$\begin{aligned} F(A + B) &= (A + B)M + M(A + B) = AM + BM + MA + MB \\ &= (AM + MA) + (BM + MB) = F(A) + F(B) \end{aligned}$$

and

$$F(kA) = (kA)M + M(kA) = k(AM) + k(MA) = k(AM + MA) = kF(A)$$

Thus, F is linear.

5.13. Prove [Theorem 5.2](#): Let V and U be vector spaces over a field K . Let $\{v_1, v_2, \dots, v_n\}$ be a basis of V and let

$u_1, u_2, \dots, u_n$ be any vectors in U . Then there exists a unique linear mapping $F: V \rightarrow U$ such that $F(v_1) = u_1, F(v_2) = u_2, \dots, F(v_n) = u_n$.

There are three steps to the proof of the theorem: (1) Define the mapping $F: V \rightarrow U$ such that $F(v_i) = u_i, i = 1, \dots, n$. (2) Show that F is linear. (3) Show that F is unique.

Step 1. Let $v \in V$. Because $\{v_1, \dots, v_n\}$ is a basis of V , there exist unique scalars $a_1, \dots, a_n \in K$ for which $v = a_1v_1 + a_2v_2 + \dots + a_nv_n$. We define $F: V \rightarrow U$ by

$$F(v) = a_1u_1 + a_2u_2 + \dots + a_nu_n$$

(Because the a_i are unique, the mapping F is well defined.) Now, for $i = 1, \dots, n$,

$$v_i = 0v_1 + \dots + 1v_i + \dots + 0v_n$$

Hence,

$$F(v_i) = 0u_1 + \dots + 1u_i + \dots + 0u_n = u_i$$

Thus, the first step of the proof is complete.

Step 2. Suppose $v = a_1v_1 + a_2v_2 + \dots + a_nv_n$ and $w = b_1v_1 + b_2v_2 + \dots + b_nv_n$. Then

$$v + w = (a_1 + b_1)v_1 + (a_2 + b_2)v_2 + \dots + (a_n + b_n)v_n$$

and, for any $k \in K, kv = ka_1v_1 + ka_2v_2 + \dots + ka_nv_n$. By definition of the mapping F ,

$$F(v) = a_1u_1 + a_2u_2 + \dots + a_nu_n \quad \text{and} \quad F(w) = b_1u_1 + b_2u_2 + \dots + b_nu_n$$

Hence

$$\begin{aligned}
F(v+w) &= (a_1 + b_1)u_1 + (a_2 + b_2)u_2 + \cdots + (a_n + b_n)u_n \\
&= (a_1u_1 + a_2u_2 + \cdots + a_nu_n) + (b_1u_1 + b_2u_2 + \cdots + b_nu_n) \\
&= F(v) + F(w)
\end{aligned}$$

and

$$F(kv) = k(a_1u_1 + a_2u_2 + \cdots + a_nu_n) = kF(v)$$

Thus, F is linear.

Step 3. Suppose $G : V \rightarrow U$ is linear and $G(v_i) = u_i$, $i = 1, \dots, n$. Let

$$v = a_1v_1 + a_2v_2 + \cdots + a_nv_n$$

Then

$$\begin{aligned}
G(v) &= G(a_1v_1 + a_2v_2 + \cdots + a_nv_n) = a_1G(v_1) + a_2G(v_2) + \cdots + a_nG(v_n) \\
&= a_1u_1 + a_2u_2 + \cdots + a_nu_n = F(v)
\end{aligned}$$

Because $G(v) = F(v)$ for every $v \in V$, $G = F$. Thus, F is unique and the theorem is proved.

5.14. Let $F : \mathbf{R}^2 \rightarrow \mathbf{R}^2$ be the linear mapping for which $F(1, 2) = (2, 3)$ and $F(0, 1) = (1, 4)$. [Note that $\{(1, 2), (0, 1)\}$ is a basis of $\mathbf{R}^2$, so such a linear map F exists and is unique by [Theorem 5.2](#).] Find a formula for F ; that is, find $F(a, b)$.

Write (a, b) as a linear combination of $(1, 2)$ and $(0, 1)$ using unknowns x and y ,

$$(a, b) = x(1, 2) + y(0, 1) = (x, 2x + y), \quad \text{so} \quad a = x, \quad b = 2x + y$$

Solve for x and y in terms of a and b to get $x = a$, $y = -2a + b$. Then

$$F(a, b) = xF(1, 2) + yF(0, 1) = a(2, 3) + (-2a + b)(1, 4) = (b, -5a + 4b)$$

5.15. Suppose a linear mapping $F: V \rightarrow U$ is one-to-one and onto. Show that the inverse mapping $F^{-1}: U \rightarrow V$ is also linear.

Suppose $u, u' \in U$. Because F is one-to-one and onto, there exist unique vectors $v, v' \in V$ for which $F(v) = u$ and $F(v') = u'$. Because F is linear, we also have

$$F(v + v') = F(v) + F(v') = u + u' \quad \text{and} \quad F(kv) = kF(v) = ku$$

By definition of the inverse mapping,

$$F^{-1}(u) = v, \quad F^{-1}(u') = v', \quad F^{-1}(u + u') = v + v', \quad F^{-1}(ku) = kv.$$

Then

$$F^{-1}(u + u') = v + v' = F^{-1}(u) + F^{-1}(u') \quad \text{and} \quad F^{-1}(ku) = kv = kF^{-1}(u)$$

Thus, F^{-1} is linear.

Kernel and Image of Linear Mappings

5.16. Let $F: \mathbf{R}^4 \rightarrow \mathbf{R}^3$ be the linear mapping defined by

$$F(x, y, z, t) = (x - y + z + t, \quad x + 2z - t, \quad x + y + 3z - 3t)$$

Find a basis and the dimension of (a) the image of F ,
(b) the kernel of F .

(a) Find the images of the usual basis of $\mathbf{R}^4$:

$$F(1, 0, 0, 0) = (1, 1, 1), \quad F(0, 0, 1, 0) = (1, 2, 3)$$

$$F(0, 1, 0, 0) = (-1, 0, 1), \quad F(0, 0, 0, 1) = (1, -1, -3)$$

By Proposition 5.4, the image vectors span $\text{Im } F$. Hence, form the matrix whose rows are these image vectors, and row reduce to echelon form:

$$\begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 1 \\ 1 & 2 & 3 \\ 1 & -1 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \\ 0 & -2 & -4 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Thus, $(1, 1, 1)$ and $(0, 1, 2)$ form a basis for $\text{Im } F$;
hence, $\dim(\text{Im } F) = 2$.

(b) Set $F(v) = 0$, where $v = (x, y, z, t)$; that is, set

$$F(x, y, z, t) = (x - y + z + t, x + 2z - t, x + y + 3z - 3t) = (0, 0, 0)$$

Set corresponding entries equal to each other to form the following homogeneous system whose solution space is $\text{Ker } F$:

$$\begin{array}{lll} x - y + z + t = 0 & & x - y + z + t = 0 \\ x + 2z - t = 0 & \text{or} & y + z - 2t = 0 \\ x + y + 3z - 3t = 0 & & 2y + 2z - 4t = 0 \end{array} \quad \text{or} \quad \begin{array}{l} x - y + z + t = 0 \\ y + z - 2t = 0 \end{array}$$

The free variables are z and t . Hence, $\dim(\text{Ker } F) = 2$.

(i) Set $z = -1, t = 0$ to obtain the solution $(2, 1, -1, 0)$.

(ii) Set $z = 0, t = 1$ to obtain the solution $(1, 2, 0, 1)$.

Thus, $(2, 1, 1, 0)$ and $(1, 2, 0, 1)$ form a basis of $\text{Ker } F$.

[As expected, $\dim(\text{Im } F) + \dim(\text{Ker } F) = 2 + 2 = 4 = \dim \mathbf{R}^4$, the domain of F .]

5.17. Let $G : \mathbf{R}^3 \rightarrow \mathbf{R}^3$ be the linear mapping defined by

$$G(x, y, z) = (x + 2y - z, \quad y + z, \quad x + y - 2z)$$

Find a basis and the dimension of (a) the image of G ,
(b) the kernel of G .

(a) Find the images of the usual basis of $\mathbf{R}^3$:

$$G(1,0,0) = (1,0,1), \quad G(0,1,0) = (2,1,1), \quad G(0,0,1) = (-1,1,-2)$$

By Proposition 5.4, the image vectors span $\text{Im } G$. Hence, form the matrix M whose rows are these image vectors, and row reduce to echelon form:

$$M = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \\ -1 & 1 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

Thus, $(1, 0, 1)$ and $(0, 1, 1)$ form a basis for $\text{Im } G$; hence, $\dim(\text{Im } G) = 2$.

(b) Set $G(v) = 0$, where $v = (x, y, z)$; that is,

$$G(x, y, z) = (x + 2y - z, \quad y + z, \quad x + y - 2z) = (0, 0, 0)$$

Set corresponding entries equal to each other to form the following homogeneous system whose solution space is $\text{Ker } G$:

$$\begin{array}{lcl} \begin{array}{l} x + 2y - z = 0 \\ y + z = 0 \\ x + y - 2z = 0 \end{array} & \text{or} & \begin{array}{l} x + 2y - z = 0 \\ y + z = 0 \\ -y - z = 0 \end{array} & \text{or} & \begin{array}{l} x + 2y - z = 0 \\ y + z = 0 \end{array} \end{array}$$

The only free variable is z ; hence, $\dim(\text{Ker } G) = 1$. Set $z = -1$; then $y = -1$ and $x = 3$. Thus, $(3, -1, 1)$ forms a basis of $\text{Ker } G$. [As expected, $\dim(\text{Im } G) + \dim(\text{Ker } G) = 2 + 1 = 3 = \dim \mathbf{R}^3$, the domain of G .]

5.18. Consider the matrix mapping $A : \mathbf{R}^4 \rightarrow \mathbf{R}^3$, where

$$A = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 3 & 5 & -2 \\ 3 & 8 & 13 & -3 \end{bmatrix}. \text{ Find a basis and the dimension of}$$

(a) the image of A , (b) the kernel of A .

(a) The column space of A is equal to $\text{Im } A$. Now reduce A^T to echelon form:

$$A^T = \begin{bmatrix} 1 & 1 & 3 \\ 2 & 3 & 8 \\ 3 & 5 & 13 \\ 1 & -2 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & 2 \\ 0 & 2 & 4 \\ 0 & -3 & -6 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Thus, $\{(1, 1, 3), (0, 1, 2)\}$ is a basis of $\text{Im } A$, and $\dim(\text{Im } A) = 2$.

(b) Here $\text{Ker } A$ is the solution space of the homogeneous system $AX = 0$, where $X = \{x, y, z, t\}^T$. Thus, reduce the matrix A of coefficients to echelon form:

$$\begin{bmatrix} 1 & 2 & 3 & 1 \\ 0 & 1 & 2 & -3 \\ 0 & 2 & 4 & -6 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 & 1 \\ 0 & 1 & 2 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{or} \quad \begin{aligned} x + 2y + 3z + t &= 0 \\ y + 2z - 3t &= 0 \end{aligned}$$

The free variables are z and t . Thus, $\dim(\text{Ker } A) = 2$.

(i) Set $z = 1, t = 0$ to get the solution $(1, -2, 1, 0)$.

(ii) Set $z = 0, t = 1$ to get the solution $(-7, 3, 0, 1)$.

Thus, $(1, -2, 1, 0)$ and $(-7, 3, 0, 1)$ form a basis for $\text{Ker } A$.

5.19. Find a linear map $F: \mathbf{R}^3 \rightarrow \mathbf{R}^4$ whose image is spanned by $(1, 2, 0, -4)$ and $(2, 0, -1, -3)$.

Form a 4×3 matrix whose columns consist only of the given vectors, say

$$A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 0 & 0 \\ 0 & -1 & -1 \\ -4 & -3 & -3 \end{bmatrix}$$

Recall that A determines a linear map $A : \mathbf{R}^3 \rightarrow \mathbf{R}^4$ whose image is spanned by the columns of A . Thus, A satisfies the required condition.

5.20. Suppose $f: V \rightarrow U$ is linear with kernel W , and that $f(v) = u$. Show that the “coset” $v + W = \{v + w : w \in W\}$ is the preimage of u ; that is, $f^{-1}(u) = v + W$.

We must prove that (i) $f^{-1}(u) \subseteq v + W$ and (ii) $v + W \subseteq f^{-1}(u)$.

We first prove (i). Suppose $v' \in f^{-1}(u)$. Then $f(v') = u$, and so

$$f(v' - v) = f(v') - f(v) = u - u = 0$$

that is, $v' - v \in W$. Thus, $v' = v + (v' - v) \in v + W$, and hence $f^{-1}(u) \subseteq v + W$.

Now we prove (ii). Suppose $v' \in v + W$. Then $v' = v + w$, where $w \in W$. Because W is the kernel of f , we have $f(w) = 0$. Accordingly,

$$f(v') = f(v + w) = f(v) + f(w) = f(v) + 0 = f(v) = u$$

Thus, $v' \in f^{-1}(u)$.

Both inclusions imply $f^{-1}(u) = v + W$.

5.21. Suppose $F: V \rightarrow U$ and $G: U \rightarrow W$ are linear. Prove

(a) $\text{rank}(G \circ F) \leq \text{rank}(G)$,

(b) $\text{rank}(G \circ F) \leq \text{rank}(F)$.

(a) Because $F(V) \subseteq U$, we also have $G(F(V)) \subseteq G(U)$, and so $\dim [G(F(V))] \leq \dim [G(U)]$. Then $\text{rank}(G \circ F) = \dim[(G \circ F)(V)] = \dim [G(F(V))] \leq \dim [G(U)] = \text{rank}(G)$.

(b) We have $\dim [G(F(V)) \subseteq \dim [F(V)]$. Hence, $\text{rank}(G \circ F) = \dim[(G \circ F)(V)] = \dim [G(F(V))] \subseteq \dim [F(V)] = \text{rank}(F)$

5.22. Prove [Theorem 5.3](#): Let $F: V \rightarrow U$ be linear. Then,

(a) $\text{Im } F$ is a subspace of U , (b) $\text{Ker } F$ is a subspace of V .

(a) Because $F(0) = 0$, we have $0 \in \text{Im } F$. Now suppose $u, u' \in \text{Im } F$ and $a, b \in K$. Because u and u' belong to the image of F , there exist vectors $v, v' \in V$ such that $F(v) = u$ and $F(v') = u'$. Then

$$F(av + bv') = aF(v) + bF(v') = au + bu' \in \text{Im } F$$

Thus, the image of F is a subspace of U .

(b) Because $F(0) = 0$, we have $0 \in \text{Ker } F$. Now suppose $v, w \in \text{Ker } F$ and $a, b \in K$. Because v and w belong to the kernel of F , $F(v) = 0$ and $F(w) = 0$. Thus,

$$F(av + bw) = aF(v) + bF(w) = a0 + b0 = 0 + 0 = 0, \quad \text{and so} \quad av + bw \in \text{Ker } F$$

Thus, the kernel of F is a subspace of V .

5.23. Prove [Theorem 5.6](#): Suppose V has finite dimension and $F: V \rightarrow U$ is linear. Then

$$\dim V = \dim(\text{Ker } F) + \dim(\text{Im } F) = \text{nullity}(F) + \text{rank}(F)$$

Suppose $\dim(\text{Ker } F) = r$ and $\{w_1, \dots, w_r\}$ is a basis of $\text{Ker } F$, and suppose $\dim(\text{Im } F) = s$ and $\{u_1, \dots, u_s\}$ is a basis of $\text{Im } F$. (By Proposition 5.4, $\text{Im } F$ has finite dimension.) Because every $u_j \in \text{Im } F$, there exist vectors $v_1, \dots, v_s$ in V such that $F(v_1) = u_1, \dots, F(v_s) = u_s$. We claim that the set

$$B = \{w_1, \dots, w_r, v_1, \dots, v_s\}$$

is a basis of V ; that is, (i) B spans V , and (ii) B is linearly independent. Once we prove (i) and (ii), then $\dim V = r + s = \dim(\text{Ker } F) + \dim(\text{Im } F)$.

(i) B spans V . Let $v \in V$. Then $F(v) \in \text{Im } F$. Because the u_j span $\text{Im } F$, there exist scalars $a_1, \dots, a_s$ such that $F(v) = a_1 u_1 + \dots + a_s u_s$. Set $\hat{v} = a_1 v_1 + \dots + a_s v_s - v$. Then

$$\begin{aligned} F(\hat{v}) &= F(a_1 v_1 + \dots + a_s v_s - v) = a_1 F(v_1) + \dots + a_s F(v_s) - F(v) \\ &= a_1 u_1 + \dots + a_s u_s - F(v) = 0 \end{aligned}$$

Thus, $\hat{v} \in \text{Ker } F$. Because the w_i span $\text{Ker } F$, there exist scalars $b_1, \dots, b_r$ such that

$$\hat{v} = b_1 w_1 + \dots + b_r w_r = a_1 v_1 + \dots + a_s v_s - v$$

Accordingly,

$$v = a_1 v_1 + \dots + a_s v_s - b_1 w_1 - \dots - b_r w_r$$

Thus, B spans V .

(ii) B is linearly independent. Suppose

$$x_1 w_1 + \dots + x_r w_r + y_1 v_1 + \dots + y_s v_s = 0 \tag{1}$$

where $x_i, y_j \in K$. Then

$$\begin{aligned} 0 &= F(0) = F(x_1 w_1 + \dots + x_r w_r + y_1 v_1 + \dots + y_s v_s) \\ &= x_1 F(w_1) + \dots + x_r F(w_r) + y_1 F(v_1) + \dots + y_s F(v_s) \end{aligned} \tag{2}$$

But $F(w_i) = 0$, since $w_i \in \text{Ker } F$, and $F(v_j) = u_j$.

Substituting into (2), we will obtain $y_1 u_1 + \dots + y_s u_s = 0$. Since the u_j are linearly independent, each $y_j = 0$.

Substitution into (1) gives $x_1 w_1 + \dots + x_r w_r = 0$. Since

the w_i are linearly independent, each $x_i = 0$. Thus B is linearly independent.

Singular and Nonsingular Linear Maps, Isomorphisms

5.24. Determine whether or not each of the following linear maps is nonsingular. If not, find a nonzero vector v whose image is 0.

(a) $F: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ defined by $F(x, y) = (x - y, x - y)$.

(b) $G: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ defined by $G(x, y) = (2x - 4y, 3x - 6y)$.

(a) Find $\text{Ker } F$ by setting $F(v) = 0$, where $v = (x, y)$,

$$(x - y, x - 2y) = (0, 0) \quad \text{or} \quad \begin{matrix} x - y = 0 \\ x - 2y = 0 \end{matrix} \quad \text{or} \quad \begin{matrix} x - y = 0 \\ -y = 0 \end{matrix}$$

The only solution is $x = 0, y = 0$. Hence, F is nonsingular.

(b) Set $G(x, y) = (0, 0)$ to find $\text{Ker } G$:

$$(2x - 4y, 3x - 6y) = (0, 0) \quad \text{or} \quad \begin{matrix} 2x - 4y = 0 \\ 3x - 6y = 0 \end{matrix} \quad \text{or} \quad x - 2y = 0$$

The system has nonzero solutions, because y is a free variable. Hence, G is singular. Let $y = 1$ to obtain the solution $v = (2, 1)$, which is a nonzero vector, such that $G(v) = 0$.

5.25. The linear map $F: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ defined by $F(x, y) = (x - y, x - 2y)$ is nonsingular by the previous [Problem 5.24](#). Find a formula for F^{-1}

Set $F(x, y) = (a, b)$, so that $F^{-1}(a, b) = (x, y)$. We have

$$(x - y, x - 2y) = (a, b) \quad \text{or} \quad \begin{matrix} x - y = a \\ x - 2y = b \end{matrix} \quad \text{or} \quad \begin{matrix} x - y = a \\ y = a - b \end{matrix}$$

Solve for x and y in terms of a and b to get $x = 2a - b$, $y = a - b$. Thus,

$$F^{-1}(a, b) = (2a - b, a - b) \quad \text{or} \quad F^{-1}(x, y) = (2x - y, x - y)$$

(The second equation is obtained by replacing a and b by x and y , respectively.)

5.26. Let $G : \mathbf{R}^2 \rightarrow \mathbf{R}^3$ be defined by $G(x, y) = (x + y, x - 2y, 3x + y)$.

(a) Show that G is nonsingular. (b) Find a formula for G^{-1} .

(a) Set $G(x, y) = (0, 0, 0)$ to find $\text{Ker } G$. We have

$$(x + y, x - 2y, 3x + y) = (0, 0, 0) \quad \text{or} \quad \begin{matrix} x + y = 0 \\ x - 2y = 0 \\ 3x + y = 0 \end{matrix}$$

The only solution is $x = 0$, $y = 0$; hence, G is nonsingular.

(b) Although G is nonsingular, it is not invertible, because $\mathbf{R}^2$ and $\mathbf{R}^3$ have different dimensions. (Thus, [Theorem 5.9](#) does not apply.) Accordingly, G^{-1} does not exist.

5.27. Suppose that $F : V \rightarrow U$ is linear and that V is of finite dimension. Show that V and the image of F have the same dimension if and only if F is nonsingular.

Determine all nonsingular linear mappings $T : \mathbf{R}^4 \rightarrow \mathbf{R}^3$.

By [Theorem 5.6](#), $\dim V = \dim(\text{Im } F) + \dim(\text{Ker } F)$. Hence, V and $\text{Im } F$ have the same dimension if and only if $\dim(\text{Ker } F) = 0$ or $\text{Ker } F = \{0\}$ (i.e., if and only if F is nonsingular).

Because $\dim \mathbf{R}^3$ is less than $\dim \mathbf{R}^4$, we have that $\dim(\text{Im } T)$ is less than the dimension of the domain $\mathbf{R}^4$ of T . Accordingly no linear mapping $T: \mathbf{R}^4 \rightarrow \mathbf{R}^3$ can be nonsingular.

5.28. Prove [Theorem 5.7](#): Let $F: V \rightarrow U$ be a nonsingular linear mapping. Then the image of any linearly independent set is linearly independent.

Suppose $v_1, v_2, \dots, v_n$ are linearly independent vectors in V . We claim that $F(v_1), F(v_2), \dots, F(v_n)$ are also linearly independent. Suppose $a_1 F(v_1) + a_2 F(v_2) + \dots + a_n F(v_n) = 0$, where $a_i \in K$. Because F is linear, $F(a_1 v_1 + a_2 v_2 + \dots + a_n v_n) = 0$. Hence,

$$a_1 v_1 + a_2 v_2 + \dots + a_n v_n \in \text{Ker } F$$

But F is nonsingular—that is, $\text{Ker } F = \{0\}$. Hence, $a_1 v_1 + a_2 v_2 + \dots + a_n v_n = 0$. Because the v_i are linearly independent, all the a_i are 0. Accordingly, the $F(v_i)$ are linearly independent. Thus, the theorem is proved.

5.29. Prove [Theorem 5.9](#): Suppose V has finite dimension and $\dim V = \dim U$. Suppose $F: V \rightarrow U$ is linear. Then F is an isomorphism if and only if F is nonsingular.

If F is an isomorphism, then only 0 maps to 0; hence, F is nonsingular. Conversely, suppose F is nonsingular. Then $\dim(\text{Ker } F) = 0$. By [Theorem 5.6](#), $\dim V = \dim(\text{Ker } F) + \dim(\text{Im } F)$. Thus,

$$\dim U = \dim V = \dim(\text{Im } F)$$

Because U has finite dimension, $\text{Im } F = U$. This means F maps V onto U . Thus, F is one-to-one and onto; that is, F is an isomorphism.

Operations with Linear Maps

5.30. Define $F : \mathbf{R}^3 \rightarrow \mathbf{R}^2$ and $G : \mathbf{R}^3 \rightarrow \mathbf{R}^2$ by $F(x, y, z) = (2x, y + z)$ and $G(x, y, z) = (x - z, y)$. Find formulas defining the maps: (a) $F + G$, (b) $3F$, (c) $2F - 5G$.

(a) $(F + G)(x, y, z) = F(x, y, z) + G(x, y, z) = (2x, y + z) + (x - z, y) = (3x - z, 2y + z)$

(b) $(3F)(x, y, z) = 3F(x, y, z) = 3(2x, y + z) = (6x, 3y + 3z)$

(c) $(2F - 5G)(x, y, z) = 2F(x, y, z) - 5G(x, y, z) = 2(2x, y + z) - 5(x - z, y)$
 $= (4x, 2y + 2z) + (-5x + 5z, -5y) = (-x + 5z, -3y + 2z)$

5.31. Let $F : \mathbf{R}^3 \rightarrow \mathbf{R}^2$ and $G : \mathbf{R}^2 \rightarrow \mathbf{R}^2$ be defined by $F(x, y, z) = (2x, y + z)$ and $G(x, y) = (y, x)$. Derive formulas defining the mappings: (a) $G \circ F$, (b) $F \circ G$.

(a) $(G \circ F)(x, y, z) = G(F(x, y, z)) = G(2x, y + z) = (y + z, 2x)$

(b) The mapping $F \circ G$ is not defined, because the image of G is not contained in the domain of F .

5.32. Prove: (a) The zero mapping $\mathbf{0}$, defined by $\mathbf{0}(v) = 0 \in U$ for every $v \in V$, is the zero element of $\text{Hom}(V, U)$. (b) The negative of $F \in \text{Hom}(V, U)$ is the mapping $(-1)F$, that is, $F = (-1)(-1)F$.

Let $F \in \text{Hom}(V, U)$. Then, for every $v \in V$:

(a) $(F + \mathbf{0})(v) = F(v) + \mathbf{0}(v) = F(v) + 0 = F(v)$

Because $(F + \mathbf{0})(v) = F(v)$ for every $v \in V$, we have $F + \mathbf{0} = F$. Similarly, $\mathbf{0} + F = F$:

(b) $(F + (-1)F)(v) = F(v) + (-1)F(v) = F(v) - F(v) = 0 = \mathbf{0}(v)$

Thus, $F + (-1)F = \mathbf{0}$: Similarly $(1)F + (-1)F = \mathbf{0}$: Hence, $-F = (-1)F$:

5.33. Suppose $F_1, F_2, \dots, F_n$ are linear maps from V into U . Show that, for any scalars $a_1, a_2, \dots, a_n$, and for any $v \in V$,

$$(a_1F_1 + a_2F_2 + \cdots + a_nF_n)(v) = a_1F_1(v) + a_2F_2(v) + \cdots + a_nF_n(v)$$

The mapping a_1F_1 is defined by $(a_1F_1)(v) = a_1F(v)$. Hence, the theorem holds for $n = 1$. Accordingly, by induction,

$$\begin{aligned} (a_1F_1 + a_2F_2 + \cdots + a_nF_n)(v) &= (a_1F_1)(v) + (a_2F_2 + \cdots + a_nF_n)(v) \\ &= a_1F_1(v) + a_2F_2(v) + \cdots + a_nF_n(v) \end{aligned}$$

5.34. Consider linear mappings $F: \mathbf{R}^3 \rightarrow \mathbf{R}^2$, $G: \mathbf{R}^3 \rightarrow \mathbf{R}^2$, $H: \mathbf{R}^3 \rightarrow \mathbf{R}^2$ defined by

$$F(x, y, z) = (x + y + z, x + y), \quad G(x, y, z) = (2x + z, x + y), \quad H(x, y, z) = (2y, x)$$

Show that F, G, H are linearly independent [as elements of $\text{Hom}(\mathbf{R}^3; \mathbf{R}^2)$].

Suppose, for scalars $a; b; c \in K$,

$$aF + bG + cH = \mathbf{0} \tag{1}$$

(Here $\mathbf{0}$ is the zero mapping.) For $e_1 = (1; 0; 0) \in \mathbf{R}^3$, we have $\mathbf{0}(e_1) = (0; 0)$ and

$$\begin{aligned} (aF + bG + cH)(e_1) &= aF(1, 0, 0) + bG(1, 0, 0) + cH(1, 0, 0) \\ &= a(1, 1) + b(2, 1) + c(0, 1) = (a + 2b, a + b + c) \end{aligned}$$

Thus by (1), $(a + 2b; a + b + c) = (0; 0)$ and so

$$a + 2b = 0 \quad \text{and} \quad a + b + c = 0 \tag{2}$$

Similarly for $e_2 = (0; 1; 0) \in \mathbf{R}^3$, we have $\mathbf{0}(e_2) = (0; 0)$ and

$$\begin{aligned}(aF + bG + cH)(e_2) &= aF(0, 1, 0) + bG(0, 1, 0) + cH(0, 1, 0) \\ &= a(1, 1) + b(0, 1) + c(2, 0) = (a + 2c, \quad a + b)\end{aligned}$$

Thus,

$$a + 2c = 0 \quad \text{and} \quad a + b = 0 \tag{3}$$

Using (2) and (3), we obtain

$$a = 0, \quad b = 0, \quad c = 0 \tag{4}$$

Because (1) implies (4), the mappings F, G, H are linearly independent.

5.35. Let k be a nonzero scalar. Show that a linear map T is singular if and only if kT is singular. Hence, T is singular if and only if T is singular.

Suppose T is singular. Then $T(v) = 0$ for some vector $v \neq 0$. Hence,

$$(kT)(v) = kT(v) = k0 = 0$$

and so kT is singular.

Now suppose kT is singular. Then $(kT)(w) = 0$ for some vector $w \neq 0$. Hence,

$$T(kw) = kT(w) = (kT)(w) = 0$$

But $k \neq 0$ and $w \neq 0$ implies $kw \neq 0$. Thus, T is also singular.

5.36. Find the dimension d of:

- (a) $\text{Hom}(\mathbf{R}^3; \mathbf{R}^4)$, (b) $\text{Hom}(\mathbf{R}^5; \mathbf{R}^3)$, (c) $\text{Hom}(\mathbf{P}_3(t); \mathbf{R}^2)$,
 (d) $\text{Hom}(\mathbf{M}_{2;3}; \mathbf{R}^4)$.

Use $\dim [\text{Hom}(V; U)] = mn$, where $\dim V = m$ and $\dim U = n$.

(a) $d = 3(4) = 12$.

(b) $d = 5(3) = 15$.

(c) Because $\dim \mathbf{P}_3(t) = 4$, $d = 4(2) = 8$.

(d) Because $\dim \mathbf{M}_{2;3} = 6$, $d = 6(4) = 24$.

5.38. Prove [Theorem 5.11](#). Suppose $\dim V = m$ and $\dim U = n$. Then $\dim [\text{Hom}(V; U)] = mn$.

Suppose $fv_1; \dots, v_m$ is a basis of V and $\{u_1; \dots, u_n\}$ is a basis of U . By [Theorem 5.2](#), a linear mapping in $\text{Hom}(V; U)$ is uniquely determined by arbitrarily assigning elements of U to the basis elements v_i of V . We define

$$F_{ij} \in \text{Hom}(V, U), \quad i = 1, \dots, m, \quad j = 1, \dots, n$$

to be the linear mapping for which $F_{ij}(v_i) = u_j$, and $F_{ij}(v_k) = 0$ for $k \neq i$. That is, F_{ij} maps v_i into u_j and the other v 's into 0. Observe that $\{F_{ij}\}$ contains exactly mn elements; hence, the theorem is proved if we show that it is a basis of $\text{Hom}(V; U)$.

Proof that $\{F_{ij}\}$ generates $\text{Hom}(V; U)$. Consider an arbitrary function $F \in \text{Hom}(V; U)$. Suppose $F(v_1) = w_1$; $F(v_2) = w_2$; $\dots$, $F(v_m) = w_m$. Because $w_k \in U$, it is a linear combination of the u 's; say,

$$w_k = a_{k1}u_1 + a_{k2}u_2 + \dots + a_{kn}u_n, \quad k = 1, \dots, m, \quad a_{ij} \in K \quad (1)$$

Consider the linear mapping $G = \sum_{i=1}^m \sum_{j=1}^n a_{ij} F_{ij}$. Because G is a linear combination of the F_{ij} , the proof that $\{F_{ij}\}$ generates $\text{Hom}(V; U)$ is complete if we show that $F = G$.

We now compute $G(v_k)$; $k = 1; \dots, m$. Because $F_{ij}(v_k) = 0$ for $k \neq i$ and $F_{ki}(v_k) = u_i$,

$$\begin{aligned} G(v_k) &= \sum_{i=1}^m \sum_{j=1}^n a_{ij} F_{ij}(v_k) = \sum_{j=1}^n a_{kj} F_{kj}(v_k) = \sum_{j=1}^n a_{kj} u_j \\ &= a_{k1} u_1 + a_{k2} u_2 + \dots + a_{kn} u_n \end{aligned}$$

Thus, by (1), $G(v_k) = w_k$ for each k . But $F(v_k) = w_k$ for each k . Accordingly, by [Theorem 5.2](#), $F = G$; hence, $\{F_{ij}\}$ generates $\text{Hom}(V; U)$.

Proof that $\{F_{ij}\}$ is linearly independent. Suppose, for scalars $c_{ij} \in K$,

$$\sum_{i=1}^m \sum_{j=1}^n c_{ij} F_{ij} = \mathbf{0}$$

For v_k ; $k = 1; \dots, m$,

$$\begin{aligned} 0 = \mathbf{0}(v_k) &= \sum_{i=1}^m \sum_{j=1}^n c_{ij} F_{ij}(v_k) = \sum_{j=1}^n c_{kj} F_{kj}(v_k) = \sum_{j=1}^n c_{kj} u_j \\ &= c_{k1} u_1 + c_{k2} u_2 + \dots + c_{kn} u_n \end{aligned}$$

But the u_j are linearly independent; hence, for $k = 1; \dots, m$, we have $c_{k1} = 0$; $c_{k2} = 0$; $\dots$, $c_{kn} = 0$. In other words, all the $c_{ij} = 0$, and so $\{F_{ij}\}$ is linearly independent.

- 5.38.** Prove [Theorem 5.12](#): (i) $G \circ (F + F') = G \circ F + G \circ F'$.
(ii) $(G + G') \circ F = G \circ F + G' \circ F$. (iii) $k(G \circ F) = (kG) \circ F = G \circ (kF)$.

(i) For every $v \in V$,

$$\begin{aligned}(G \circ (F + F'))(v) &= G((F + F')(v)) = G(F(v) + F'(v)) \\ &= G(F(v)) + G(F'(v)) = (G \circ F)(v) + (G \circ F')(v) = (G \circ F + G \circ F')(v)\end{aligned}$$

Thus, $G(F + F') = G \circ F + G \circ F'$.

(ii) For every $v \in V$,

$$\begin{aligned}((G + G') \circ F)(v) &= (G + G')(F(v)) = G(F(v)) + G'(F(v)) \\ &= (G \circ F)(v) + (G' \circ F)(v) = (G \circ F + G' \circ F)(v)\end{aligned}$$

Thus, $(G + G') \circ F = G \circ F + G' \circ F$.

(iii) For every $v \in V$,

$$(k(G \circ F))(v) = k(G \circ F)(v) = k(G(F(v))) = (kG)(F(v)) = (kG \circ F)(v)$$

and

$$(k(G \circ F))(v) = k(G \circ F)(v) = k(G(F(v))) = G(kF(v)) = G((kF)(v)) = (G \circ kF)(v)$$

Accordingly, $k(G \circ F) = (kG) \circ F = G \circ (kF)$. (We emphasize that two mappings are shown to be equal by showing that each of them assigns the same image to each point in the domain.)

Algebra of Linear Maps

5.39. Let F and G be the linear operators on $\mathbf{R}^2$ defined by $F(x; y) = (y; x)$ and $G(x; y) = (0; x)$. Find formulas defining the following operators:

(a) $F + G$, (b) $2F - 3G$, (c) FG , (d) GF , (e) F^2 , (f) G^2 .

(a) $(F + G)(x; y) = F(x; y) + G(x; y) = (y; x) + (0; x) = (y; 2x)$.

- (b) $(2F - 3G)(x; y) = 2F(x; y) - 3G(x; y) = 2(y; x) - 3(0; x) = (2y; x)$.
- (c) $(FG)(x; y) = F(G(x; y)) = F(0; x) = (x; 0)$.
- (d) $(GF)(x; y) = G(F(x; y)) = G(y; x) = (0; y)$.
- (e) $F^2(x; y) = F(F(x; y)) = F(y; x) = (x; y)$. (Note that $F^2 = I$, the identity mapping.)
- (f) $G^2(x; y) = G(G(x; y)) = G(0; x) = (0; 0)$. (Note that $G^2 = \mathbf{0}$, the zero mapping.)

5.40. Consider the linear operator T on $\mathbf{R}^3$ defined by $T(x; y; z) = (2x; 4x - y; 2x + 3y - z)$.

- (a) Show that T is invertible. Find formulas for (b) T^{-1} , (c) T^2 , (d) T^{-2} .
- (a) Let $W = \text{Ker } T$. We need only show that T is nonsingular (i.e., that $W = \{0\}$). Set $T(x; y; z) = (0; 0; 0)$, which yields

$$T(x, y, z) = (2x, 4x - y, 2x + 3y - z) = (0, 0, 0)$$

Thus, W is the solution space of the homogeneous system

$$2x = 0, \quad 4x - y = 0, \quad 2x + 3y - z = 0$$

which has only the trivial solution $(0, 0, 0)$. Thus, $W = \{0\}$. Hence, T is nonsingular, and so T is invertible.

- (b) Set $T(x; y; z) = (r; s; t)$ [and so $T^{-1}(r; s; t) = (x; y; z)$]. We have

$$(2x, 4x - y, 2x + 3y - z) = (r, s, t) \quad \text{or} \quad 2x = r, \quad 4x - y = s, \quad 2x + 3y - z = t$$

Solve for x, y, z in terms of r, s, t to get $x = \frac{1}{2}r, y = 2r - s, z = 7r - 3s - t$. Thus,

$$T^{-1}(r, s, t) = (\frac{1}{2}r, 2r - s, 7r - 3s - t) \quad \text{or} \quad T^{-1}(x, y, z) = (\frac{1}{2}x, 2x - y, 7x - 3y - z)$$

(c) Apply T twice to get

$$\begin{aligned} T^2(x, y, z) &= T(2x, 4x - y, 2x + 3y - z) \\ &= [4x, 4(2x) - (4x - y), 2(2x) + 3(4x - y) - (2x + 3y - z)] \\ &= (4x, 4x + y, 14x - 6y + z) \end{aligned}$$

(d) Apply T

$$\begin{aligned} T^{-1}(x, y, z) &= T^{-1}(\frac{1}{2}x, 2x - y, 7x - 3y - z) \\ &= [\frac{1}{4}x, 2(\frac{1}{2}x) - (2x - y), 7(\frac{1}{2}x) - 3(2x - y) - (7x - 3y - z)] \\ &= (\frac{1}{4}x, -x + y, -\frac{19}{2}x + 6y + z) \end{aligned}$$

5.41. Let V be of finite dimension and let T be a linear operator on V for which $TR = I$, for some operator R on V . (We call R a *right inverse* of T .)

(a) Show that T is invertible. (b) Show that $R = T^{-1}$.

(c) Give an example showing that the above need not hold if V is of infinite dimension.

(a) Let $\dim V = n$. By [Theorem 5.14](#), T is invertible if and only if T is onto; hence, T is invertible if and only if $\text{rank}(T) = n$. We have $n = \text{rank}(I) = \text{rank}(TR) \leq \text{rank}(T) \leq n$. Hence, $\text{rank}(T) = n$ and T is invertible.

(b) $TT^{-1} = T^{-1}T = I$. Then $R = IR = (T^{-1}T)R = T^{-1}(TR) = T^{-1}I = T^{-1}$.

(c) Let V be the space of polynomials in t over K ; say, $p(t) = a_0 + a_1t + a_2t^2 + \dots + a_st^s$. Let T and R be the

operators on V defined by

$$T(p(t)) = 0 + a_1 + a_2t + \cdots + a_st^{s-1} \quad \text{and} \quad R(p(t)) = a_0t + a_1t^2 + \cdots + a_st^{s+1}$$

We have

$$(TR)(p(t)) = T(R(p(t))) = T(a_0t + a_1t^2 + \cdots + a_st^{s+1}) = a_0 + a_1t + \cdots + a_st^s = p(t)$$

and so $TR = I$, the identity mapping. On the other hand, if $k \in K$ and $k \neq 0$, then

$$(RT)(k) = R(T(k)) = R(0) = 0 \neq k$$

Accordingly, $RT \neq I$.

5.42. Let F and G be linear operators on $\mathbf{R}^2$ defined by $F(x; y) = (0; x)$ and $G(x; y) = (x; 0)$. Show that

(a) $GF = \mathbf{0}$, the zero mapping, but $FG \neq \mathbf{0}$.

(b) $G^2 = G$. (a) $(GF)(x; y) = G(F(x; y)) = G(0; x) = (0; 0)$. Because GF assigns $0 = (0; 0)$ to every vector $(x; y)$ in $\mathbf{R}^2$, it is the zero mapping; that is, $GF = \mathbf{0}$.

On the other hand, $(FG)(x; y) = F(G(x; y)) = F(x; 0) = (0; x)$. For example, $(FG)(2; 3) = (0; 2)$. Thus, $FG \neq \mathbf{0}$, as it does not assign $0 = (0; 0)$ to every vector in $\mathbf{R}^2$.

(b) For any vector $(x; y)$ in $\mathbf{R}^2$, we have $G^2(x; y) = G(G(x; y)) = G(x; 0) = (x; 0) = G(x; y)$. Hence, $G^2 = G$.

5.43. Find the dimension of (a) $A(\mathbf{R}^4)$, (b) $A(\mathbf{P}_2(t))$, (c) $A(\mathbf{M}_{2;3})$.

Use $\dim [A(V)] = n^2$ where $\dim V = n$. Hence, (a) $\dim [A(\mathbf{R}^4)] = 4^2 = 16$, (b) $\dim [A(\mathbf{P}_2(t))] = 3^2 = 9$, (c) $\dim [A(\mathbf{M}_{2;3})] = 6^2 = 36$.

5.44. Let E be a linear operator on V for which $E^2 = E$. (Such an operator is called a *projection*.) Let U be the image of E , and let W be the kernel. Prove

- (a) If $u \in U$, then $E(u) = u$ (i.e., E is the identity mapping on U).
- (b) If $E \neq I$, then E is singular—that is, $E(v) = 0$ for some $v \neq 0$.
- (c) $V = U \oplus W$.

(a) If $u \in U$, the image of E , then $E(v) = u$ for some $v \in V$. Hence, using $E^2 = E$, we have

$$u = E(v) = E^2(v) = E(E(v)) = E(u)$$

(b) If $E \neq I$, then for some $v \in V$, $E(v) = u$, where $v \neq u$. By (i), $E(u) = u$. Thus,

$$E(v - u) = E(v) - E(u) = u - u = 0, \quad \text{where} \quad v - u \neq 0$$

(c) We first show that $V = U + W$. Let $v \in V$. Set $u = E(v)$ and $w = v - E(v)$. Then

$$v = E(v) + v - E(v) = u + w$$

By definition, $u = E(v) \in U$, the image of E . We now show that $w \in W$, the kernel of E ,

$$E(w) = E(v - E(v)) = E(v) - E^2(v) = E(v) - E(v) = 0$$

and thus $w \in W$. Hence, $V = U + W$.

We next show that $U \cap W = \{0\}$. Let $v \in U \cap W$. Because $v \in U$, $E(v) = v$ by part (a). Because $v \in W$, $E(v) = 0$. Thus, $v = E(v) = 0$ and so $U \cap W = \{0\}$.

The above two properties imply that $V = U \oplus W$.

SUPPLEMENTARY PROBLEMS

Mappings

- 5.45.** Determine the number of different mappings from (a) $\{1; 2\}$ into $\{1; 2; 3\}$; (b) $\{1; 2; \dots, r\}$ into $\{1; 2; \dots, s\}$:
- 5.46.** Let $f: \mathbf{R} \rightarrow \mathbf{R}$ and $g: \mathbf{R} \rightarrow \mathbf{R}$ be defined by $f(x) = x^2 + 3x + 1$ and $g(x) = 2x - 3$. Find formulas defining the composition mappings: (a) $f \circ g$; (b) $g \circ f$; (c) $g \circ g$; (d) $f \circ f$.
- 5.47.** For each mappings $f: \mathbf{R} \rightarrow \mathbf{R}$ find a formula for its inverse: (a) $f(x) = 3x + 7$, (b) $f(x) = x^3 + 2$.
- 5.48.** For any mapping $f: A \rightarrow B$, show that $\mathbf{1}_B \circ f = f = f \circ \mathbf{1}_A$.

Linear Mappings

- 5.49.** Show that the following mappings are linear:
- (a) $F: \mathbf{R}^3 \rightarrow \mathbf{R}^2$ defined by $F(x; y; z) = (x + 2y - 3z; 4x - 5y + 6z)$.
- (b) $F: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ defined by $F(x; y) = (ax + by; cx + dy)$, where a, b, c, d belong to $\mathbf{R}$.
- 5.50.** Show that the following mappings are not linear:
- (a) $F: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ defined by $F(x; y) = (x^2; y^2)$.
- (b) $F: \mathbf{R}^3 \rightarrow \mathbf{R}^2$ defined by $F(x; y; z) = (x + 1; y + z)$.
- (c) $F: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ defined by $F(x; y) = (xy; y)$.
- (d) $F: \mathbf{R}^3 \rightarrow \mathbf{R}^2$ defined by $F(x; y; z) = (|x|; y + z)$.

- 5.51.** Find $F(a; b)$, where the linear map $F: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ is defined by $F(1; 2) = (3; -1)$ and $F(0; 1) = (2; 1)$.
- 5.52.** Find a 2×2 matrix A that maps
- (a) $(1; 3)^T$ and $(1; 4)^T$ into $(-2; 5)^T$ and $(3; -1)^T$, respectively.
 - (b) $(2; -4)^T$ and $(-1; 2)^T$ into $(1; 1)^T$ and $(1; 3)^T$, respectively.
- 5.53.** Find a 2×2 singular matrix B that maps $(1; 1)^T$ into $(1; 3)^T$.
- 5.54.** Let V be the vector space of real n -square matrices, and let M be a fixed nonzero matrix in V . Show that the first two of the following mappings $T: V \rightarrow V$ are linear, but the third is not:
- (a) $T(A) = MA$, (b) $T(A) = AM + MA$, (c) $T(A) = M + A$.
- 5.55.** Give an example of a nonlinear map $F: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ such that $F^{-1}(0) = \{0\}$ but F is not one-to-one.
- 5.56.** Let $F: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ be defined by $F(x; y) = (3x + 5y; 2x + 3y)$, and let S be the unit circle in $\mathbf{R}^2$. (S consists of all points satisfying $x^2 + y^2 = 1$.) Find (a) the image $F(S)$, (b) the preimage $F^{-1}(S)$.
- 5.57.** Consider the linear map $G: \mathbf{R}^3 \rightarrow \mathbf{R}^3$ defined by $G(x; y; z) = (x + y + z; y - 2z; y - 3z)$ and the unit sphere S_2 in $\mathbf{R}^3$, which consists of the points satisfying $x^2 + y^2 + z^2 = 1$. Find (a) $G(S_2)$, (b) $G^{-1}(S_2)$.
- 5.58.** Let H be the plane $x + 2y + 3z = 4$ in $\mathbf{R}^3$ and let G be the linear map in [Problem 5.57](#). Find (a) $G(H)$, (b) $G^{-1}(H)$.

5.59. Let W be a subspace of V . The *inclusion* map, denoted by $i: W \rightarrow V$, is defined by $i(w) = w$ for every $w \in W$. Show that the inclusion map is linear.

5.60. Suppose $F: V \rightarrow U$ is linear. Show that $F(-v) = -F(v)$.

Kernel and Image of Linear Mappings

5.61. For each linear map F find a basis and the dimension of the kernel and the image of F :

(a) $F: \mathbf{R}^3 \rightarrow \mathbf{R}^3$ defined by $F(x; y; z) = (x + 2y - 3z; 2x + 5y - 4z; x + 4y + z)$,

(b) $F: \mathbf{R}^4 \rightarrow \mathbf{R}^3$ defined by $F(x; y; z; t) = (x + 2y + 3z + 2t; 2x + 4y + 7z + 5t; x + 2y + 6z + 5t)$.

5.62. For each linear map G , find a basis and the dimension of the kernel and the image of G :

(a) $G: \mathbf{R}^3 \rightarrow \mathbf{R}^2$ defined by $G(x; y; z) = (x + y + z; 2x + 2y + 2z)$,

(b) $G: \mathbf{R}^3 \rightarrow \mathbf{R}^2$ defined by $G(x; y; z) = (x + y; y + z)$,

(c) $G: \mathbf{R}^5 \rightarrow \mathbf{R}^3$ defined by

$$G(x, y, z, s, t) = (x + 2y + 2z + s + t, \quad x + 2y + 3z + 2s - t, \quad 3x + 6y + 8z + 5s - t).$$

5.63. Each of the following matrices determines a linear map from $\mathbf{R}^4$ into $\mathbf{R}^3$:

$$(a) \quad A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 2 & -1 & 2 & -1 \\ 1 & -3 & 2 & -2 \end{bmatrix}, \quad (b) \quad B = \begin{bmatrix} 1 & 0 & 2 & -1 \\ 2 & 3 & -1 & 1 \\ -2 & 0 & -5 & 3 \end{bmatrix}.$$

Find a basis as well as the dimension of the kernel and the image of each linear map.

- 5.64.** Find a linear mapping $F: \mathbf{R}^3 \rightarrow \mathbf{R}^3$ whose image is spanned by $(1, 2, 3)$ and $(4, 5, 6)$.
- 5.65.** Find a linear mapping $G: \mathbf{R}^4 \rightarrow \mathbf{R}^3$ whose kernel is spanned by $(1, 2, 3, 4)$ and $(0, 1, 1, 1)$.
- 5.66.** Let $V = \mathbf{P}_{10}(t)$, the vector space of polynomials of degree ≤ 10 . Consider the linear map $\mathbf{D}^4$ denotes the fourth derivative $d^4(f) = dt^4$. Find a basis and the dimension of (a) the image of $\mathbf{D}^4$; (b) the kernel of $\mathbf{D}^4$.
- 5.67.** Suppose $F: V \rightarrow U$ is linear. Show that (a) the image of any subspace of V is a subspace of U ; (b) the preimage of any subspace of U is a subspace of V .
- 5.68.** Show that if $F: V \rightarrow U$ is onto, then $\dim U \leq \dim V$. Determine all linear maps $F: \mathbf{R}^3 \rightarrow \mathbf{R}^4$ that are onto.
- 5.69.** Consider the zero mapping $\mathbf{0}: V \rightarrow U$ defined by $\mathbf{0}(v) = 0; \forall v \in V$. Find the kernel and the image of $\mathbf{0}$.

Operations with linear Mappings

- 5.70.** Let $F: \mathbf{R}^3 \rightarrow \mathbf{R}^2$ and $G: \mathbf{R}^3 \rightarrow \mathbf{R}^2$ be defined by $F(x; y; z) = (y; x + z)$ and $G(x; y; z) = (2z; x - y)$. Find formulas defining the mappings $F + G$ and $3F - 2G$.
- 5.71.** Let $H: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ be defined by $H(x; y) = (y; 2x)$. Using the maps F and G in [Problem 5.70](#), find formulas defining the mappings: (a) $H \circ F$ and $H \circ G$, (b) $F \circ H$ and $G \circ H$, (c) $H \circ (F + G)$ and $H \circ F + H \circ G$.
- 5.72.** Show that the following mappings F, G, H are linearly independent:
- (a) $F; G; H \in \text{Hom}(\mathbf{R}^2; \mathbf{R}^2)$ defined by $F(x; y) = (x; 2y)$, $G(x; y) = (y; x + y)$, $H(x; y) = (0; x)$,

(b) $F, G, H \in \text{Hom}(\mathbf{R}^3; \mathbf{R})$ defined by $F(x; y; z) = x + y + z$, $G(x; y; z) = y + z$, $H(x; y; z) = x - z$.

5.73. For $F, G \in \text{Hom}(V; U)$, show that $\text{rank}(F + G) \leq \text{rank}(F) + \text{rank}(G)$. (Here V has finite dimension.)

5.74. Let $F: V \rightarrow U$ and $G: U \rightarrow V$ be linear. Show that if F and G are nonsingular, then $G \circ F$ is nonsingular. Give an example where $G \circ F$ is nonsingular but G is not. [Hint: Let $\dim V < \dim U$.]

5.75. Find the dimension d of (a) $\text{Hom}(\mathbf{R}^2; \mathbf{R}^8)$, (b) $\text{Hom}(\mathbf{P}_4(t); \mathbf{R}^3)$, (c) $\text{Hom}(\mathbf{M}_{2;4}; \mathbf{P}_2(t))$.

5.76. Determine whether or not each of the following linear maps is nonsingular. If not, find a nonzero vector v whose image is 0; otherwise find a formula for the inverse map:

(a) $F: \mathbf{R}^3 \rightarrow \mathbf{R}^3$ defined by $F(x; y; z) = (x + y + z; 2x + 3y + 5z; x + 3y + 7z)$,

(b) $G: \mathbf{R}^3 \rightarrow \mathbf{P}_2(t)$ defined by $G(x; y; z) = (x + y)t^2 + (x + 2y + 2z)t + y + z$,

(c) $H: \mathbf{R}^2 \rightarrow \mathbf{P}_2(t)$ defined by $H(x; y) = (x + 2y)t^2 + (x + y)t + x + y$.

5.77. When can $\dim [\text{Hom}(V; U)] = \dim V$?

Algebra of Linear Operators

5.78. Let F and G be the linear operators on $\mathbf{R}^2$ defined by $F(x; y) = (x + y; 0)$ and $G(x; y) = (y; x)$. Find formulas defining the linear operators: (a) $F + G$, (b) $5F - 3G$, (c) FG , (d) GF , (e) F^2 , (f) G^2 .

- 5.79.** Show that each linear operator T on $\mathbf{R}^2$ is nonsingular and find a formula for T^{-1} , where
- (a) $T(x; y) = (x + 2y; 2x + 3y)$, (b) $T(x; y) = (2x - 3y; 3x + 4y)$.
- 5.80.** Show that each of the following linear operators T on $\mathbf{R}^3$ is nonsingular and find a formula for T^{-1} , where
- (a) $T(x; y; z) = (x - 3y + 2z; y - 4z; z)$; (b) $T(x; y; z) = (x + z; x - y; y)$.
- 5.81.** Find the dimension of $A(V)$, where (a) $V = \mathbf{R}^7$, (b) $V = \mathbf{P}_5(t)$, (c) $V = \mathbf{M}_{3;4}$.
- 5.82.** Which of the following integers can be the dimension of an algebra $A(V)$ of linear maps: 5, 9, 12, 25, 28, 36, 45, 64, 88, 100?
- 5.83.** Let T be the linear operator on $\mathbf{R}^2$ defined by $T(x; y) = (x + 2y; 3x + 4y)$. Find a formula for $f(T)$, where
- (a) $f(t) = t^2 + 2t - 3$, (b) $f(t) = t^2 - 5t - 2$.

Miscellaneous Problems

- 5.84.** Suppose $F: V \rightarrow U$ is linear and k is a nonzero scalar. Prove that the maps F and kF have the same kernel and the same image.
- 5.85.** Suppose F and G are linear operators on V and that F is nonsingular. Assume that V has finite dimension. Show that $\text{rank}(FG) = \text{rank}(GF) = \text{rank}(G)$.
- 5.86.** Suppose V has finite dimension. Suppose T is a linear operator on V such that $\text{rank}(T^2) = \text{rank}(T)$. Show that $\text{Ker } T \cap \text{Im } T = \{0\}$.

5.87. Suppose $V = U \oplus W$. Let E_1 and E_2 be the linear operators on V defined by $E_1(v) = u$, $E_2(v) = w$, where $v = u + w$, $u \in U$, $w \in W$. Show that (a) $E_1^2 = E_1$ and $E_2^2 = E_2$ (i.e., that E_1 and E_2 are projections); (b) $E_1 + E_2 = I$, the identity mapping; (c) $E_1 E_2 = \mathbf{0}$ and $E_2 E_1 = \mathbf{0}$.

5.88. Let E_1 and E_2 be linear operators on V satisfying parts (a), (b), (c) of [Problem 5.88](#). Prove

$$V = \text{Im } E_1 \oplus \text{Im } E_2$$

5.89. Let v and w be elements of a real vector space V . The *line segment* L from v to $v + w$ is defined to be the set of vectors $v + tw$ for $0 \leq t \leq 1$. (See Fig. 5.6.)

(a) Show that the line segment L between vectors v and u consists of the points:

(i) $(1 - t)v + tw$ for $0 \leq t \leq 1$, (ii) $t_1 v + t_2 u$ for $t_1 + t_2 = 1$, $t_1 \geq 0$, $t_2 \geq 0$.

(b) Let $F: V \rightarrow U$ be linear. Show that the image $F(L)$ of a line segment L in V is a line segment in U .

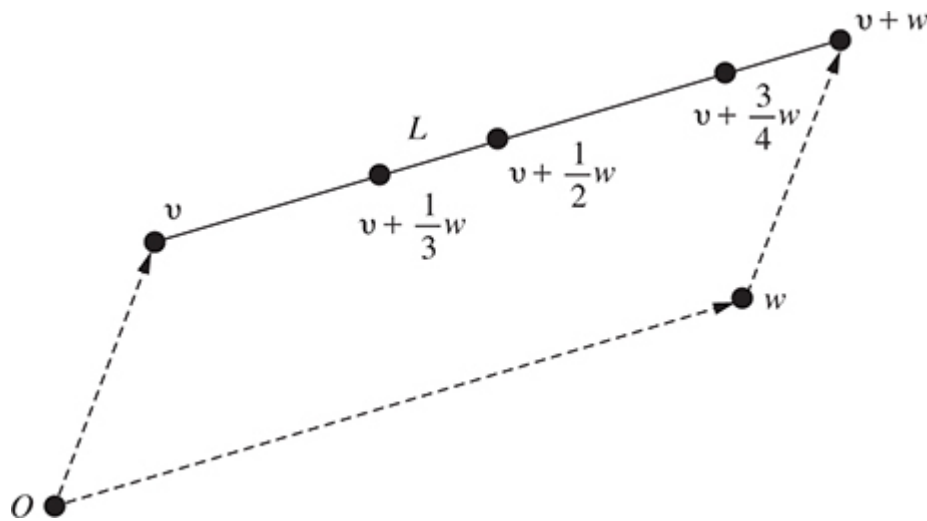


Figure 5-6

- 5.90.** Let $F: V \rightarrow U$ be linear and let W be a subspace of V . The *restriction* of F to W is the map $F|W: W \rightarrow U$ defined by $F|W(v) = F(v)$ for every v in W . Prove the following:
- (a) $F|W$ is linear; (b) $\text{Ker}(F|W) = (\text{Ker } F) \cap W$; (c) $\text{Im}(F|W) = F(W)$.
- 5.91.** A subset X of a vector space V is said to be *convex* if the line segment L between any two points (vectors) $P, Q \in X$ is contained in X . (a) Show that the intersection of convex sets is convex; (b) suppose $F: V \rightarrow U$ is linear and X is convex. Show that $F(X)$ is convex.

ANSWERS TO SUPPLEMENTARY PROBLEMS

- 5.45.** (a) $3^2 = 9$; (b) s^r
- 5.46.** (a) $(f \circ g)(x) = 4x^2 + 1$, (b) $(g \circ f)(x) = 2x^2 + 6x - 1$,
 (c) $(g \circ g)(x) = 4x - 9$,
 (d) $(f \circ f)(x) = x^4 + 6x^3 + 14x^2 + 15x + 5$
- 5.47.** (a) $f^{-1}(x) = \frac{1}{3}(x + 7)$, (b) $f^{-1}(x) = \sqrt[3]{x - 2}$
- 5.49.** $F(x; y; z) = A(x; y; z)^T$, where (a)
 $A = \begin{bmatrix} 1 & 2 & -3 \\ 4 & -5 & 6 \end{bmatrix}$, (b) $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$
- 5.50.** (a) $u = (2; 2)$, $k = 3$; then $F(ku) = (36; 36)$ but $kF(u) = (12; 12)$; (b) $F(0) \neq 0$;
 (c) $u = (1; 2)$, $v = (3; 4)$; then $F(u + v) = (24; 6)$ but $F(u) + F(v) = (14; 6)$;
 (d) $u = (1; 2; 3)$, $k = 2$; then $F(ku) = (2; 10)$ but $kF(u) = (2; 10)$.

5.51. $F(-a; b) = (a + 2b; -3a + b)$

5.52. (a) $A = \begin{bmatrix} -17 & 5 \\ 23 & -6 \end{bmatrix}$; (b) None. $(2; 4)$ and $(1; 2)$ are linearly dependent but not $(1, 1)$ and $(1, 3)$.

5.53. $B = \begin{bmatrix} 1 & 0 \\ 3 & 0 \end{bmatrix}$ [Hint: Send $(0; 1)^T$ into $(0; 0)^T$.]

5.55. $F(x; y) = (x^2; y^2)$

5.56. (a) $13x^2 - 42xy + 34y^2 = 1$, (b) $13x^2 + 42xy + 34y^2 = 1$

5.57. (a) $x^2 - 8xy + 26y^2 + 6xz - 38yz + 14z^2 = 1$, (b) $x^2 + 2xy + 3y^2 + 2xz - 8yz + 14z^2 = 1$

5.58. (a) $x - y + 2z = 4$, (b) $x + 6z = 4$

5.61. (a) $\dim(\text{Ker } F) = 1$, $\{(7; 2; 1)\}$; $\dim(\text{Im } F) = 2$, $\{(1; 2; 1); (0; 1; 2)\}$;

(b) $\dim(\text{Ker } F) = 2$, $\{(2; 1; 0; 0); (1; 0; 1; 1)\}$; $\dim(\text{Im } F) = 2$, $\{(1; 2; 1); (0; 1; 3)\}$

5.62. (a) $\dim(\text{Ker } G) = 2$, $\{(1; 0; 1); (1; 1; 0)\}$; $\dim(\text{Im } G) = 1$, $\{(1; 2)\}$;

(b) $\dim(\text{Ker } G) = 1$, $\{(1; 1; 1)\}$; $\text{Im } G = \mathbf{R}^2$, $\{(1; 0); (0; 1)\}$;

(c) $\dim(\text{Ker } G) = 3$, $\{(2; 1; 0; 0; 0); (1; 0; 1; 1; 0); (5; 0; 2; 0; 1)\}$; $\dim(\text{Im } G) = 2$, $\{(1; 1; 3); (0; 1; 2)\}$

5.63. (a) $\dim(\text{Ker } A) = 2$, $\{(4; 2; 5; 0); (1; 3; 0; 5)\}$; $\dim(\text{Im } A) = 2$, $\{(1; 2; 1); (0; 1; 1)\}$;

(b) $\dim(\text{Ker } B) = 1$, $\{(1; {}^2_3, 1; 1)\}$; $\text{Im } B = \mathbf{R}^3$

5.64. $F(x; y; z) = (x + 4y; 2x + 5y; 3x + 6y)$

5.65. $F(x; y; z; t) = (x + y - z; 2x + y - t; 0)$

5.66. (a) $\{1; t; t^2; \dots, t^6\}$, (b) $\{1; t; t^2; t^3\}$

5.68. None, because $\dim \mathbf{R}^4 > \dim \mathbf{R}^3$:

5.69. $\text{Ker } \mathbf{0} = V, \text{Im } \mathbf{0} = \{0\}$

5.70. $(F + G)(x; y; z) = (y + 2z; 2x y + z), (3F - 2G)(x; y; z) = (3y - 4z; x + 2y + 3z)$

5.71. (a) $(H \circ F)(x; y; z) = (x + z; 2y), (H \circ G)(x; y; z) = (x y; 4z);$

(b) not defined; (c) $(H \circ (F + G))(x; y; z) = (H \circ F + H \circ G)(x; y; z) = (2x - y + z; 2y + 4z)$

5.74. $F(x; y) = (x; y; y); G(x; y; z) = (x; y)$

5.75. (a) 16, (b) 15, (c) 24

5.76. (a) $v = (2; 3; 1);$ (b) $G^1(at^2 + bt + c) = (b - 2c; a - b + 2c; a + b - c);$

(c) H is nonsingular, but not invertible, because $\dim \mathbf{P}_2(t) > \dim \mathbf{R}^2$.

5.77. $\dim U = 1$; that is, $U = K$.

5.78. (a) $(F + G)(x; y) = (x; x);$ (b) $(5F + 3G)(x; y) = (5x + 8y; 3x);$ (c) $(FG)(x; y) = (x y; 0);$

(d) $(GF)(x; y) = (0; x + y);$ (e) $F^2(x; y) = (x + y; 0)$ (note that $F^2 = F$); (f) $G^2(x; y) = (x; -y).$

[Note that $G^2 + I = 0$; hence, G is a zero of $f(t) = t^2 + 1$.]

5.79. (a) $T^{-1}(x; y) = (3x + 2y; 2x y),$ (b) $T^{-1}(x; y) = (4x + 3y; 3x + 2y)$

5.80. (a) $T^{-1}(x; y; z) = (x + 3y + 14z; y - 4z; z)$, (b) $T^{-1}(x; y; z) = (y + z; y; x y - z)$

5.81. (a) 49, (b) 36, (c) 144

5.82. Squares: 9, 25, 36, 64, 100

5.83. (a) $T(x; y) = (6x + 14y; 21x + 27y)$; (b) $T(x; y) = (0; 0)$ —that is, $f(T) = 0$

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CHAPTER 6

Linear Mappings and Matrices

6.1 Introduction

Consider a basis $S = \{u_1, u_2, \dots, u_n\}$ of a vector space V over a field K . For any vector $v \in V$, suppose $v = a_1 u_1 + a_2 u_2 + \dots + a_n u_n$

$$v = a_1 u_1 + a_2 u_2 + \dots + a_n u_n$$

Then the coordinate vector of v relative to the basis S , which we assume to be a column vector (unless otherwise stated or implied), is denoted and defined by

$$[v]_S = [a_1, a_2, \dots, a_n]^T$$

Recall (Section 4.11) that the mapping $v \mapsto [v]_S$, determined by the basis S , is an isomorphism between V and K^n .

This chapter shows that there is also an isomorphism, determined by the basis S , between the algebra $A(V)$ of

linear operators on V and the algebra M of n -square matrices over K . Thus, every linear mapping $F: V \rightarrow V$ will correspond to an n -square matrix $[F]_S$ determined by the basis S . We will also show how our matrix representation changes when we choose another basis.

6.2 Matrix Representation of a Linear Operator

Let T be a linear operator (transformation) from a vector space V into itself, and suppose $S = \{u_1, u_2, \dots, u_n\}$ is a basis of V . Now $T(u_1), T(u_2), \dots, T(u_n)$ are vectors in V , and so each is a linear combination of the vectors in the basis S ; say,

$$\begin{aligned} T(u_1) &= a_{11}u_1 + a_{12}u_2 + \cdots + a_{1n}u_n \\ T(u_2) &= a_{21}u_1 + a_{22}u_2 + \cdots + a_{2n}u_n \\ &\vdots \\ T(u_n) &= a_{n1}u_1 + a_{n2}u_2 + \cdots + a_{nn}u_n \end{aligned}$$

The following definition applies.

DEFINITION: The transpose of the above matrix of coefficients, denoted by $m_S(T)$ or $[T_S]$, is called the *matrix representation* of T relative to the basis S , or simply the matrix of T in the basis S . (The subscript S may be omitted if the basis S is understood.)

Using the coordinate (column) vector notation, the matrix representation of T may be written in the form

$$m_S(T) = [T]_S = [[T(u_1)]_S, [T(u_2)]_S, \dots, [T(u_1)]_S]$$

That is, the columns of $m(T)$ are the coordinate vectors of $T(u_1), T(u_2), \dots, T(u_n)$, respectively.

EXAMPLE 6.1 Let $F: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ be the linear operator defined by $F(x, y) = (2x + 3y, 4x + 5y)$.

(a) Find the matrix representation of F relative to the basis $S = \{u_1, u_2\} = \{(1, 2), (2, 5)\}$.

(1) First find $F(u_1)$, and then write it as a linear combination of the basis vectors u_1 and u_2 . (For notational convenience, we use column vectors.) We have

$$F(u_1) = F\left(\begin{bmatrix} 1 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} 8 \\ -6 \end{bmatrix} = x \begin{bmatrix} 1 \\ 2 \end{bmatrix} + y \begin{bmatrix} 2 \\ 5 \end{bmatrix} \quad \text{and} \quad \begin{array}{l} x + 2y = 8 \\ 2x + 5y = -6 \end{array}$$

Solve the system to obtain $x = 52, y = 22$. Hence, $F(u_1) = 52 u_1 - 22 u_2$.

(2) Next find $F(u_2)$, and then write it as a linear combination of u_1 and u_2 :

$$F(u_2) = F\left(\begin{bmatrix} 2 \\ 5 \end{bmatrix}\right) = \begin{bmatrix} 19 \\ -17 \end{bmatrix} = x \begin{bmatrix} 1 \\ 2 \end{bmatrix} + y \begin{bmatrix} 2 \\ 5 \end{bmatrix} \quad \text{and} \quad \begin{array}{l} x + 2y = 19 \\ 2x + 5y = -17 \end{array}$$

Solve the system to get $x = 129, y = -55$. Thus, $F(u_2) = 129 u_1 - 55 u_2$.

Now write the coordinates of $F(u_1)$ and $F(u_2)$ as columns to obtain the matrix

$$[F]_S = \begin{bmatrix} 52 & 129 \\ -22 & -55 \end{bmatrix}$$

(b) Find the matrix representation of F relative to the (usual) basis $E = \{e_1, e_2\} = \{(1, 0), (0, 1)\}$.

Find $F(e_1)$ and write it as a linear combination of the usual basis vectors e_1 and e_2 , and then find $F(e_2)$ and write it as a linear combination of e_1 and e_2 . We have

$$\begin{aligned} F(e_1) &= F(1, 0) = (2, 2) = 2e_1 + 2e_2 \\ F(e_2) &= F(0, 1) = (3, -5) = 3e_1 - 5e_2 \end{aligned} \quad \text{and so} \quad [F]_E = \begin{bmatrix} 2 & 3 \\ 2 & -5 \end{bmatrix}$$

Note that the coordinates of $F(e_1)$ and $F(e_2)$ form the columns, not the rows, of $[F]_E$. Also, note that the arithmetic is much simpler using the usual basis of $\mathbf{R}^2$.

EXAMPLE 6.2 Let V be the vector space of functions with basis $S = \{\sin t, \cos t, e^{3t}\}$, and let $\mathbf{D} : V \rightarrow V$ be the differential operator defined by $\mathbf{D}(f(t)) = d(f(t))/dt$. We compute the matrix representing $\mathbf{D}$ in the basis S :

$$\mathbf{D}(\sin t) = \cos t = 0(\sin t) + 1(\cos t) + 0(e^{3t})$$

$$\mathbf{D}(\cos t) = -\sin t = -1(\sin t) + 0(\cos t) + 0(e^{3t})$$

$$\mathbf{D}(e^{3t}) = 3e^{3t} = 0(\sin t) + 0(\cos t) + 3(e^{3t})$$

and so

$$[\mathbf{D}] = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

Note that the coordinates of $\mathbf{D}(\sin t)$, $\mathbf{D}(\cos t)$, $\mathbf{D}(e^{3t})$ form the columns, not the rows, of $[\mathbf{D}]$.

Matrix Mappings and Their Matrix Representation

Consider the following matrix A , which may be viewed as a linear operator on $\mathbf{R}^2$, and basis S of $\mathbf{R}^2$:

$$A = \begin{bmatrix} 3 & -2 \\ 4 & -5 \end{bmatrix} \quad \text{and} \quad S = \{u_1, u_2\} = \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \end{bmatrix} \right\}$$

(We write vectors as columns, because our map is a matrix.) We find the matrix representation of A relative to the basis S .

(1) First we write $A(u_1)$ as a linear combination of u_1 and u_2 . We have

$$A(u_1) = \begin{bmatrix} 3 & -2 \\ 4 & -5 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 \\ -6 \end{bmatrix} = x \begin{bmatrix} 1 \\ 2 \end{bmatrix} + y \begin{bmatrix} 2 \\ 5 \end{bmatrix} \quad \text{and so} \quad \begin{array}{l} x + 2y = -1 \\ 2x + 5y = -6 \end{array}$$

Solving the system yields $x = 7$, $y = 4$. Thus, $A(u_1) = 7u_1 + 4u_2$.

(2) Next we write $A(u_2)$ as a linear combination of u_1 and u_2 . We have

$$A(u_2) = \begin{bmatrix} 3 & -2 \\ 4 & -5 \end{bmatrix} \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} -4 \\ -17 \end{bmatrix} = x \begin{bmatrix} 1 \\ 2 \end{bmatrix} + y \begin{bmatrix} 2 \\ 5 \end{bmatrix} \quad \text{and so} \quad \begin{array}{l} x + 2y = -4 \\ 2x + 5y = -17 \end{array}$$

Solving the system yields $x = 14$, $y = -9$. Thus, $A(u_2) = 14u_1 - 9u_2$. Writing the coordinates of $A(u_1)$ and $A(u_2)$ as columns gives us the following matrix representation of A :

$$[A]_S = \begin{bmatrix} 7 & 14 \\ -4 & -9 \end{bmatrix}$$

Remark: Suppose we want to find the matrix representation of A relative to the usual basis $E = \{e_1, e_2\} = \{[1, 0]^T, [0, 1]^T\}$ of $\mathbf{R}^2$: We have

$$\begin{aligned} A(e_1) &= \begin{bmatrix} 3 & -2 \\ 4 & -5 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix} = 3e_1 + 4e_2 \\ A(e_2) &= \begin{bmatrix} 3 & -2 \\ 4 & -5 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ -5 \end{bmatrix} = -2e_1 - 5e_2 \end{aligned} \quad \text{and so} \quad [A]_E = \begin{bmatrix} 3 & -2 \\ 4 & -5 \end{bmatrix}$$

Note that $[A]_E$ is the original matrix A . This result is true in general:

The matrix representation of any $n \times n$ square matrix A over a field K relative to the usual basis E of K^n is the matrix A itself; that is,

$$[A]_E = A$$

Algorithm for Finding Matrix Representations

Next follows an algorithm for finding matrix representations. The first Step 0 is optional. It may be useful to use it in Step 1 (b), which is repeated for each basis vector.

ALGORITHM 6.1: The input is a linear operator T on a vector space V and a basis $S = \{u_1, u_2, \dots, u_n\}$ of V . The output is the matrix representation $[T]_S$.

Step 0. Find a formula for the coordinates of an arbitrary vector v relative to the basis S .

Step 1. Repeat for each basis vector u_k in S :

(a) Find $T(u_k)$.

(b) Write $T(u_k)$ as a linear combination of the basis vectors $u_1, u_2, \dots, u_n$.

Step 2. Form the matrix $[T]_S$ whose columns are the coordinate vectors in Step 1 (b).

EXAMPLE 6.3 Let $F: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ be defined by $F(x, y) = (2x + 3y, 4x - 5y)$. Find the matrix representation $[F]_S$ of F relative to the basis $S = \{u_1, u_2\} = \{(1, 2), (2, 5)\}$.

(Step 0) First find the coordinates of $(a, b) \in \mathbf{R}^2$ relative to the basis S . We have

$$\begin{bmatrix} a \\ b \end{bmatrix} = x \begin{bmatrix} 1 \\ -2 \end{bmatrix} + y \begin{bmatrix} 2 \\ -5 \end{bmatrix} \quad \text{or} \quad \begin{array}{l} x + 2y = a \\ -2x - 5y = b \end{array} \quad \text{or} \quad \begin{array}{l} x + 2y = a \\ -y = 2a + b \end{array}$$

Solving for x and y in terms of a and b yields $x = 5a + 2b$, $y = -2a - b$. Thus,

$$(a, b) = (5a + 2b)u_1 + (-2a - b)u_2$$

(Step 1) Now we find $F(u_1)$ and write it as a linear combination of u_1 and u_2 using the above formula for $(a; b)$, and then we repeat the process for $F(u_2)$. We have

$$\begin{aligned} F(u_1) &= F(1, -2) = (-4, 14) = 8u_1 - 6u_2 \\ F(u_2) &= F(2, -5) = (-11, 33) = 11u_1 - 11u_2 \end{aligned}$$

(Step 2) Finally, we write the coordinates of $F(u_1)$ and $F(u_2)$ as columns to obtain the required matrix:

$$[F]_S = \begin{bmatrix} 8 & 11 \\ -6 & -11 \end{bmatrix}$$

Properties of Matrix Representations

This subsection gives the main properties of the matrix representations of linear operators T on a vector space V . We emphasize that we are always given a particular basis S of V .

Our first theorem, proved in [Problem 6.9](#), tells us that the “action” of a linear operator T on a vector v is preserved by its matrix representation.

THEOREM 6.1: Let $T: V \rightarrow V$ be a linear operator, and let S be a (finite) basis of V . Then, for any vector v in V , $[T]_S[v]_S = [T(v)]_S$.

EXAMPLE 6.4 Consider the linear operator F on R^2 and the basis S of [Example 6.3](#); that is,

$$F(x, y) = (2x + 3y, \quad 4x - 5y) \quad \text{and} \quad S = \{u_1, u_2\} = \{(1, -2), \quad (2, -5)\}$$

Let

$$v = (5, -7), \quad \text{and so} \quad F(v) = (-11, 55)$$

Using the formula from [Example 6.3](#), we get

$$[v] = [11, -3]^T \quad \text{and} \quad [F(v)] = [55, -33]^T$$

We verify [Theorem 6.1](#) for this vector v (where $[F]$ is obtained from [Example 6.3](#)):

$$[F][v] = \begin{bmatrix} 8 & 11 \\ -6 & -11 \end{bmatrix} \begin{bmatrix} 11 \\ -3 \end{bmatrix} = \begin{bmatrix} 55 \\ -33 \end{bmatrix} = [F(v)]$$

Given a basis S of a vector space V , we have associated a matrix $[T]$ to each linear operator T in the algebra $A(V)$ of linear operators on V . [Theorem 6.1](#) tells us that the “action” of an individual linear operator T is preserved by this representation. The next two theorems (proved in Problems

6.10 and 6.11) tell us that the three basic operations in $A(V)$ with these operators—namely (i) addition, (ii) scalar multiplication, and (iii) composition—are also preserved.

THEOREM 6.2: Let V be an n -dimensional vector space over K , let S be a basis of V , and let $\mathbf{M}$ be the algebra of $n \times n$ matrices over K . Then the mapping

$$m: A(V) \rightarrow \mathbf{M} \quad \text{defined by} \quad m(T) = [T]_S$$

is a vector space isomorphism. That is, for any $F, G \in A(V)$ and any $k \in K$,

- (i) $m(F + G) = m(F) + m(G)$ or $[F + G] = [F] + [G]$
- (ii) $m(kF) = km(F)$ or $[kF] = k[F]$
- (iii) m is bijective (one-to-one and onto).

THEOREM 6.3: For any linear operators $F, G \in A(V)$,

$$m(G \circ F) = m(G)m(F) \quad \text{or} \quad [G \circ F] = [G][F]$$

(Here $G \circ F$ denotes the composition of the maps G and F .)

6.3 Change of Basis

Let V be an n -dimensional vector space over a field K . We have shown that once we have selected a basis S of V , every vector $v \in V$ can be represented by means of an n -tuple $[v]_S$ in K^n , and every linear operator T in $A(V)$ can be represented by an $n \times n$ matrix over K . We ask the following natural question:

How do our representations change if we select another basis?

In order to answer this question, we first need a definition.

DEFINITION: Let $S = \{u_1, u_2, \dots, u_n\}$ be a basis of a vector space V , and let $S' = \{v_1, v_2, \dots, v_n\}$ be another basis. (For reference, we will call S the “old” basis and S' the “new” basis.) Because S' is a basis, each vector in the “new” basis S' can be written uniquely as a linear combination of the vectors in S ; say,

$$v_1 = a_{11}u_1 + a_{12}u_2 + \cdots + a_{1n}u_n$$

$$v_2 = a_{21}u_1 + a_{22}u_2 + \cdots + a_{2n}u_n$$

.....

$$v_n = a_{n1}u_1 + a_{n2}u_2 + \cdots + a_{nn}u_n$$

Let P be the transpose of the above matrix of coefficients; that is, let $P = [p_{ij}]$, where $p_{ij} = a_{ji}$. Then P is called the *change-of-basis matrix* (or *transition matrix*) from the “old” basis S to the “new” basis S' .

The following remarks are in order.

Remark 1: The above change-of-basis matrix P may also be viewed as the matrix whose columns are, respectively, the coordinate column vectors of the “new” basis vectors u_i relative to the “old” basis S ; namely,

$$P = [[v_1]_S, [v_2]_S, \dots, [v_n]_S]$$

Remark 2: Analogously, there is a change-of-basis matrix Q from the “new” basis S' to the “old” basis S .

Similarly, Q may be viewed as the matrix whose columns are, respectively, the coordinate column vectors of the “old” basis vectors u_i relative to the “new” basis S' ; namely,

$$Q = [[u_1]_{S'}, [u_2]_{S'}, \dots, [u_n]_{S'}]$$

Remark 3: Because the vectors $u_1, u_2, \dots, u_n$ in the new basis S are linearly independent, the matrix P is invertible ([Problem 6.18](#)). Similarly, Q is invertible. In fact, we have the following proposition (proved in [Problem 6.18](#)).

PROPOSITION 6.4: Let P and Q be the above change-of-basis matrices. Then $Q = P^{-1}$.

Now suppose $S = \{u_1, u_2, \dots, u_n\}$ is a basis of a vector space V , and suppose $P = [p_{ij}]$ is any nonsingular matrix. Then the n vectors

$$v_i = p_{1i}u_1 + p_{2i}u_2 + \dots + p_{ni}u_n, \quad i = 1, 2, \dots, n$$

corresponding to the columns of P , are linearly independent [[Problem 6.21](#) (a)]. Thus, they form another basis S' of V . Moreover, P will be the change-of-basis matrix from S to the new basis S' .

EXAMPLE 6.5 Consider the following two bases of $\mathbf{R}^2$:

$$S = \{u_1, u_2\} = \{(1, 2), (3, 5)\} \quad \text{and} \quad S' = \{v_1, v_2\} = \{(1, -1), (1, -2)\}$$

(a) Find the change-of-basis matrix P from S to the “new” basis S' .

Write each of the new basis vectors of S' as a linear combination of the original basis vectors u_1 and u_2 of S . We have

$$\begin{aligned} \begin{bmatrix} 1 \\ -2 \end{bmatrix} &= x \begin{bmatrix} 1 \\ 2 \end{bmatrix} + y \begin{bmatrix} 3 \\ 5 \end{bmatrix} && \text{or} && \begin{aligned} x + 3y &= 1 \\ 2x + 5y &= -2 \end{aligned} && \text{yielding} && \begin{aligned} x &= -8, & y &= 3 \end{aligned} \\ \begin{bmatrix} 1 \\ -1 \end{bmatrix} &= x \begin{bmatrix} 1 \\ 2 \end{bmatrix} + y \begin{bmatrix} 3 \\ 5 \end{bmatrix} && \text{or} && \begin{aligned} x + 3y &= 1 \\ 2x + 5y &= -1 \end{aligned} && \text{yielding} && \begin{aligned} x &= -11, & y &= 4 \end{aligned} \end{aligned}$$

Thus,

$$\begin{aligned} v_1 &= -8u_1 + 3u_2 \\ v_2 &= -11u_1 + 4u_2 \end{aligned} \quad \text{and hence,} \quad P = \begin{bmatrix} -8 & -11 \\ 3 & 4 \end{bmatrix}.$$

Note that the coordinates of v_1 and v_2 are the columns, not rows, of the change-of-basis matrix P .

(b) Find the change-of-basis matrix Q from the “new” basis S back to the “old” basis S .

Here we write each of the “old” basis vectors u_1 and u_2 of S as a linear combination of the “new” basis vectors v_1 and v_2 of S . This yields

$$\begin{aligned} u_1 &= 4v_1 - 3v_2 \\ u_2 &= 11v_1 - 8v_2 \end{aligned} \quad \text{and hence,} \quad Q = \begin{bmatrix} 4 & 11 \\ -3 & -8 \end{bmatrix}$$

As expected from Proposition 6.4, $Q = P^{-1}$. (In fact, we could have obtained Q by simply finding P^{-1} .)

EXAMPLE 6.6 Consider the following two bases of $\mathbf{R}^3$:

$$E = \{e_1, e_2, e_3\} = \{(1, 0, 0), \quad (0, 1, 0), \quad (0, 0, 1)\}$$

and

$$S = \{u_1, u_2, u_3\} = \{(1, 0, 1), \quad (2, 1, 2), \quad (1, 2, 2)\}$$

(a) Find the change-of-basis matrix P from the basis E to the basis S .

Because E is the usual basis, we can immediately write each basis element of S as a linear combination of the basis elements of E . Specifically,

$$\begin{aligned} u_1 &= (1, 0, 1) = e_1 + 0 + e_3 \\ u_2 &= (2, 1, 2) = 2e_1 + e_2 + 2e_3 \\ u_3 &= (1, 2, 2) = e_1 + 2e_2 + 2e_3 \end{aligned} \quad \text{and hence,} \quad P = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 1 & 2 & 2 \end{bmatrix}$$

Again, the coordinates of u_1, u_2, u_3 appear as the columns in P . Observe that P is simply the matrix whose columns are the basis vectors of S . This is true only because the original basis was the usual basis E .

(b) Find the change-of-basis matrix Q from the basis S to the basis E .

The definition of the change-of-basis matrix Q tells us to write each of the (usual) basis vectors in E as a linear combination of the basis elements of S . This yields

$$\begin{aligned} e_1 &= (1, 0, 0) = -2u_1 + 2u_2 - u_3 \\ e_2 &= (0, 1, 0) = -2u_1 + u_2 \\ e_3 &= (0, 0, 1) = 3u_1 - 2u_2 + u_3 \end{aligned} \quad \text{and hence,} \quad Q = \begin{bmatrix} -2 & -2 & 3 \\ 2 & 1 & -2 \\ -1 & 0 & 1 \end{bmatrix}$$

We emphasize that to find Q , we need to solve three 3×3 systems of linear equations—one 3×3 system for each of e_1, e_2, e_3 .

Alternatively, we can find $Q = P^{-1}$ by forming the matrix $M = [P, I]$ and row reducing M to row canonical form:

$$M = \begin{bmatrix} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 & 1 & 0 \\ 1 & 2 & 2 & 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & -2 & -2 & 3 \\ 0 & 1 & 0 & 2 & 1 & -2 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{bmatrix} = [I, P^{-1}]$$

thus,

$$Q = P^{-1} = \begin{bmatrix} -2 & -2 & 3 \\ 2 & 1 & -2 \\ -1 & 0 & 1 \end{bmatrix}$$

(Here we have used the fact that Q is the inverse of P .)

The result in [Example 6.6](#) (a) is true in general. We state this result formally, because it occurs often.

PROPOSITION 6.5: The change-of-basis matrix from the usual basis E of K^n to any basis S of K^n is the matrix P whose columns are, respectively, the basis vectors of S .

Applications of Change-of-Basis Matrix

First we show how a change of basis affects the coordinates of a vector in a vector space V . The following theorem is proved in [Problem 6.22](#).

THEOREM 6.6: Let P be the change-of-basis matrix from a basis S to a basis S' in a vector space V . Then, for any vector $v \in V$, we have

$$P[v]_{S'} = [v]_S \quad \text{and hence,} \quad P^{-1}[v]_S = [v]_{S'}$$

Namely, if we multiply the coordinates of v in the original basis S by P^{-1} , we get the coordinates of v in the new basis S' .

Remark 1: Although P is called the change-of-basis matrix from the old basis S to the new basis S' , we emphasize that P^{-1} transforms the coordinates of v in the original basis S into the coordinates of v in the new basis S' .

Remark 2: Because of the above theorem, many texts call $Q = P^{-1}$, not P , the transition matrix from the old basis S to the new basis S' . Some texts also refer to Q as the *change-of-coordinates* matrix.

We now give the proof of the above theorem for the special case that $\dim V = 3$. Suppose P is the change-of-basis matrix from the basis $S = \{u_1, u_2, u_3\}$ to the basis $S' = \{v_1, v_2, v_3\}$; say,

$$\begin{aligned} v_1 &= a_1u_1 + a_2u_2 + a_3u_3 \\ v_2 &= b_1u_1 + b_2u_2 + b_3u_3 \\ v_3 &= c_1u_1 + c_2u_2 + c_3u_3 \end{aligned} \quad \text{and hence,} \quad P = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$$

Now suppose $v \in V$ and, say, $v = k_1v_1 + k_2v_2 + k_3v_3$.

Then, substituting for v_1, v_2, v_3 from above, we obtain

$$\begin{aligned} v &= k_1(a_1u_1 + a_2u_2 + a_3u_3) + k_2(b_1u_1 + b_2u_2 + b_3u_3) + k_3(c_1u_1 + c_2u_2 + c_3u_3) \\ &= (a_1k_1 + b_1k_2 + c_1k_3)u_1 + (a_2k_1 + b_2k_2 + c_2k_3)u_2 + (a_3k_1 + b_3k_2 + c_3k_3)u_3 \end{aligned}$$

Thus,

$$[v]_{S'} = \begin{bmatrix} k_1 \\ k_2 \\ k_3 \end{bmatrix} \quad \text{and} \quad [v]_S = \begin{bmatrix} a_1k_1 + b_1k_2 + c_1k_3 \\ a_2k_1 + b_2k_2 + c_2k_3 \\ a_3k_1 + b_3k_2 + c_3k_3 \end{bmatrix}$$

Accordingly,

$$P[v]_{S'} = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \\ k_3 \end{bmatrix} = \begin{bmatrix} a_1k_1 + b_1k_2 + c_1k_3 \\ a_2k_1 + b_2k_2 + c_2k_3 \\ a_3k_1 + b_3k_2 + c_3k_3 \end{bmatrix} = [v]_S$$

Finally, multiplying the equation $[v]_S = P[v]_{S'}$ by P^{-1} , we get

$$P^{-1}[v]_S = P^{-1}P[v]_{S'} = I[v]_{S'} = [v]_{S'}$$

The next theorem (proved in [Problem 6.26](#)) shows how a change of basis affects the matrix representation of a linear operator.

THEOREM 6.7: Let P be the change-of-basis matrix from a basis S to a basis S' in a vector space V . Then, for any linear operator T on V ,

$$[T]_{S'} = P^{-1}[T]_S P$$

That is, if A and B are the matrix representations of T relative, respectively, to S and S' , then

$$B = P^{-1}AP$$

EXAMPLE 6.7 Consider the following two bases of $\mathbf{R}^3$:

$$\begin{aligned} E &= \{e_1, e_2, e_3\} = \{(1, 0, 0), \quad (0, 1, 0), \quad (0, 0, 1)\} \\ S &= \{u_1, u_2, u_3\} = \{(1, 0, 1), \quad (2, 1, 2), \quad (1, 2, 2)\} \end{aligned}$$

and

The change-of-basis matrix P from E to S and its inverse P^{-1} were obtained in [Example 6.6](#).

(a) Write $v = (1, 3, 5)$ as a linear combination of u_1, u_2, u_3 , or, equivalently, find $[v]_S$.

One way to do this is to directly solve the vector equation $v = xu_1 + yu_2 + zu_3$; that is,

$$\begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} = x \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + y \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} + z \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \quad \text{or} \quad \begin{aligned} x + 2y + z &= 1 \\ y + 2z &= 3 \\ x + 2y + 2z &= 5 \end{aligned}$$

The solution is $x = 7, y = 5, z = 4$, so $v = 7u_1 + 5u_2 + 4u_3$.

On the other hand, we know that $[v]_E = [1, 3, 5]^T$, because E is the usual basis, and we already know P^{-1} .

Therefore, by [Theorem 6.6](#),

$$[v]_S = P^{-1}[v]_E = \begin{bmatrix} -2 & -2 & 3 \\ 2 & 1 & -2 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 7 \\ -5 \\ 4 \end{bmatrix}$$

Thus, again, $v = 7u_1 + 5u_2 + 4u_3$.

(b) Let $A = \begin{bmatrix} 1 & 3 & -2 \\ 2 & -4 & 1 \\ 3 & -1 & 2 \end{bmatrix}$, which may be viewed as a linear operator on $\mathbf{R}^3$. Find the matrix B that represents A relative to the basis S .

The definition of the matrix representation of A relative to the basis S tells us to write each of $A(u_1)$, $A(u_2)$, $A(u_3)$ as a linear combination of the basis vectors u_1, u_2, u_3 of S . This yields

$$\begin{aligned} A(u_1) &= (-1, 3, 5) = 11u_1 - 9u_2 + 6u_3 \\ A(u_2) &= (1, 2, 9) = 21u_1 - 14u_2 + 8u_3 \\ A(u_3) &= (3, -4, 5) = 17u_1 - 8u_2 + 2u_3 \end{aligned} \quad \text{and hence,} \quad B = \begin{bmatrix} 11 & 21 & 17 \\ -9 & -14 & -8 \\ 6 & 8 & 2 \end{bmatrix}$$

We emphasize that to find B , we need to solve three 3×3 systems of linear equations—one 3×3 system for each of $A(u_1)$, $A(u_2)$, $A(u_3)$.

On the other hand, because we know P and P^{-1} , we can use [Theorem 6.7](#). That is,

$$B = P^{-1}AP = \begin{bmatrix} -2 & -2 & 3 \\ 2 & 1 & -2 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & -2 \\ 2 & -4 & 1 \\ 3 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 1 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 11 & 21 & 17 \\ -9 & -14 & -8 \\ 6 & 8 & 2 \end{bmatrix}$$

This, as expected, gives the same result.

6.4 Similarity

Suppose A and B are square matrices for which there exists an invertible matrix P such that $B = P^{-1}AP$; then B is said to be *similar* to A , or B is said to be obtained from A by a *similarity transformation*. We show ([Problem 6.29](#)) that similarity of matrices is an equivalence relation.

By [Theorem 6.7](#) and the above remark, we have the following basic result.

THEOREM 6.8: Two matrices represent the same linear operator if and only if the matrices are similar.

That is, all the matrix representations of a linear operator T form an equivalence class of similar matrices.

A linear operator T is said to be *diagonalizable* if there exists a basis S of V such that T is represented by a diagonal matrix; the basis S is then said to *diagonalize* T . The preceding theorem gives us the following result.

THEOREM 6.9: Let A be the matrix representation of a linear operator T . Then T is diagonalizable if and only if there exists an invertible matrix P such that $P^{-1}AP$ is a diagonal matrix.

That is, T is diagonalizable if and only if its matrix representation can be diagonalized by a similarity transformation.

We emphasize that not every operator is diagonalizable. However, we will show ([Chapter 10](#)) that every linear operator can be represented by certain “standard” matrices called its *normal* or *canonical* forms. Such a discussion will require some theory of fields, polynomials, and determinants.

Functions and Similar Matrices

Suppose f is a function on square matrices that assigns the same value to similar matrices; that is, $f(A) = f(B)$ whenever A is similar to B . Then f induces a function, also denoted by f , on linear operators T in the following natural way. We define

$$f(T) = f([T]_S)$$

where S is any basis. By [Theorem 6.8](#), the function is well defined.

The determinant ([Chapter 8](#)) is perhaps the most important example of such a function. The trace (Section 2.7) is another important example of such a function.

EXAMPLE 6.8 Consider the following linear operator F and bases E and S of $\mathbf{R}^2$:

$$F(x, y) = (2x + 3y, 4x - 5y), \quad E = \{(1, 0), (0, 1)\}, \quad S = \{(1, 2), (2, 5)\}$$

By [Example 6.1](#), the matrix representations of F relative to the bases E and S are, respectively,

$$A = \begin{bmatrix} 2 & 3 \\ 4 & -5 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 52 & 129 \\ -22 & -55 \end{bmatrix}$$

(ii) Trace of $F = \text{tr}(A) = 2 - 5 = -3$.

(ii) Trace of $F = \text{tr}(B) = 52 \ 55 = 3$.

As expected, both matrices yield the same result.

6.5 Matrices and General Linear Mappings

$$S = \{v_1, v_2, \dots, v_m\} \quad \text{and} \quad S' = \{u_1, u_2, \dots, u_n\}$$

Suppose $F: V \rightarrow U$ is a linear mapping. Then the vectors $F(v_1), F(v_2), \dots, F(v_m)$ belong to U , and so each is a linear combination of the basis vectors in S ; say,

$$\begin{aligned} F(v_1) &= a_{11}u_1 + a_{12}u_2 + \cdots + a_{1n}u_n \\ F(v_2) &= a_{21}u_1 + a_{22}u_2 + \cdots + a_{2n}u_n \\ &\vdots \\ F(v_m) &= a_{m1}u_1 + a_{m2}u_2 + \cdots + a_{mn}u_n \end{aligned}$$

DEFINITION: The transpose of the above matrix of coefficients, denoted by $m_{S;S_0}(F)$ or $[F_S;S_0$, is called the *matrix representation* of F relative

to the bases S and S' . [We will use the simple notation $m(F)$ and $[F$ when the bases are understood.]

The following theorem is analogous to [Theorem 6.1](#) for linear operators ([Problem 6.67](#)).

THEOREM 6.10: For any vector $v \in V$, $[F]_{S',S}[v]_S = [F(v)]_{S'}$.

That is, multiplying the coordinates of v in the basis S of V by $[F]$, we obtain the coordinates of $F(v)$ in the basis S' of U .

Recall that for any vector spaces V and U , the collection of all linear mappings from V into U is a vector space and is denoted by $\text{Hom}(V, U)$. The following theorem is analogous to [Theorem 6.2](#) for linear operators, where now we let $\mathbf{M} = \mathbf{M}_{m,n}$ denote the vector space of all $m \times n$ matrices ([Problem 6.67](#)).

THEOREM 6.11: The mapping $m: \text{Hom}(V, U) \rightarrow \mathbf{M}$ defined by $m(F) = [F]$ is a vector space isomorphism. That is, for any $F, G \in \text{Hom}(V, U)$ and any scalar k ,

$$(i) m(F + G) = m(F) + m(G) \text{ or } [F + G] = [F] + [G]$$

$$(ii) m(kF) = km(F) \text{ or } [kF] = k[F]$$

(iii) m is bijective (one-to-one and onto).

Our next theorem is analogous to [Theorem 6.3](#) for linear operators ([Problem 6.67](#)).

THEOREM 6.12: Let S, S', S'' be bases of vector spaces V, U, W , respectively. Let $F: V \rightarrow U$ and $G: U \rightarrow W$ be linear mappings. Then

$$[G \circ F]_{S,S''} = [G]_{S',S''}[F]_{S,S'}$$

That is, relative to the appropriate bases, the matrix representation of the composition of two mappings is the matrix product of the matrix representations of the individual mappings.

Next we show how the matrix representation of a linear mapping $F: V \rightarrow U$ is affected when new bases are selected ([Problem 6.67](#)).

THEOREM 6.13: Let P be the change-of-basis matrix from a basis e to a basis e' in V , and let Q be the change-of-basis matrix from a basis f to a basis f' in U . Then, for any linear map $F: V \rightarrow U$,

$$[F]_{e',f'} = Q^{-1}[F]_{e,f}P$$

In other words, if A is the matrix representation of a linear mapping F relative to the bases e and f , and B is the matrix representation of F relative to the bases e' and f' , then

$$B = Q^{-1}AP$$

Our last theorem, proved in [Problem 6.36](#), shows that any linear mapping from one vector space V into another vector space U can be represented by a very simple matrix. We note that this theorem is analogous to [Theorem 3.18](#) for $m \times n$ matrices.

THEOREM 6.14: Let $F: V \rightarrow U$ be linear and, say, $\text{rank}(F) = r$. Then there exist bases of V and U such that the matrix representation of F has the form

$$A = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$$

where I_r is the r -square identity matrix.

The above matrix A is called the *normal* or *canonical* form of the linear map F .

SOLVED PROBLEMS

Matrix Representation of Linear Operators

6.1. Consider the linear mapping $F: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ defined by $F(x, y) = (3x + 4y, 2x - 5y)$ and the following bases of $\mathbf{R}^2$:

$$E = \{e_1, e_2\} = \{(1, 0), (0, 1)\} \quad \text{and} \quad S = \{u_1, u_2\} = \{(1, 2), (2, 3)\}$$

(a) Find the matrix A representing F relative to the basis E .

(b) Find the matrix B representing F relative to the basis S .

(a) Because E is the usual basis, the rows of A are simply the coefficients in the components of $F(x, y)$; that is, using $(a, b) = ae_1 + be_2$, we have

$$\begin{aligned} F(e_1) &= F(1, 0) = (3, 2) = 3e_1 + 2e_2 \\ F(e_2) &= F(0, 1) = (4, -5) = 4e_1 - 5e_2 \end{aligned} \quad \text{and so} \quad A = \begin{bmatrix} 3 & 4 \\ 2 & -5 \end{bmatrix}$$

Note that the coefficients of the basis vectors are written as columns in the matrix representation.

(b) First find $F(u_1)$ and write it as a linear combination of the basis vectors u_1 and u_2 . We have

$$F(u_1) = F(1, 2) = (11, -8) = x(1, 2) + y(2, 3), \quad \text{and so} \quad \begin{aligned} x + 2y &= 11 \\ 2x + 3y &= -8 \end{aligned}$$

Solve the system to obtain $x = -49$, $y = 30$. Therefore,

$$F(u_1) = -49u_1 + 30u_2$$

Next find $F(u_2)$ and write it as a linear combination of the basis vectors u_1 and u_2 . We have

$$F(u_2) = F(2, 3) = (18, -11) = x(1, 2) + y(2, 3), \quad \text{and so} \quad \begin{aligned} x + 2y &= 18 \\ 2x + 3y &= -11 \end{aligned}$$

Solve for x and y to obtain $x = -76$, $y = 47$. Hence,

$$F(u_2) = -76u_1 + 47u_2$$

Write the coefficients of u_1 and u_2 as columns to obtain

$$B = \begin{bmatrix} -49 & -76 \\ 30 & 47 \end{bmatrix}$$

(b') Alternatively, one can first find the coordinates of an arbitrary vector (a, b) in $\mathbf{R}^2$ relative to the basis S .

We have

$$(a, b) = x(1, 2) + y(2, 3) = (x + 2y, 2x + 3y), \quad \text{and so} \quad \begin{aligned} x + 2y &= a \\ 2x + 3y &= b \end{aligned}$$

Solve for x and y in terms of a and b to get $x = -3a + 2b$, $y = 2a - b$. Thus,

$$(a, b) = (-3a + 2b)u_1 + (2a - b)u_2$$

Then use the formula for $(a; b)$ to find the coordinates of $F(u_1)$ and $F(u_2)$ relative to S :

$$\begin{aligned} F(u_1) &= F(1, 2) = (11, -8) = -49u_1 + 30u_2 \\ F(u_2) &= F(2, 3) = (18, -11) = -76u_1 + 47u_2 \end{aligned} \quad \text{and so} \quad B = \begin{bmatrix} -49 & -76 \\ 30 & 47 \end{bmatrix}$$

6.2. Consider the following linear operator G on $\mathbf{R}^2$ and basis S :

$$G(x, y) = (2x - 7y, 4x + 3y) \quad \text{and} \quad S = \{u_1, u_2\} = \{(1, 3), (2, 5)\}$$

(a) Find the matrix representation $[G]_S$ of G relative to S .

(b) Verify $[G]_S[v]_S = [G(v)]_S$ for the vector $v = (4, 3)$ in $\mathbf{R}^2$.

First find the coordinates of an arbitrary vector $v = (a, b)$ in $\mathbf{R}^2$ relative to the basis S .

We have

$$\begin{bmatrix} a \\ b \end{bmatrix} = x \begin{bmatrix} 1 \\ 3 \end{bmatrix} + y \begin{bmatrix} 2 \\ 5 \end{bmatrix}, \quad \text{and so} \quad \begin{aligned} x + 2y &= a \\ 3x + 5y &= b \end{aligned}$$

Solve for x and y in terms of a and b to get $x = -5a + 2b$, $y = 3a - b$. Thus,

$$(a, b) = (-5a + 2b)u_1 + (3a - b)u_2, \quad \text{and so} \quad [v] = [-5a + 2b, 3a - b]^T$$

(a) Using the formula for (a, b) and $G(x, y) = (2x - 7y, 4x + 3y)$, we have

$$\begin{aligned} G(u_1) &= G(1, 3) = (-19, 13) = 121u_1 - 70u_2 \\ G(u_2) &= G(2, 5) = (-31, 23) = 201u_1 - 116u_2 \end{aligned} \quad \text{and so} \quad [G]_S = \begin{bmatrix} 121 & 201 \\ -70 & -116 \end{bmatrix}$$

(We emphasize that the coefficients of u_1 and u_2 are written as columns, not rows, in the matrix representation.)

(b) Use the formula $(a, b) = (-5a + 2b)u_1 + (3a - b)u_2$ to get

$$\begin{aligned} v &= (4, -3) = -26u_1 + 15u_2 \\ G(v) &= G(4, -3) = (20, 7) = -131u_1 + 80u_2 \end{aligned}$$

Then

$$[v]_S = [-26, 15]^T \quad \text{and} \quad [G(v)]_S = [-131, 80]^T$$

Accordingly,

$$[G]_S[v]_S = \begin{bmatrix} 121 & 201 \\ -70 & -116 \end{bmatrix} \begin{bmatrix} -26 \\ 15 \end{bmatrix} = \begin{bmatrix} -131 \\ 80 \end{bmatrix} = [G(v)]_S$$

(This is expected from [Theorem 6.1.](#))

6.3. Consider the following 2×2 matrix A and basis S of $\mathbf{R}^2$:

$$A = \begin{bmatrix} 2 & 4 \\ 5 & 6 \end{bmatrix} \quad \text{and} \quad S = \{u_1, u_2\} = \left\{ \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \begin{bmatrix} 3 \\ -7 \end{bmatrix} \right\}$$

The matrix A defines a linear operator on $\mathbf{R}^2$. Find the matrix B that represents the mapping A relative to the basis S .

First find the coordinates of an arbitrary vector $(a, b)^T$ with respect to the basis S . We have

$$\begin{bmatrix} a \\ b \end{bmatrix} = x \begin{bmatrix} 1 \\ -2 \end{bmatrix} + y \begin{bmatrix} 3 \\ -7 \end{bmatrix} \quad \text{or} \quad \begin{cases} x + 3y = a \\ -2x - 7y = b \end{cases}$$

Solve for x and y in terms of a and b to obtain $x = 7a + 3b$, $y = -2a - b$. Thus,

$$(a, b)^T = (7a + 3b)u_1 + (-2a - b)u_2$$

Then use the formula for $(a, b)^T$ to find the coordinates of Au_1 and Au_2 relative to the basis S :

$$Au_1 = \begin{bmatrix} 2 & 4 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \end{bmatrix} = \begin{bmatrix} -6 \\ -7 \end{bmatrix} = -63u_1 + 19u_2$$

$$Au_2 = \begin{bmatrix} 2 & 4 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} 3 \\ -7 \end{bmatrix} = \begin{bmatrix} -22 \\ -27 \end{bmatrix} = -235u_1 + 71u_2$$

Writing the coordinates as columns yields

$$B = \begin{bmatrix} -63 & -235 \\ 19 & 71 \end{bmatrix}$$

6.4. Find the matrix representation of each of the following linear operators F on $\mathbf{R}^3$ relative to the usual basis $E = \{e_1, e_2, e_3\}$ of $\mathbf{R}^3$; that is, find $[F] = [F]_E$:

(a) F defined by $F(x, y, z) = (x + 2y - 3z, 4x - 5y + 6z, 7x + 8y + 9z)$.

(b) F defined by the 3×3 matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 4 \\ 5 & 5 & 5 \end{bmatrix}$

(c) F defined by $F(e_1) = (1, 3, 5)$, $F(e_2) = (2, 4, 6)$, $F(e_3) = (7, 7, 7)$. ([Theorem 5.2](#) states that a linear map is completely defined by its action on the vectors in a basis.)

(a) Because E is the usual basis, simply write the coefficients of the components of $F(x, y, z)$ as rows:

$$[F] = \begin{bmatrix} 1 & 2 & -3 \\ 4 & -5 & -6 \\ 7 & 8 & 9 \end{bmatrix}$$

(b) Because E is the usual basis, $[F] = A$, the matrix A itself.

(c) Here

$$\begin{aligned} F(e_1) &= (1, 3, 5) = e_1 + 3e_2 + 5e_3 \\ F(e_2) &= (2, 4, 6) = 2e_1 + 4e_2 + 6e_3 \\ F(e_3) &= (7, 7, 7) = 7e_1 + 7e_2 + 7e_3 \end{aligned} \quad \text{and so} \quad [F] = \begin{bmatrix} 1 & 2 & 7 \\ 3 & 4 & 7 \\ 5 & 6 & 7 \end{bmatrix}$$

That is, the columns of $[F]$ are the images of the usual basis vectors.

6.5. Let G be the linear operator on $\mathbf{R}^3$ defined by $G(x, y, z) = (2y + z, x + 4y, 3x)$.

(a) Find the matrix representation of G relative to the basis

$$S = \{w_1, w_2, w_3\} = \{(1, 1, 1), (1, 1, 0), (1, 0, 0)\}$$

(b) Verify that $[G]v = G(v)$ for any vector v in $\mathbf{R}^3$.

First find the coordinates of an arbitrary vector $(a, b, c) \in \mathbf{R}^3$ with respect to the basis S . Write (a, b, c) as a linear combination of w_1, w_2, w_3 using unknown scalars x, y , and z :

$$(a, b, c) = x(1, 1, 1) + y(1, 1, 0) + z(1, 0, 0) = (x + y + z, x + y, x)$$

Set corresponding components equal to each other to obtain the system of equations

$$x + y + z = a, \quad x + y = b, \quad x = c$$

Solve the system for x, y, z in terms of a, b, c to find $x = c, y = b - c, z = a - b$. Thus,

$$(a, b, c) = cw_1 + (b - c)w_2 + (a - b)w_3, \quad \text{or equivalently,} \quad [(a, b, c)] = [c, b - c, a - b]^T$$

(a) Because $G(x, y, z) = (2y + z, x + 4y, 3x)$,

$$G(w_1) = G(1, 1, 1) = (3, -3, 3) = 3w_1 - 6w_2 + 6w_3$$

$$G(w_2) = G(1, 1, 0) = (2, -3, 3) = 3w_1 - 6w_2 + 5w_3$$

$$G(w_3) = G(1, 0, 0) = (0, 1, 3) = 3w_1 - 2w_2 - w_3$$

Write the coordinates $G(w_1)$, $G(w_2)$, $G(w_3)$ as columns to get

$$[G] = \begin{bmatrix} 3 & 3 & 3 \\ -6 & -6 & -2 \\ 6 & 5 & -1 \end{bmatrix}$$

(b) Write $G(v)$ as a linear combination of w_1, w_2, w_3 , where $v = (a, b, c)$ is an arbitrary vector in $\mathbf{R}^3$,

$$G(v) = G(a, b, c) = (2b + c, a - 4b, 3a) = 3aw_1 + (-2a - 4b)w_2 + (-a + 6b + c)w_3$$

or equivalently,

$$[G(v)] = [3a, -2a - 4b, -a + 6b + c]^T$$

Accordingly,

$$[G][v] = \begin{bmatrix} 3 & 3 & 3 \\ -6 & -6 & -2 \\ 6 & 5 & -1 \end{bmatrix} \begin{bmatrix} c \\ b - c \\ a - b \end{bmatrix} = \begin{bmatrix} 3a \\ -2a - 4b \\ -a + 6b + c \end{bmatrix} = [G(v)]$$

6.6. Consider the following 3×3 matrix A and basis S of $\mathbf{R}^3$:

$$A = \begin{bmatrix} 1 & -2 & 1 \\ 3 & -1 & 0 \\ 1 & 4 & -2 \end{bmatrix} \quad \text{and} \quad S = \{u_1, u_2, u_3\} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \right\}$$

The matrix A defines a linear operator on $\mathbf{R}^3$. Find the matrix B that represents the mapping A relative to

the basis S . (Recall that A represents itself relative to the usual basis of $\mathbf{R}^3$.)

First find the coordinates of an arbitrary vector (a, b, c) in $\mathbf{R}^3$ with respect to the basis S . We have

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = x \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + y \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + z \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad \text{or} \quad \begin{array}{rcl} x + z & = & a \\ x + y + 2z & = & b \\ x + y + 3z & = & c \end{array}$$

Solve for x, y, z in terms of a, b, c to get

$$x = a + b - c, \quad y = -a + 2b - c, \quad z = c - b$$

thus,

$$(a, b, c)^T = (a + b - c)u_1 + (-a + 2b - c)u_2 + (c - b)u_3$$

Then use the formula for $(a, b, c)^T$ to find the coordinates of Au_1, Au_2, Au_3 relative to the basis S :

$$\begin{aligned} A(u_1) &= A(1, 1, 1)^T = (0, 2, 3)^T = -u_1 + u_2 + u_3 \\ A(u_2) &= A(1, 1, 0)^T = (-1, -1, 2)^T = -4u_1 - 3u_2 + 3u_3 \\ A(u_3) &= A(1, 2, 3)^T = (0, 1, 3)^T = -2u_1 - u_2 + 2u_3 \end{aligned} \quad \text{so} \quad B = \begin{bmatrix} -1 & -4 & -2 \\ 1 & -3 & -1 \\ 1 & 3 & 2 \end{bmatrix}$$

6.7. For each of the following linear transformations (operators) L on $\mathbf{R}^2$, find the matrix A that represents L (relative to the usual basis of $\mathbf{R}^2$):

(a) L is defined by $L(1, 0) = (2, 4)$ and $L(0, 1) = (5, 8)$.

(b) L is the rotation in $\mathbf{R}^2$ counterclockwise by 90° .

(c) L is the reflection in $\mathbf{R}^2$ about the line $y = x$.

(a) Because $\{(1, 0), (0, 1)\}$ is the usual basis of $\mathbf{R}^2$, write their images under L as columns to get

$$A = \begin{bmatrix} 2 & 5 \\ 4 & 8 \end{bmatrix}$$

(b) Under the rotation L , we have $L(1, 0) = (0, 1)$ and $L(0, 1) = (1, 0)$. Thus,

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

(c) Under the reflection L , we have $L(1, 0) = (0, -1)$ and $L(0, 1) = (-1, 0)$. Thus,

$$A = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$$

6.8. The set $S = \{e^{3t}, te^{3t}, t^2 e^{3t}\}$ is a basis of a vector space V of functions $f: \mathbf{R} \rightarrow \mathbf{R}$. Let $\mathbf{D}$ be the differential operator on V ; that is, $\mathbf{D}(f) = df/dt$. Find the matrix representation of $\mathbf{D}$ relative to the basis S .

Find the image of each basis function:

$$\begin{aligned} \mathbf{D}(e^{3t}) &= 3e^{3t} = 3(e^{3t}) + 0(te^{3t}) + 0(t^2 e^{3t}) \\ \mathbf{D}(te^{3t}) &= e^{3t} + 3te^{3t} = 1(e^{3t}) + 3(te^{3t}) + 0(t^2 e^{3t}) \\ \mathbf{D}(t^2 e^{3t}) &= 2te^{3t} + 3t^2 e^{3t} = 0(e^{3t}) + 2(te^{3t}) + 3(t^2 e^{3t}) \end{aligned} \quad \text{and thus,} \quad [\mathbf{D}] = \begin{bmatrix} 3 & 1 & 0 \\ 0 & 3 & 2 \\ 0 & 0 & 3 \end{bmatrix}$$

6.9. Prove [Theorem 6.1](#): Let $T: V \rightarrow V$ be a linear operator, and let S be a (finite) basis of V . Then, for any vector v in V , $[T]_S[v]_S = [T(v)]_S$.

Suppose $S = \{u_1, u_2, \dots, u_n\}$, and suppose, for $i = 1, \dots, n$,

$$T(u_i) = a_{i1}u_1 + a_{i2}u_2 + \cdots + a_{in}u_n = \sum_{j=1}^n a_{ij}u_j$$

Then $[T]_S$ is the n -square matrix whose j th row is

$$(a_{1j}, a_{2j}, \dots, a_{nj}) \quad (1)$$

Now suppose

$$v = k_1 u_1 + k_2 u_2 + \dots + k_n u_n = \sum_{i=1}^n k_i u_i$$

Writing a column vector as the transpose of a row vector, we have

$$[v]_S = [k_1, k_2, \dots, k_n]^T \quad (2)$$

Furthermore, using the linearity of T ,

$$\begin{aligned} T(v) &= T\left(\sum_{i=1}^n k_i u_i\right) = \sum_{i=1}^n k_i T(u_i) = \sum_{i=1}^n k_i \left(\sum_{j=1}^n a_{ij} u_j\right) \\ &= \sum_{j=1}^n \left(\sum_{i=1}^n a_{ij} k_i\right) u_j = \sum_{j=1}^n (a_{1j} k_1 + a_{2j} k_2 + \dots + a_{nj} k_n) u_j \end{aligned}$$

Thus, $[T(v)]_S$ is the column vector whose j th entry is

$$a_{1j} k_1 + a_{2j} k_2 + \dots + a_{nj} k_n \quad (3)$$

On the other hand, the j th entry of $[T]_S[v]_S$ is obtained by multiplying the j th row of $[T]_S$ by $[v]_S$ —that is (1) by (2). But the product of (1) and (2) is (3). Hence, $[T]_S[v]_S$ and $[T(v)]_S$ have the same entries. Thus, $[T]_S[v]_S = [T(v)]_S$.

6.10. Prove [Theorem 6.2](#): Let $S = \{u_1, u_2, \dots, u_n\}$ be a basis for V over K , and let $\mathbf{M}$ be the algebra of n -square matrices over K . Then the mapping $m: A(V) \rightarrow \mathbf{M}$ defined by $m(T) = [T]_S$ is a vector space isomorphism. That is, for any $F, G \in A(V)$ and any $k \in K$, we have

(i) $[F + G] = [F] + [G]$,

- (ii) $[kF = k[F]$,
 (iii) m is one-to-one and onto.
 (i) Suppose, for $i = 1, \dots, n$,

$$F(u_i) = \sum_{j=1}^n a_{ij}u_j \quad \text{and} \quad G(u_i) = \sum_{j=1}^n b_{ij}u_j$$

Consider the matrices $A = [a_{ij}]$ and $B = [b_{ij}]$. Then $[F] = A^T$ and $[G] = B^T$. We have, for $i = 1, \dots, n$,

$$(F + G)(u_i) = F(u_i) + G(u_i) = \sum_{j=1}^n (a_{ij} + b_{ij})u_j$$

Because $A + B$ is the matrix $(a_{ij} + b_{ij})$, we have

$$[F + G] = (A + B)^T = A^T + B^T = [F] + [G]$$

- (ii) Also, for $i = 1, \dots, n$,

$$(kF)(u_i) = kF(u_i) = k \sum_{j=1}^n a_{ij}u_j = \sum_{j=1}^n (ka_{ij})u_j$$

Because kA is the matrix (ka_{ij}) , we have

$$[kF] = (kA)^T = kA^T = k[F]$$

- (iii) Finally, m is one-to-one, because a linear mapping is completely determined by its values on a basis. Also, m is onto, because matrix $A = [a_{ij}]$ in $\mathbf{M}$ is the image of the linear operator,

$$F(u_i) = \sum_{j=1}^n a_{ij}u_j, \quad i = 1, \dots, n$$

Thus, the theorem is proved.

6.11. Prove [Theorem 6.3](#): For any linear operators $G, F \in A(V)$, $[G \circ F] = [G][F]$.

Using the notation in [Problem 6.10](#), we have

$$\begin{aligned}(G \circ F)(u_i) &= G(F(u_i)) = G\left(\sum_{j=1}^n a_{ij}u_j\right) = \sum_{j=1}^n a_{ij}G(u_j) \\ &= \sum_{j=1}^n a_{ij}\left(\sum_{k=1}^n b_{jk}u_k\right) = \sum_{k=1}^n \left(\sum_{j=1}^n a_{ij}b_{jk}\right)u_k\end{aligned}$$

Recall that AB is the matrix $AB = [c_{ik}]$, where $c_{ik} = \sum_{j=1}^n a_{ij}b_{jk}$. Accordingly,

$$[G \circ F] = (AB)^T = B^T A^T = [G][F]$$

The theorem is proved.

6.12. Let A be the matrix representation of a linear operator T . Prove that, for any polynomial $f(t)$, we have that $f(A)$ is the matrix representation of $f(T)$. [Thus, $f(T) = 0$ if and only if $f(A) = 0$.]

Let ϕ be the mapping that sends an operator T into its matrix representation A . We need to prove that $\phi(f(T)) = f(A)$. Suppose $f(t) = a_n t^n + \dots + a_1 t + a_0$. The proof is by induction on n , the degree of $f(t)$. Suppose $n = 0$. Recall that $\phi(I) = I$, where I is the identity mapping and I is the identity matrix. Thus,

$$\phi(f(T)) = \phi(a_0 I) = a_0 \phi(I) = a_0 I = f(A)$$

and so the theorem holds for $n = 0$.

Now assume the theorem holds for polynomials of degree less than n . Then, because ϕ is an algebra isomorphism,

$$\begin{aligned}
\phi(f(T)) &= \phi(a_n T^n + a_{n-1} T^{n-1} + \cdots + a_1 T + a_0 I') \\
&= a_n \phi(T) \phi(T^{n-1}) + \phi(a_{n-1} T^{n-1} + \cdots + a_1 T + a_0 I') \\
&= a_n A A^{n-1} + (a_{n-1} A^{n-1} + \cdots + a_1 A + a_0 I) = f(A)
\end{aligned}$$

and the theorem is proved.

Change of Basis

The coordinate vector $[v]_S$ in this section will always denote a column vector; that is,

$$[v]_S = [a_1, a_2, \dots, a_n]^T$$

6.13. Consider the following bases of $\mathbf{R}^2$:

$$E = \{e_1, e_2\} = \{(1, 0), (0, 1)\} \quad \text{and} \quad S = \{u_1, u_2\} = \{(1, 3), (1, 4)\}$$

- (a) Find the change-of-basis matrix P from the usual basis E to S .
- (b) Find the change-of-basis matrix Q from S back to E .
- (c) Find the coordinate vector $[v]_S$ of $v = (5, 3)$ relative to S .
- (a) Because E is the usual basis, simply write the basis vectors in S as columns: $P = \begin{bmatrix} 1 & 1 \\ 3 & 4 \end{bmatrix}$
- (b) **Method 1.** Use the definition of the change-of-basis matrix. That is, express each vector in E as a linear combination of the vectors in S . We do this by first finding the coordinates of an arbitrary vector $v = (a, b)$ relative to S . We have

$$(a, b) = x(1, 3) + y(1, 4) = (x + y, 3x + 4y) \quad \text{or} \quad \begin{cases} x + y = a \\ 3x + 4y = b \end{cases}$$

Solve for x and y to obtain $x = 4a - b$, $y = -3a + b$.
Thus,

$$v = (4a - b)u_1 + (-3a + b)u_2 \quad \text{and} \quad [v]_S = [(a, b)]_S = [4a - b, -3a + b]^T$$

Using the above formula for $[v]_S$ and writing the coordinates of the e_i as columns yields

$$\begin{aligned} e_1 = (1, 0) &= 4u_1 - 3u_2 \\ e_2 = (0, 1) &= -u_1 + u_2 \end{aligned} \quad \text{and} \quad Q = \begin{bmatrix} 4 & -1 \\ -3 & 1 \end{bmatrix}$$

Method 2. Because $Q = P^{-1}$, find P^{-1} , say by using the formula for the inverse of a 2×2 matrix.

Thus,

$$P^{-1} = \begin{bmatrix} 4 & -1 \\ -3 & 1 \end{bmatrix}$$

(c) **Method 1.** Write v as a linear combination of the vectors in S , say by using the above formula for $v = (a, b)$. We have $v = (5, 3) = 23u_1 - 18u_2$, and so $[v]_S = [23, -18]^T$.

Method 2. Use, from [Theorem 6.6](#), the fact that $[v]_S = P^{-1}[v]_E$ and the fact that $[v]_E = [5, -3]^T$:

$$[v]_S = P^{-1}[v]_E = \begin{bmatrix} 4 & -1 \\ -3 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ -3 \end{bmatrix} = \begin{bmatrix} 23 \\ -18 \end{bmatrix}$$

6.14. The vectors $u_1 = (1, 2, 0)$, $u_2 = (1, 3, 2)$, $u_3 = (0, 1, 3)$ form a basis S of $\mathbf{R}^3$. Find

(a) The change-of-basis matrix P from the usual basis $E = \{e_1, e_2, e_3\}$ to S .

(b) The change-of-basis matrix Q from S back to E .

(a) Because E is the usual basis, simply write the basis

$$\text{vectors of } S \text{ as columns: } P = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 3 & 1 \\ 0 & 2 & 3 \end{bmatrix}$$

(b) **Method 1.** Express each basis vector of E as a linear combination of the basis vectors of S by first finding the coordinates of an arbitrary vector $v = (a, b, c)$ relative to the basis S . We have

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = x \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + y \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} + z \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix} \quad \text{or} \quad \begin{array}{rcl} x + y & = & a \\ 2x + 3y + z & = & b \\ 2y + 3z & = & c \end{array}$$

Solve for x, y, z to get $x = 7a - 3b + c$, $y = 6a + 3b - c$, $z = 4a - 2b + c$. Thus,

$$v = (a, b, c) = (7a - 3b + c)u_1 + (-6a + 3b - c)u_2 + (4a - 2b + c)u_3$$

or

$$[v]_S = [(a, b, c)]_S = [7a - 3b + c, -6a + 3b - c, 4a - 2b + c]^T$$

Using the above formula for $[v]_S$ and then writing the coordinates of the e_i as columns yields

$$\begin{array}{lcl} e_1 = (1, 0, 0) = 7u_1 - 6u_2 + 4u_3 \\ e_2 = (0, 1, 0) = -3u_1 + 3u_2 - 2u_3 \\ e_3 = (0, 0, 1) = u_1 - u_2 + u_3 \end{array} \quad \text{and} \quad Q = \begin{bmatrix} 7 & -3 & 1 \\ -6 & 3 & -1 \\ 4 & -2 & 1 \end{bmatrix}$$

Method 2. Find P^{-1} by row reducing $M = [P, I]$ to the form $[I, P^{-1}]$:

$$\begin{aligned}
 M &= \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 2 & 3 & 1 & 0 & 1 & 0 \\ 0 & 2 & 3 & 0 & 0 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & -2 & 1 & 0 \\ 0 & 2 & 3 & 0 & 0 & 1 \end{array} \right] \\
 &\sim \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & -2 & 1 & 0 \\ 0 & 0 & 1 & 4 & -2 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 7 & -3 & 1 \\ 0 & 1 & 0 & -6 & 3 & -1 \\ 0 & 0 & 1 & 4 & -2 & 1 \end{array} \right] = [I, P^{-1}]
 \end{aligned}$$

$$\text{Thus, } Q = P^{-1} = \begin{bmatrix} 7 & -3 & 1 \\ -6 & 3 & -1 \\ 4 & -2 & 1 \end{bmatrix}$$

6.15. Suppose the x -axis and y -axis in the plane $\mathbf{R}^2$ are rotated counterclockwise 45° so that the new x^0 -axis and y^0 -axis are along the line $y = x$ and the line $y = -x$, respectively.

- (a) Find the change-of-basis matrix P .
- (b) Find the coordinates of the point $A(5, 6)$ under the given rotation.
- (a) The unit vectors in the direction of the new x^0 - and y^0 -axes are

$$u_1 = \left(\frac{1}{2}\sqrt{2}, \frac{1}{2}\sqrt{2}\right) \quad \text{and} \quad u_2 = \left(-\frac{1}{2}\sqrt{2}, \frac{1}{2}\sqrt{2}\right)$$

(The unit vectors in the direction of the original x and y axes are the usual basis of $\mathbf{R}^2$.) Thus, write the coordinates of u_1 and u_2 as columns to obtain

$$P = \begin{bmatrix} \frac{1}{2}\sqrt{2} & -\frac{1}{2}\sqrt{2} \\ \frac{1}{2}\sqrt{2} & \frac{1}{2}\sqrt{2} \end{bmatrix}$$

- (b) Multiply the coordinates of the point by P^{-1} :

$$\begin{bmatrix} \frac{1}{2}\sqrt{2} & \frac{1}{2}\sqrt{2} \\ -\frac{1}{2}\sqrt{2} & \frac{1}{2}\sqrt{2} \end{bmatrix} \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} \frac{11}{2}\sqrt{2} \\ \frac{1}{2}\sqrt{2} \end{bmatrix}$$

(Because P is orthogonal, P^{-1} is simply the transpose of P .)

6.16. The vectors $u_1 = (1, 1, 0)$, $u_2 = (0, 1, 1)$, $u_3 = (1, 2, 2)$ form a basis S of $\mathbf{R}^3$. Find the coordinates of an arbitrary vector $v = (a, b, c)$ relative to the basis S .

Method 1. Express v as a linear combination of u_1 , u_2 , u_3 using unknowns x , y , z . We have

$$(a, b, c) = x(1, 1, 0) + y(0, 1, 1) + z(1, 2, 2) = (x + z, x + y + 2z, y + 2z)$$

this yields the system

$$\begin{array}{lcl} \begin{array}{l} x + z = a \\ x + y + 2z = b \\ y + 2z = c \end{array} & \text{or} & \begin{array}{l} x + z = a \\ y + z = -a + b \\ y + 2z = c \end{array} & \text{or} & \begin{array}{l} x + z = a \\ y + z = -a + b \\ z = a - b + c \end{array} \end{array}$$

Solving by back-substitution yields $x = b - c$, $y = 2a - 2b + c$, $z = a - b + c$. Thus,

$$[v]_S = [b - c, 2a - 2b + c, a - b + c]^T$$

Method 2. Find P^{-1} by row reducing $M = [P, I]$ to the form $[I, P^{-1}]$, where P is the change-of-basis matrix from the usual basis E to S or, in other words, the matrix whose columns are the basis vectors of S .

We have

$$\begin{aligned}
 M &= \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 2 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array} \right] \\
 &\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1 & -1 \\ 0 & 1 & 0 & -2 & 2 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{array} \right] = [I, P^{-1}]
 \end{aligned}$$

Thus,

$$P^{-1} = \begin{bmatrix} 0 & 1 & -1 \\ -2 & 2 & -1 \\ 1 & -1 & 1 \end{bmatrix} \quad \text{and} \quad [v]_S = P^{-1}[v]_E = \begin{bmatrix} 0 & 1 & -1 \\ -2 & 2 & -1 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} b - c \\ -2a + 2b - c \\ a - b + c \end{bmatrix}$$

6.17. Consider the following bases of $\mathbf{R}^2$:

$$S = \{u_1, u_2\} = \{(1, -2), (3, -4)\} \quad \text{and} \quad S' = \{v_1, v_2\} = \{(1, 3), (3, 8)\}$$

- Find the coordinates of $u = (a, b)$ relative to the basis S .
- Find the change-of-basis matrix P from S to S' .
- Find the coordinates of $u = (a, b)$ relative to the basis S' .
- Find the change-of-basis matrix Q from S' back to S .
- Verify $Q = P^{-1}$.
- Show that, for any vector $u = (a, b)$ in $\mathbf{R}^2$, $P^{-1}[u]_S = [u]_{S'}$. (See [Theorem 6.6](#).)

(a) Let $u = xu_1 + yu_2$ for unknowns x and y ; that is,

$$\begin{bmatrix} a \\ b \end{bmatrix} = x \begin{bmatrix} 1 \\ -2 \end{bmatrix} + y \begin{bmatrix} 3 \\ -4 \end{bmatrix} \quad \text{or} \quad \begin{array}{l} x + 3y = a \\ -2x - 4y = b \end{array} \quad \text{or} \quad \begin{array}{l} x + 3y = a \\ 2y = 2a + b \end{array}$$

Solve for x and y in terms of a and b to get $x = 2a - 3b$ and $y = a + \frac{1}{2}b$. Thus,

$$(a, b) = (-2a - \frac{3}{2}b)u_1 + (a + \frac{1}{2}b)u_2 \quad \text{or} \quad [(a, b)]_S = [-2a - \frac{3}{2}b, \quad a + \frac{1}{2}b]^T$$

(b) Use part (a) to write each of the basis vectors v_1 and v_2 of S as a linear combination of the basis vectors u_1 and u_2 of S ; that is,

$$v_1 = (1, 3) = (-2 - \frac{9}{2})u_1 + (1 + \frac{3}{2})u_2 = -\frac{13}{2}u_1 + \frac{5}{2}u_2$$

$$v_2 = (3, 8) = (-6 - 12)u_1 + (3 + 4)u_2 = -18u_1 + 7u_2$$

Then P is the matrix whose columns are the coordinates of v_1 and v_2 relative to the basis S ; that is,

$$P = \begin{bmatrix} -\frac{13}{2} & -18 \\ \frac{5}{2} & 7 \end{bmatrix}$$

(c) Let $v = xv_1 + yv_2$ for unknown scalars x and y :

$$\begin{bmatrix} a \\ b \end{bmatrix} = x \begin{bmatrix} 1 \\ 3 \end{bmatrix} + y \begin{bmatrix} 3 \\ 8 \end{bmatrix} \quad \text{or} \quad \begin{array}{l} x + 3y = a \\ 3x + 8y = b \end{array} \quad \text{or} \quad \begin{array}{l} x + 3y = a \\ -y = b - 3a \end{array}$$

Solve for x and y to get $x = 8a + 3b$ and $y = 3a - b$. Thus,

$$(a, b) = (-8a + 3b)v_1 + (3a - b)v_2 \quad \text{or} \quad [(a, b)]_{S'} = [-8a + 3b, \quad 3a - b]^T$$

(d) Use part (c) to express each of the basis vectors u_1 and u_2 of S as a linear combination of the basis vectors v_1 and v_2 of S' :

$$u_1 = (1, -2) = (-8 - 6)v_1 + (3 + 2)v_2 = -14v_1 + 5v_2$$

$$u_2 = (3, -4) = (-24 - 12)v_1 + (9 + 4)v_2 = -36v_1 + 13v_2$$

Write the coordinates of u_1 and u_2 relative to S as

columns to obtain $Q = \begin{bmatrix} -14 & -36 \\ 5 & 13 \end{bmatrix}$.

$$(e) \quad QP = \begin{bmatrix} -14 & -36 \\ 5 & 13 \end{bmatrix} \begin{bmatrix} -\frac{13}{2} & -18 \\ \frac{5}{2} & 7 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

(f) Use parts (a), (c), and (d) to obtain

$$P^{-1}[v]_S = Q[v]_S = \begin{bmatrix} -14 & -36 \\ 5 & 13 \end{bmatrix} \begin{bmatrix} -2a - \frac{3}{2}b \\ a + \frac{1}{2}b \end{bmatrix} = \begin{bmatrix} -8a + 3b \\ 3a - b \end{bmatrix} = [v]_S$$

6.18. Suppose P is the change-of-basis matrix from a basis $\{u_i\}$ to a basis $\{w_i\}$, and suppose Q is the change-of-basis matrix from the basis $\{w_i\}$ back to $\{u_i\}$. Prove that P is invertible and that $Q = P^{-1}$.

Suppose, for $i = 1, 2, \dots, n$, that

$$w_i = a_{i1}u_1 + a_{i2}u_2 + \dots + a_{in}u_n = \sum_{j=1}^n a_{ij}u_j \quad (1)$$

and, for $j = 1, 2, \dots, n$,

$$u_j = b_{j1}w_1 + b_{j2}w_2 + \dots + b_{jn}w_n = \sum_{k=1}^n b_{jk}w_k \quad (2)$$

Let $A = [a_{ij}]$ and $B = [b_{jk}]$. Then $P = A^T$ and $Q = B^T$.

Substituting (2) into (1) yields

$$w_i = \sum_{j=1}^n a_{ij} \left(\sum_{k=1}^n b_{jk}w_k \right) = \sum_{k=1}^n \left(\sum_{j=1}^n a_{ij}b_{jk} \right) w_k$$

Because $\{w_i\}$ is a basis, $\sum a_{ij} b_{jk} = \delta_{ik}$, where δ_{ik} is the Kronecker delta; that is, $\delta_{ik} = 1$ if $i = k$ but $\delta_{ik} = 0$ if $i \neq k$. Suppose $AB = [c_{ik}]$. Then $c_{ik} = \delta_{ik}$. Accordingly, $AB = I$, and so $QP = B^T A^T = (AB)^T = I^T = I$.

$$QP = B^T A^T = (AB)^T = I^T = I$$

Thus, $Q = P^{-1}$

6.19. Consider a finite sequence of vectors $S = \{u_1, u_2, \dots, u_n\}$. Let S' be the sequence of vectors obtained from S by one of the following “elementary operations”:

- (1) Interchange two vectors.
- (2) Multiply a vector by a nonzero scalar.
- (3) Add a multiple of one vector to another vector.

Show that S and S' span the same subspace W . Also, show that S' is linearly independent if and only if S is linearly independent.

Observe that, for each operation, the vectors S' are linear combinations of vectors in S . Also, because each operation has an inverse of the same type, each vector in S is a linear combination of vectors in S' . Thus, S and S' span the same subspace W . Moreover, S' is linearly independent if and only if $\dim W = n$, and this is true if and only if S is linearly independent.

6.20. Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be row equivalent $m \times n$ matrices over a field K , and let $v_1, v_2, \dots, v_n$ be any vectors in a vector space V over K . For $i = 1, 2, \dots, m$, let u_i and w_i be defined by

$$u_i = a_{i1}v_1 + a_{i2}v_2 + \cdots + a_{in}v_n \quad \text{and} \quad w_i = b_{i1}v_1 + b_{i2}v_2 + \cdots + b_{in}v_n$$

Show that $\{u_i\}$ and $\{w_i\}$ span the same subspace of V .

Applying an “elementary operation” of [Problem 6.19](#) to $\{u_i\}$ is equivalent to applying an elementary row operation to the matrix A . Because A and B are row equivalent, B can be obtained from A by a sequence of elementary row operations. Hence, $\{w_i\}$ can be obtained from $\{u_i\}$ by the corresponding sequence of operations. Accordingly, $\{u_i\}$ and $\{w_i\}$ span the same space.

6.21. Suppose $u_1, u_2, \dots, u_n$ belong to a vector space V over a field K , and suppose $P = [a_{ij}]$ is an n -square matrix over K . For $i = 1, 2, \dots, n$, let $v_i = a_{i1} u_1 + a_{i2} u_2 + \dots + a_{in} u_n$.

- (a) Suppose P is invertible. Show that $\{u_i\}$ and $\{v_i\}$ span the same subspace of V . Hence, $\{u_i\}$ is linearly independent if and only if $\{v_i\}$ is linearly independent.
 - (b) Suppose P is singular (not invertible). Show that $\{v_i\}$ is linearly dependent.
 - (c) Suppose $\{v_i\}$ is linearly independent. Show that P is invertible.
- (a) Because P is invertible, it is row equivalent to the identity matrix I . Hence, by [Problem 6.19](#), $\{v_i\}$ and $\{u_i\}$ span the same subspace of V . Thus, one is linearly independent if and only if the other is linearly independent.
 - (b) Because P is not invertible, it is row equivalent to a matrix with a zero row. This means $\{v_i\}$ spans a subspace that has a spanning set with less than n elements. Thus, $\{v_i\}$ is linearly dependent.

(c) This is the contrapositive of the statement of part (b), and so it follows from part (b).

6.22. Prove [Theorem 6.6](#): Let P be the change-of-basis matrix from a basis S to a basis S' in a vector space V . Then, for any vector $v \in V$, we have $P[v]_S = [v]_{S'}$, and hence, $P^{-1}[v]_{S'} = [v]_S$.

Suppose $S = \{u_1, \dots, u_n\}$ and $S' = \{w_1, \dots, w_n\}$, and suppose, for $i = 1, \dots, n$,

$$w_i = a_{i1}u_1 + a_{i2}u_2 + \cdots + a_{in}u_n = \sum_{j=1}^n a_{ij}u_j$$

Then P is the n -square matrix whose j th row is

$$(a_{1j}, a_{2j}, \dots, a_{nj}) \quad (1)$$

Also suppose $v = k_1w_1 + k_2w_2 + \cdots + k_nw_n = \sum_{i=1}^n k_iw_i$. Then

$$[v]_{S'} = [k_1, k_2, \dots, k_n]^T \quad (2)$$

Substituting for w_i in the equation for v , we obtain

$$\begin{aligned} v &= \sum_{i=1}^n k_iw_i = \sum_{i=1}^n k_i \left(\sum_{j=1}^n a_{ij}u_j \right) = \sum_{j=1}^n \left(\sum_{i=1}^n a_{ij}k_i \right) u_j \\ &= \sum_{j=1}^n (a_{1j}k_1 + a_{2j}k_2 + \cdots + a_{nj}k_n) u_j \end{aligned}$$

Accordingly, $[v]_S$ is the column vector whose j th entry is

$$a_{1j}k_1 + a_{2j}k_2 + \cdots + a_{nj}k_n \quad (3)$$

On the other hand, the j th entry of $P[v]_{S'}$ is obtained by multiplying the j th row of P by $[v]_{S'}$ —that is, (1) by

(2). However, the product of (1) and (2) is (3).
Hence, $P[v]_S$ and $[v]_S$ have the same entries. Thus,
 $P[v]_S = [v]_S$, as claimed.

Furthermore, multiplying the above by P^{-1} gives $\hat{v}$.

Linear Operators and Change of Basis

6.23. Consider the linear transformation F on $\mathbf{R}^2$ defined by
 $F(x, y) = (5x - y, 2x + y)$ and the following bases of $\mathbf{R}^2$:

$$E = \{e_1, e_2\} = \{(1, 0), (0, 1)\} \quad \text{and} \quad S = \{u_1, u_2\} = \{(1, 4), (2, 7)\}$$

- (a) Find the change-of-basis matrix P from E to S and the change-of-basis matrix Q from S back to E .
- (b) Find the matrix A that represents F in the basis E .
- (c) Find the matrix B that represents F in the basis S .
- (a) Because E is the usual basis, simply write the vectors in S as columns to obtain the change-of-basis matrix P . Recall, also, that $Q = P^{-1}$. Thus,

$$P = \begin{bmatrix} 1 & 2 \\ 4 & 7 \end{bmatrix} \quad \text{and} \quad Q = P^{-1} = \begin{bmatrix} -7 & 2 \\ 4 & -1 \end{bmatrix}$$

- (b) Write the coefficients of x and y in $F(x, y) = (5x - y, 2x + y)$ as rows to get

$$A = \begin{bmatrix} 5 & -1 \\ 2 & 1 \end{bmatrix}$$

- (c) **Method 1.** Find the coordinates of $F(u_1)$ and $F(u_2)$ relative to the basis S . This may be done by first finding the coordinates of an arbitrary vector (a, b) in $\mathbf{R}^2$ relative to the basis S . We have

$$(a, b) = x(1, 4) + y(2, 7) = (x + 2y, 4x + 7y), \quad \text{and so} \quad \begin{array}{l} x + 2y = a \\ 4x + 7y = b \end{array}$$

Solve for x and y in terms of a and b to get $x = -7a + 2b$, $y = 4a - b$. Then

$$(a, b) = (-7a + 2b)u_1 + (4a - b)u_2$$

Now use the formula for (a, b) to obtain

$$\begin{array}{l} F(u_1) = F(1, 4) = (1, 6) = 5u_1 - 2u_2 \\ F(u_2) = F(2, 7) = (3, 11) = u_1 + u_2 \end{array} \quad \text{and so} \quad B = \begin{bmatrix} 5 & 1 \\ -2 & 1 \end{bmatrix}$$

Method 2. By [Theorem 6.7](#), $B = P^{-1}AP$. Thus,

$$B = P^{-1}AP = \begin{bmatrix} -7 & 2 \\ 4 & -1 \end{bmatrix} \begin{bmatrix} 5 & -1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 4 & 7 \end{bmatrix} = \begin{bmatrix} 5 & 1 \\ -2 & 1 \end{bmatrix}$$

6.24. Let $A = \begin{bmatrix} 2 & 3 \\ 4 & -1 \end{bmatrix}$. Find the matrix B that represents the linear operator A relative to the basis $S = \{u_1, u_2\} = \{[1, 3]^T, [2, 5]^T\}$. [Recall A defines a linear operator $A: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ relative to the usual basis E of $\mathbf{R}^2$].

Method 1. Find the coordinates of $A(u_1)$ and $A(u_2)$ relative to the basis S by first finding the coordinates of an arbitrary vector $[a, b]^T$ in $\mathbf{R}^2$ relative to the basis S . By [Problem 6.2](#),

$$[a, b]^T = (-5a + 2b)u_1 + (3a - b)u_2$$

Using the formula for $[a, b]^T$, we obtain

$$A(u_1) = \begin{bmatrix} 2 & 3 \\ 4 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 11 \\ 1 \end{bmatrix} = -53u_1 + 32u_2$$

and

$$A(u_2) = \begin{bmatrix} 2 & 3 \\ 4 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 19 \\ 3 \end{bmatrix} = -89u_1 + 54u_2$$

Thus,

$$B = \begin{bmatrix} -53 & -89 \\ 32 & 54 \end{bmatrix}$$

Method 2. Use $B = P^{-1}AP$, where P is the change-of-basis matrix from the usual basis E to S . Thus, simply write the vectors in S (as columns) to obtain the change-of-basis matrix P and then use the formula

for P^{-1} . This gives

$$P = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix} \quad \text{and} \quad P^{-1} = \begin{bmatrix} -5 & 2 \\ 3 & -1 \end{bmatrix}$$

Then

$$S = \{u_1, u_2, u_3\} = \{[1, 1, 0]^T, [0, 1, 1]^T, [1, 2, 2]^T\}$$

6.25. Let $A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 5 & -4 \\ 1 & -2 & 2 \end{bmatrix}$. Find the matrix B that

represents the linear operator A relative to the basis

$$S = \{u_1, u_2, u_3\} = \{[1, 1, 0]^T, [0, 1, 1]^T, [1, 2, 2]^T\}$$

[Recall A that defines a linear operator $A: \mathbf{R}^3 \rightarrow \mathbf{R}^3$ relative to the usual basis E of $\mathbf{R}^3$.]

Method 1. Find the coordinates of $A(u_1)$, $A(u_2)$, $A(u_3)$ relative to the basis S by first finding the coordinates

of an arbitrary vector $v = (a, b, c)$ in $\mathbf{R}^3$ relative to the basis S . By [Problem 6.16](#),

$$[v]_S = (b - c)u_1 + (-2a + 2b - c)u_2 + (a - b + c)u_3$$

Using this formula for $[a, b, c]^T$, we obtain

$$\begin{aligned} A(u_1) &= [4, 7, -1]^T = 8u_1 + 7u_2 - 4u_3, & A(u_2) &= [4, 1, 0]^T = u_1 - 6u_2 + 3u_3 \\ A(u_3) &= [9, 4, 1]^T = 3u_1 - 11u_2 + 6u_3 \end{aligned}$$

Writing the coefficients of u_1, u_2, u_3 as columns yields

$$B = \begin{bmatrix} 8 & 1 & 3 \\ 7 & -6 & -11 \\ -4 & 3 & 6 \end{bmatrix}$$

Method 2. Use $B = P^{-1}AP$, where P is the change-of-basis matrix from the usual basis E to S . The matrix P (whose columns are simply the vectors in S) and P^{-1} appear in [Problem 6.16](#). Thus,

$$B = P^{-1}AP = \begin{bmatrix} 0 & 1 & -1 \\ -2 & 2 & -1 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 1 \\ 2 & 5 & -4 \\ 1 & -2 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 8 & 1 & 3 \\ 7 & -6 & -11 \\ -4 & 3 & 6 \end{bmatrix}$$

6.26. Prove [Theorem 6.7](#): Let P be the change-of-basis matrix from a basis S to a basis S' in a vector space V . Then, for any linear operator T on V , $[T]_{S'} = P^{-1}[T]_S P$. Let v be a vector in V . Then, by [Theorem 6.6](#), $P[v]_{S'} = [v]_S$. Therefore,

$$P^{-1}[T]_{S'}P[v]_{S'} = P^{-1}[T]_S[v]_S = P^{-1}[T(v)]_S = [T(v)]_{S'}$$

But $[T]_{S'}[v]_{S'} = [T(v)]_{S'}$. Hence,

$$P^{-1}[T]_S P[v]_{S'} = [T]_{S'}[v]_{S'}$$

Because the mapping $v \mapsto [v]_{S'}$. Thus, $P^{-1} [T]_S P = [T]_{S'}$, as claimed.

Similarity of Matrices

6.27. Let $A = \begin{bmatrix} 4 & -2 \\ 3 & 6 \end{bmatrix}$.

- (a) Find $B = P^{-1}AP$. (b) Verify $\text{tr}(B) = \text{tr}(A)$. (c) Verify $\det(B) = \det(A)$.

(a) First find P^{-1} using the formula for the inverse of a 2×2 matrix. We have

$$P^{-1} = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix}$$

Then

$$B = P^{-1}AP = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} 4 & -2 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 25 & 30 \\ -\frac{27}{2} & -15 \end{bmatrix}$$

(b) $\text{tr}(A) = 4 + 6 = 10$ and $\text{tr}(B) = 25 - 15 = 10$. Hence, $\text{tr}(B) = \text{tr}(A)$.

(c) $\det(A) = 4 \cdot 6 - (-2 \cdot 3) = 24 + 6 = 30$ and $\det(B) = 25 \cdot (-15) - (-\frac{27}{2} \cdot 30) = -375 + 405 = 30$. Hence, $\det(B) = \det(A)$.

6.28. Find the trace of each of the linear transformations F on $\mathbf{R}^3$ in [Problem 6.4](#).

Find the trace (sum of the diagonal elements) of any matrix representation of F such as the matrix representation $[F] = [F]_E$ of F relative to the usual basis E given in [Problem 6.4](#).

- (a) $\text{tr}(F) = \text{tr}([F] = 1 - 5 + 9 = 5.$
- (b) $\text{tr}(F) = \text{tr}([F] = 1 + 3 + 5 = 9.$
- (c) $\text{tr}(F) = \text{tr}([F] = 1 + 4 + 7 = 12.$

6.29. Write $A \approx B$ if A is similar to B —that is, if there exists an invertible matrix P such that $A = P^{-1}BP$. Prove that $\approx$ is an equivalence relation (on square matrices); that is,

- (a) $A \approx A$, for every A .
 - (b) If $A \approx B$, then $B \approx A$.
 - (c) If $A \approx B$ and $B \approx C$, then $A \approx C$.
- (a) The identity matrix I is invertible, and $I^{-1} = I$. Because $A = I^{-1}AI$, we have $A \approx A$.
- (b) Because $A \approx B$, there exists an invertible matrix P such that $A = P^{-1}BP$. Hence, $B = PAP^{-1} = (P^{-1})^{-1}AP^{-1}$ and P^{-1} is also invertible. Thus, $B \approx A$.
- (c) Because $A \approx B$, there exists an invertible matrix P such that $A = P^{-1}BP$, and as $B \approx C$, there exists an invertible matrix Q such that $B = Q^{-1}CQ$. Thus,

$$A = P^{-1}BP = P^{-1}(Q^{-1}CQ)P = (P^{-1}Q^{-1})C(QP) = (QP)^{-1}C(QP)$$

and QP is also invertible. Thus, $A \approx C$.

6.30. Suppose B is similar to A , say $B = P^{-1}AP$. Prove

- (a) $B^n = P^{-1}A^nP$, and so B^n is similar to A^n .
- (b) $f(B) = P^{-1}f(A)P$, for any polynomial $f(x)$, and so $f(B)$ is similar to $f(A)$.
- (c) B is a root of a polynomial $g(x)$ if and only if A is a root of $g(x)$.

- (a) The proof is by induction on n . The result holds for $n = 1$ by hypothesis. Suppose $n > 1$ and the result holds for $n - 1$. Then

$$B^n = BB^{n-1} = (P^{-1}AP)(P^{-1}A^{n-1}P) = P^{-1}A^nP$$

- (b) Suppose $f(x) = a_nx^n + \cdots + a_1x + a_0$. Using the left and right distributive laws and part (a), we have

$$\begin{aligned} P^{-1}f(A)P &= P^{-1}(a_nA^n + \cdots + a_1A + a_0I)P \\ &= P^{-1}(a_nA^n)P + \cdots + P^{-1}(a_1A)P + P^{-1}(a_0I)P \\ &= a_n(P^{-1}A^nP) + \cdots + a_1(P^{-1}AP) + a_0(P^{-1}IP) \\ &= a_nB^n + \cdots + a_1B + a_0I = f(B) \end{aligned}$$

- (c) By part (b), $g(B) = 0$ if and only if $P^{-1}g(A)P = 0$ if and only if $g(A) = P0P^{-1} = 0$.

Matrix Representations of General Linear Mappings

6.31. Let $F: \mathbf{R}^3 \rightarrow \mathbf{R}^2$ be the linear map defined by $F(x, y, z) = (3x + 2y - 4z, x + 5y + 3z)$.

- (a) Find the matrix of F in the following bases of $\mathbf{R}^3$ and $\mathbf{R}^2$:

$$S = \{w_1, w_2, w_3\} = \{(1, 1, 1), (1, 1, 0), (1, 0, 0)\} \quad \text{and} \quad S' = \{u_1, u_2\} = \{(1, 3), (2, 5)\}$$

- (b) Verify [Theorem 6.10](#): The action of F is preserved by its matrix representation; that is, for any v in $\mathbf{R}^3$, we have $[F]_{S,S'}[v]_S = [F(v)]_{S'}$.

- (a) From [Problem 6.2](#), $(a, b) = (-5a + 2b)u_1 + (3a - b)u_2$. Thus,

$$\begin{aligned}
F(w_1) &= F(1, 1, 1) = (1, -1) = -7u_1 + 4u_2 \\
F(w_2) &= F(1, 1, 0) = (5, -4) = -33u_1 + 19u_2 \\
F(w_3) &= F(1, 0, 0) = (3, 1) = -13u_1 + 8u_2
\end{aligned}$$

Write the coordinates of $F(w_1)$, $F(w_2)$, $F(w_3)$ as columns to get

$$[F]_{S,S'} = \begin{bmatrix} -7 & -33 & 13 \\ 4 & 19 & 8 \end{bmatrix}$$

(b) If $v = (x, y, z)$, then, by [Problem 6.5](#), $v = zw_1 + (y - z)w_2 + (x - y)w_3$. Also,

$$F(v) = (3x + 2y - 4z, x - 5y + 3z) = (-13x - 20y + 26z)u_1 + (8x + 11y - 15z)u_2$$

$$\text{Hence, } [v]_S = (z, y - z, x - y)^T \quad \text{and} \quad [F(v)]_{S'} = \begin{bmatrix} -13x - 20y + 26z \\ 8x + 11y - 15z \end{bmatrix}$$

$$\text{Thus, } [F]_{S,S'}[v]_S = \begin{bmatrix} -7 & -33 & 13 \\ 4 & 19 & 8 \end{bmatrix} \begin{bmatrix} z \\ y - z \\ x - y \end{bmatrix} = \begin{bmatrix} -13x - 20y + 26z \\ 8x + 11y - 15z \end{bmatrix} = [F(v)]_{S'}$$

6.32. Let $F: \mathbf{R}^n \rightarrow \mathbf{R}^m$ be the linear mapping defined as follows:

$$F(x_1, x_2, \dots, x_n) = (a_{11}x_1 + \dots + a_{1n}x_n, a_{21}x_1 + \dots + a_{2n}x_n, \dots, a_{m1}x_1 + \dots + a_{mn}x_n)$$

(a) Show that the rows of the matrix $[F]$ representing F relative to the usual bases of $\mathbf{R}^n$ and $\mathbf{R}^m$ are the coefficients of the x_i in the components of $F(x_1, \dots, x_n)$.

(b) Find the matrix representation of each of the following linear mappings relative to the usual basis of $\mathbf{R}^n$:

- (i) $F: \mathbf{R}^2 \rightarrow \mathbf{R}^3$ defined by $F(x, y) = (3x - y, 2x + 4y, 5x - 6y)$.
- (ii) $F: \mathbf{R}^4 \rightarrow \mathbf{R}^2$ defined by $F(x, y, s, t) = (3x - 4y + 2s - 5t, 5x + 7y - s - 2t)$.
- (iii) $F: \mathbf{R}^3 \rightarrow \mathbf{R}^4$ defined by $F(x, y, z) = (2x + 3y - 8z, x + y + z, 4x - 5z, 6y)$.

(a) We have

$$\begin{array}{l} F(1,0,\dots,0) = (a_{11}, a_{21}, \dots, a_{m1}) \\ F(0,1,\dots,0) = (a_{12}, a_{22}, \dots, a_{m2}) \\ \vdots \\ F(0,0,\dots,1) = (a_{1n}, a_{2n}, \dots, a_{mn}) \end{array} \quad \text{and thus,} \quad [F] = \left[\begin{array}{cccc} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{array} \right]$$

(b) By part (a), we need only look at the coefficients of the unknown $x, y, \dots$ in $F(x, y, \dots)$. Thus,

$$\text{(i)} \quad [F] = \begin{bmatrix} 3 & -1 \\ 2 & 4 \\ 5 & -6 \end{bmatrix}, \quad \text{(ii)} \quad [F] = \begin{bmatrix} 3 & -4 & 2 & -5 \\ 5 & 7 & -1 & -2 \end{bmatrix}, \quad \text{(iii)} \quad [F] = \begin{bmatrix} 2 & 3 & -8 \\ 1 & 1 & 1 \\ 4 & 0 & -5 \\ 0 & 6 & 0 \end{bmatrix}$$

6.33. Let $A = \begin{bmatrix} 2 & 5 & -3 \\ 1 & -4 & 7 \end{bmatrix}$. Recall that A determines a

mapping $F: \mathbf{R}^3 \rightarrow \mathbf{R}^2$ defined by $F(v) = Av$, where vectors are written as columns. Find the matrix $[F]$ that represents the mapping relative to the following bases of $\mathbf{R}^3$ and $\mathbf{R}^2$:

(a) The usual bases of $\mathbf{R}^3$ and of $\mathbf{R}^2$.

(b) $S = \{w_1, w_2, w_3\} = \{(1, 1, 1), (1, 1, 0), (1, 0, 0)\}$
and $S = \{u_1, u_2\} = f(1, 3), (2, 5)\}$.

(a) Relative to the usual bases, $[F]$ is the matrix A .

(b) From [Problem 6.2](#), $(a, b) = (-5a + 2b) u_1 + (3a - b) u_2$. Thus,

$$F(w_1) = \begin{bmatrix} 2 & 5 & -3 \\ 1 & -4 & 7 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix} = -12u_1 + 8u_2$$

$$F(w_2) = \begin{bmatrix} 2 & 5 & -3 \\ 1 & -4 & 7 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 7 \\ -3 \end{bmatrix} = -41u_1 + 24u_2$$

$$F(w_3) = \begin{bmatrix} 2 & 5 & -3 \\ 1 & -4 & 7 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} = -8u_1 + 5u_2$$

Writing the coefficients of $F(w_1)$, $F(w_2)$, $F(w_3)$ as

columns yields $[F] = \begin{bmatrix} -12 & -41 & -8 \\ 8 & 24 & 5 \end{bmatrix}$.

6.34. Consider the linear transformation T on $\mathbf{R}^2$ defined by $T(x, y) = (2x - 3y, x + 4y)$ and the following bases of $\mathbf{R}^2$:

$$E = \{e_1, e_2\} = \{(1, 0), (0, 1)\} \quad \text{and} \quad S = \{u_1, u_2\} = \{(1, 3), (2, 5)\}$$

(a) Find the matrix A representing T relative to the bases E and S .

(b) Find the matrix B representing T relative to the bases S and E .

(We can view T as a linear mapping from one space into another, each having its own basis.)

(a) From [Problem 6.2](#), $(a, b) = (-5a + 2b)u_1 + (3a - b)u_2$. Hence,

$$\begin{aligned} T(e_1) &= T(1, 0) = (2, 1) = -8u_1 + 5u_2 \\ T(e_2) &= T(0, 1) = (-3, 4) = 23u_1 - 13u_2 \end{aligned} \quad \text{and so} \quad A = \begin{bmatrix} -8 & 23 \\ 5 & -13 \end{bmatrix}$$

(b) We have

$$\begin{aligned} T(u_1) &= T(1, 3) = (-7, 13) = -7e_1 + 13e_2 \\ T(u_2) &= T(2, 5) = (-11, 22) = -11e_1 + 22e_2 \end{aligned} \quad \text{and so} \quad B = \begin{bmatrix} -7 & -11 \\ 13 & 22 \end{bmatrix}$$

6.35. How are the matrices A and B in [Problem 6.34](#) related?

By [Theorem 6.12](#), the matrices A and B are equivalent to each other; that is, there exist nonsingular matrices P and Q such that $B = Q^{-1}AP$, where P is the change-of-basis matrix from S to E , and Q is the change-of-basis matrix from E to S . Thus,

$$P = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}, \quad Q = \begin{bmatrix} -5 & 2 \\ 3 & -1 \end{bmatrix}, \quad Q^{-1} = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$$

and

$$Q^{-1}AP = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} -8 & -23 \\ 5 & -13 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix} = \begin{bmatrix} -7 & -11 \\ 13 & 22 \end{bmatrix} = B$$

6.36. Prove [Theorem 6.14](#): Let $F: V \rightarrow U$ be linear and, say, $\text{rank}(F) = r$. Then there exist bases V and of U such that the matrix representation of F has the following form, where I_r is the r -square identity matrix:

$$A = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$$

Suppose $\dim V = m$ and $\dim U = n$. Let W be the kernel of F and U the image of F . We are given that $\text{rank}(F) = r$. Hence, the dimension of the kernel of F is $m - r$. Let $\{w_1, \dots, w_{m-r}\}$ be a basis of the kernel of F and extend this to a basis of V :

$$\{v_1, \dots, v_r, w_1, \dots, w_{m-r}\}$$

Set

$$u_1 = F(v_1), \quad u_2 = F(v_2), \quad \dots, \quad u_r = F(v_r)$$

Then $\{u_1, \dots, u_r\}$ is a basis of U , the image of F . Extend this to a basis of U , say

$$\{u_1, \dots, u_r, u_{r+1}, \dots, u_n\}$$

Observe that

$$F(v_1) = u_1 = 1u_1 + 0u_2 + \dots + 0u_r + 0u_{r+1} + \dots + 0u_n$$

$$F(v_2) = u_2 = 0u_1 + 1u_2 + \dots + 0u_r + 0u_{r+1} + \dots + 0u_n$$

.....

$$F(v_r) = u_r = 0u_1 + 0u_2 + \dots + 1u_r + 0u_{r+1} + \dots + 0u_n$$

$$F(w_1) = 0 = 0u_1 + 0u_2 + \dots + 0u_r + 0u_{r+1} + \dots + 0u_n$$

.....

$$F(w_{m-r}) = 0 = 0u_1 + 0u_2 + \dots + 0u_r + 0u_{r+1} + \dots + 0u_n$$

Thus, the matrix of F in the above bases has the required form.

SUPPLEMENTARY PROBLEMS

Matrices and Linear Operators

6.37. Let $F: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ be defined by $F(x, y) = (4x + 5y, 2x - y)$.

- Find the matrix A representing F in the usual basis E .
- Find the matrix B representing F in the basis $S = \{u_1, u_2\} = \{(1, 4), (2, 9)\}$.

(c) Find P such that $B = P^{-1}AP$.

(d) For $v = (a, b)$, find $[v]_S$ and $[F(v)]_S$. Verify that $[F]_S[v]_S = [F(v)]_S$.

6.38. Let $A: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ be defined by the matrix $A = \begin{bmatrix} 5 & -1 \\ 2 & 4 \end{bmatrix}$

(a) Find the matrix B representing A relative to the basis $S = \{u_1, u_2\} = \{(1, 3), (2, 8)\}$. (Recall that A represents the mapping A relative to the usual basis E .)

(b) For $v = (a, b)$, find $[v]_S$ and $[A(v)]_S$.

6.39. For each linear transformation L on $\mathbf{R}^2$, find the matrix A representing L (relative to the usual basis of $\mathbf{R}^2$):

(a) L is the rotation in $\mathbf{R}^2$ counterclockwise by 45° .

(b) L is the reflection in $\mathbf{R}^2$ about the line $y = x$.

(c) L is defined by $L(1, 0) = (3, 5)$ and $L(0, 1) = (7, -2)$.

(d) L is defined by $L(1, 1) = (3, 7)$ and $L(1, 2) = (5, -4)$.

6.40. Find the matrix representing each linear transformation T on $\mathbf{R}^3$ relative to the usual basis of $\mathbf{R}^3$:

(a) $T(x, y, z) = (x, y, 0)$.

(b) $T(x, y, z) = (z, y + z, x + y + z)$.

(c) $T(x, y, z) = (2x - 7y + 4z, 3x + y + 4z, 6x - 8y + z)$.

6.41. Repeat [Problem 6.40](#) using the basis $S = \{u_1, u_2, u_3\} = \{(1, 1, 0), (1, 2, 3), (1, 3, 5)\}$.

6.42. Let L be the linear transformation on $\mathbf{R}^3$ defined by

$$L(1, 0, 0) = (1, 1, 1), \quad L(0, 1, 0) = (1, 3, 5), \quad L(0, 0, 1) = (2, 2, 2)$$

- (a) Find the matrix A representing L relative to the usual basis of $\mathbf{R}^3$.
- (b) Find the matrix B representing L relative to the basis S in [Problem 6.41](#).

6.43. Let $\mathbf{D}$ denote the differential operator; that is, $\mathbf{D}(f(t)) = df/dt$. Each of the following sets is a basis of a vector space V of functions. Find the matrix representing $\mathbf{D}$ in each basis:

$$(a) \quad \{e^t, e^{2t}, te^{2t}\}. \quad (b) \quad \{1, t, \sin 3t, \cos 3t\}. \quad (c) \quad \{e^{5t}, te^{5t}, t^2 e^{5t}\}.$$

6.44. Let $\mathbf{D}$ denote the differential operator on the vector space V of functions with basis $S = \{\sin \theta, \cos \theta\}$.

- (a) Find the matrix $A = [\mathbf{D}]_S$.
- (b) Use A to show that $\mathbf{D}$ is a zero of $f(t) = t^2 + 1$.

6.45. Let V be the vector space of 2×2 matrices. Consider the following matrix M and usual basis E of V :

$$M = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \text{and} \quad E = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$$

Find the matrix representing each of the following linear operators T on V relative to E :

- (a) $T(A) = MA$.
- (b) $T(A) = AM$.
- (c) $T(A) = MA - AM$.

6.46. Let $\mathbf{1}_V$ and $\mathbf{0}_V$ denote the identity and zero operators, respectively, on a vector space V . Show that, for any

basis S of V , (a) $[\mathbf{1}_V]_S = I$, the identity matrix. (b) $[\mathbf{0}_V]_S = 0$, the zero matrix.

Change of Basis

- 6.47.** Find the change-of-basis matrix P from the usual basis E of $\mathbf{R}^2$ to a basis S , the change-of-basis matrix Q from S back to E , and the coordinates of $v = (a, b)$ relative to S , for the following bases S :
- (a) $S = \{(1, 2), (3, 5)\}$.
 - (b) $S = \{(1, -3), (3, -8)\}$.
 - (c) $S = \{(2, 5), (3, 7)\}$.
 - (d) $S = \{(2, 3), (4, 5)\}$.
- 6.48.** Consider the bases $S = \{(1, 2), (2, 3)\}$ and $S' = \{(1, 3), (1, 4)\}$ of $\mathbf{R}^2$. Find the change-of-basis matrix:
- (a) P from S to S' .
 - (b) Q from S' back to S .
- 6.49.** Suppose that the x -axis and y -axis in the plane $\mathbf{R}^2$ are rotated counterclockwise 30° to yield new x' -axis and y' -axis for the plane. Find
- (a) The unit vectors in the direction of the new x' -axis and y' -axis.
 - (b) The change-of-basis matrix P for the new coordinate system.
 - (c) The new coordinates of the points $A(1, 3)$, $B(2, 5)$, $C(a, b)$.
- 6.50.** Find the change-of-basis matrix P from the usual basis E of $\mathbf{R}^3$ to a basis S , the change-of-basis matrix Q

from S back to E , and the coordinates of $v = (a, b, c)$ relative to S , where S consists of the vectors:

(a) $u_1 = (1, 1, 0)$, $u_2 = (0, 1, 2)$, $u_3 = (0, 1, 1)$.

(b) $u_1 = (1, 0, 1)$, $u_2 = (1, 1, 2)$, $u_3 = (1, 2, 4)$.

(c) $u_1 = (1, 2, 1)$, $u_2 = (1, 3, 4)$, $u_3 = (2, 5, 6)$.

6.51. Suppose S_1, S_2, S_3 are bases of V . Let P and Q be the change-of-basis matrices, respectively, from S_1 to S_2 and from S_2 to S_3 . Prove that PQ is the change-of-basis matrix from S_1 to S_3 .

Linear Operators and Change of Basis

6.52. Consider the linear operator F on $\mathbf{R}^2$ defined by $F(x, y) = (5x + y, 3x - 2y)$ and the following bases of $\mathbf{R}^2$:

$$S = \{(1, 2), (2, 3)\} \quad \text{and} \quad S' = \{(1, 3), (1, 4)\}$$

(a) Find the matrix A representing F relative to the basis S .

(b) Find the matrix B representing F relative to the basis S' .

(c) Find the change-of-basis matrix P from S to S' .

(d) How are A and B related?

6.53. Let $A: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ be defined by the matrix $A = \begin{bmatrix} 1 & -1 \\ 3 & 2 \end{bmatrix}$.

Find the matrix B that represents the linear operator A relative to each of the following bases: (a) $S = \{(1, 3)^T, (2, 5)^T\}$. (b) $S = \{(1, 3)^T, (2, 4)^T\}$.

6.54. Let $F: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ be defined by $F(x, y) = (x - 3y, 2x - 4y)$. Find the matrix A that represents F relative to each

of the following bases: (a) $S = \{(2, 5), (3, 7)\}$. (b) $S = \{(2, 3), (4, 5)\}$.

6.55. Let $A: \mathbf{R}^3 \rightarrow \mathbf{R}^3$ be defined by the matrix

$$A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 7 & 4 \\ 1 & 4 & 3 \end{bmatrix}. \text{ Find the matrix } B \text{ that represents the}$$

linear operator A relative to the basis $S = \{(1, 1, 1)^T, (0, 1, 1)^T, (1, 2, 3)^T\}$.

Similarity of Matrices

6.56. Let $A = \begin{bmatrix} 1 & 1 \\ 2 & -3 \end{bmatrix}$ and $P = \begin{bmatrix} 1 & -2 \\ 3 & -5 \end{bmatrix}$.

- (a) Find $B = P^{-1}AP$.
- (b) Verify that $\text{tr}(B) = \text{tr}(A)$.
- (c) Verify that $\det(B) = \det(A)$.

6.57. Find the trace and determinant of each of the following linear maps on $\mathbf{R}^2$:

- (a) $F(x, y) = (2x - 3y, 5x + 4y)$.
- (b) $G(x, y) = (ax + by, cx + dy)$.

6.58. Find the trace and determinant of each of the following linear maps on $\mathbf{R}^3$:

- (a) $F(x, y, z) = (x + 3y, 3x - 2z, x - 4y - 3z)$.
- (b) $G(x, y, z) = (y + 3z, 2x - 4z, 5x + 7y)$.

6.59. Suppose $S = \{u_1, u_2\}$ is a basis of V , and $T: V \rightarrow V$ is defined by $T(u_1) = 3u_1 - 2u_2$ and $T(u_2) = u_1 + 4u_2$. Suppose $S' = \{w_1, w_2\}$ is a basis of V for which $w_1 = u_1 + u_2$ and $w_2 = 2u_1 + 3u_2$.

(a) Find the matrices A and B representing T relative to the bases S and S' , respectively.

(b) Find the matrix P such that $B = P^{-1}AP$.

6.60. Let A be a 2×2 matrix such that only A is similar to itself. Show that A is a scalar matrix, that is, that

$$A = \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix}.$$

6.61. Show that all matrices similar to an invertible matrix are invertible. More generally, show that similar matrices have the same rank.

Matrix Representation of General Linear Mappings

6.62. Find the matrix representation of each of the following linear maps relative to the usual basis for $\mathbf{R}^n$:

(a) $F: \mathbf{R}^3 \rightarrow \mathbf{R}^2$ defined by $F(x, y, z) = (2x - 4y + 9z, 5x + 3y - 2z)$.

(b) $F: \mathbf{R}^2 \rightarrow \mathbf{R}^4$ defined by $F(x, y) = (3x + 4y, 5x - 2y, x + 7y, 4x)$.

(c) $F: \mathbf{R}^4 \rightarrow \mathbf{R}$ defined by $F(x_1, x_2, x_3, x_4) = 2x_1 + x_2 - 7x_3 - x_4$.

6.63. Let $G: \mathbf{R}^3 \rightarrow \mathbf{R}^2$ be defined by $G(x, y, z) = (2x + 3y, 4x + 2z)$.

(a) Find the matrix A representing G relative to the bases

$$S = \{(1, 1, 0), (1, 2, 3), (1, 3, 5)\} \quad \text{and} \quad S' = \{(1, 2), (2, 3)\}$$

(b) For any $v = (a, b, c)$ in $\mathbf{R}^3$, find $[v]_S$ and $[G(v)]_{S'}$.

(c) Verify that $A[\nu]_S = [G(\nu)]_S$.

6.64. Let $H: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ be defined by $H(x, y) = (2x + 7y, x - 3y)$ and consider the following bases of $\mathbf{R}^2$:

$$S = \{(1, 1), (1, 2)\} \quad \text{and} \quad S' = \{(1, 4), (1, 5)\}$$

(a) Find the matrix A representing H relative to the bases S and S' .

(b) Find the matrix B representing H relative to the bases S' and S .

6.65. Let $F: \mathbf{R}^3 \rightarrow \mathbf{R}^2$ be defined by $F(x, y, z) = (2x + yz, 3x - 2y + 4z)$.

(a) Find the matrix A representing F relative to the bases

$$S = \{(1, 1, 1), (1, 1, 0), (1, 0, 0)\} \quad \text{and} \quad S' = \{(1, 3), (1, 4)\}$$

(b) Verify that, for any $\nu = (a, b, c)$ in $\mathbf{R}^3$, $A[\nu]_S = [F(\nu)]_{S'}$.

6.66. Let S and S' be bases of V , and let $\mathbf{1}_V$ be the identity mapping on V . Show that the matrix A representing $\mathbf{1}_V$ relative to the bases S and S' is the inverse of the change-of-basis matrix P from S to S' ; that is, $A = P^{-1}$.

6.67. Prove (a) [Theorem 6.10](#), (b) [Theorem 6.11](#), (c) [Theorem 6.12](#), (d) [Theorem 6.13](#). [*Hint*: See the proofs of the analogous Theorems 6.1 ([Problem 6.9](#)), 6.2 ([Problem 6.10](#)), 6.3 ([Problem 6.11](#)), and 6.7 ([Problem 6.26](#)).]

Miscellaneous Problems

- 6.68.** Suppose $F: V \rightarrow V$ is linear. A subspace W of V is said to be *invariant* under F if $F(W) \subseteq W$. Suppose W is invariant under F and $\dim W = r$. Show that F has a block triangular matrix representation $M = \begin{bmatrix} A & B \\ 0 & C \end{bmatrix}$ where A is an $r \times r$ submatrix.
- 6.69.** Suppose $V = U + W$, and suppose U and W are each invariant under a linear operator $F: V \rightarrow V$. Also, suppose $\dim U = r$ and $\dim W = s$. Show that F has a block diagonal matrix representation $M = \begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix}$ where A and B are $r \times r$ and $s \times s$ submatrices.
- 6.70.** Two linear operators F and G on V are said to be *similar* if there exists an invertible linear operator T on V such that $G = T^{-1} \circ F \circ T$. Prove
- (a) F and G are similar if and only if, for any basis S of V , $[F]_S$ and $[G]_S$ are similar matrices.
 - (b) If F is diagonalizable (similar to a diagonal matrix), then any similar matrix G is also diagonalizable.

ANSWERS TO SUPPLEMENTARY PROBLEMS

Notation: $M = [R_1; R_2; \dots]$ represents a matrix M with rows $R_1, R_2, \dots$

- 6.37.** (a) $A = [4, 5; 2, -1]$; (b) $B = [220, 487; -98, -217]$; (c) $P = [1, 2; 4, 9]$; (d) $[v]_S = [9a - 2b, -4a + b]^T$ and $[F(v)]_S = [32a + 47b, -14a - 21b]^T$
- 6.38.** (a) $[\sqrt{2}, -\sqrt{2}; \sqrt{2}, \sqrt{2}]$
- 6.39.** $[A(v)]_S = [18a - 8b, \frac{1}{2}(-13a + 7b)]$

6.40. (a) $[1, 0, 0; 0, 1, 0; 0, 0, 0]$; (b) $[0, 0, 1; 0, 1, 1; 1, 1, 1]$;

(c) $[2, -7, -4; 3, 1, 4; 6, -8, 1]$

6.41. (a) $[1, 3, 5; 0, -5, -10; 0, 3, 6]$; (b) $[0, 1, 2; 1, 2, 3; 1, 0, 0]$;

(c) $[15, 65, 104; 49, -219, -351; 29, 130, 208]$

6.42. (a) $[1, 1, 2; 1, 3, 2; 1, 5, 2]$; (b) $[0; 2, 14, 22; 0, -5, -8]$

6.43. (a) $[1, 0, 0; 0, 2, 1; 0, 0, 2]$; (b) $[0, 1, 0, 0; 0; 0, 0, 0, -3; 0, 0, 3, 0]$; (c) $[5, 1, 0; 0, 5, 2; 0, 0, 5]$

6.44. (a) $A = [0, -1; 1, 0;]$ (b) $A^2 + I = 0$

6.45. (a) $[a, 0, b, 0; 0, a, 0, b; c, 0, d, 0; 0, c, 0, d]$;

(b) $[a, c, 0, 0; b, d, 0, 0; 0, 0, a, c; 0, 0, b, d]$;

(c) $[0, -c, b, 0; -b, a - d, 0, b; c, 0, d - a, -c; 0, c, -b, 0]$

6.47.

(a) $[1, 3; 2, 5], [-5, 3; 2, -1], [v] = [-5a + 3b, 2a - b]^T$;
 (b) $[1, 3; -3, -8], [-8, -3; 3, 1], [v] = [-8a - 3b, 3a + b]^T$;
 (c) $[2, 3; 5, 7], [-7, 3; 5, -2], [v] = [-7a + 3b, 5a - 2b]^T$;
 (d) $[2, 4; 3, 5], [-\frac{5}{2}, 2; \frac{3}{2}, -1], [v] = [-\frac{5}{2}a + 2b, \frac{3}{2}a - b]^T$

6.48. (a) $P = [3, 5; 1, -2 \times]$; (b) $Q = [2, 5; 1, -3]$

6.49.

Here $K = \sqrt{3}$.

(a) $\frac{1}{2}(K, 1), \frac{1}{2}(-1, K)$;

(b) $P = \frac{1}{2}[K, -1; 1, K]$;

(c) $\frac{1}{2}[K + 3, 3K - 1]^T, \frac{1}{2}[2K - 5, -5K - 2]^T, \frac{1}{2}[aK + b, bK - a]^T$

6.50. P is the matrix whose columns are u_1, u_2, u_3 , $Q = P^{-1}$, $[v] = Q[a, b, c]^T$:

- (a) $Q = [1, 0, 0; 1, -1, 1; -2, 2, -1]$, $[v] = [a, a - b + c, -2a + 2b - c]^T$;
 (b) $Q = [0, -2, 1; 2, 3, -2; -1, -1, 1]$, $[v] = [-2b + c, 2a + 3b - 2c, -a - b + c]^T$;
 (c) $Q = [-2, 2, -1; -7, 4, -1; 5, -3, 1]$, $[v] = [-2a + 2b - c, -7a + 4b - c, 5a - 3b + c]^T$

6.52. (a) $[23, -39; 15, 26]$; (b) $[35, 41; 27, -32]$; (c) $[3, 5; 1, -2]$; (d) $B = P^{-1}AP$

6.53. (a) $[28, 47; -15, -25]$; (b) $[13, 18; -\frac{15}{2}, -10]$

6.54. (a) $[43, 60; -33, -46]$; (b) $\frac{1}{2}[3, 7; -5, -9]$

6.55. $[10, 8, 20; 13, 11, 28; 5, -4, -10]$

6.56. (a) $[34, 57; 19, 32]$; (b) $\text{tr}(B) = \text{tr}(A) = -2$; (c) $\det(B) = \det(A) = -5$

6.57. (a) $\text{tr}(F) = 6$, $\det(F) = 23$; (b) $\text{tr}(G) = a + d$, $\det(G) = ad - bc$

6.58. (a) $\text{tr}(F) = -2$, $\det(F) = 13$; (b) $\text{tr}(G) = 0$, $\det(G) = 22$

6.59. (a) $A = [3, 1; -2, 4]$, $B = [8, 11; 2, -1]$; (b) $P = [1, 2; 1, 3]$

6.62. (a) $[2, -4, 9; 5, 3, -2 \times]$; (b) $[3, 5, 1, 4; 4, -2, 7, 0]$; (c) $[2, 1, -7, -1]$

6.63. (a) $[9, 1, 4; 7, 2, 1]$; (b) $[v]_S = [-a + 2b - c, 5a - 5b + 2c, -3a + 3b - c]^T$, and $[G(v)]_S = [2a - 11b + 7c, 7b - 4c]^T$

6.64. (a) $A = [47, 85; -38, -69]$; (b) $B = [71, 88; 41, -51]$

6.65. $A = [3, 11, 5; -1, -8, -3]$

CHAPTER 7

Inner Product Spaces, Orthogonality

7.1 Introduction

The definition of a vector space V involves an arbitrary field K . Here we first restrict K to be the real field $\mathbf{R}$, in which case V is called a *real vector space*; in the last sections of this chapter, we extend our results to the case where K is the complex field $\mathbf{C}$, in which case V is called a *complex vector space*. Also, we adopt the previous notation that

u, v, w	are vectors in V
a, b, c, k	are scalars in K

Furthermore, the vector spaces V in this chapter have finite dimension unless otherwise stated or implied.

Recall that the concepts of “length” and “orthogonality” did not appear in the investigation of arbitrary vector spaces V (although they did appear in Section 1.4 on the

spaces $\mathbf{R}^n$ and $\mathbf{C}^n$). Here we place an additional structure on a vector space V to obtain an inner product space, and in this context these concepts are defined.

7.2 Inner Product Spaces

We begin with a definition.

DEFINITION: Let V be a real vector space. Suppose to each pair of vectors $u, v \in V$ there is assigned a real number, denoted by $\langle u, v \rangle$. This function is called a (*real*) *inner product* on V if it satisfies the following axioms:

$$[I_1] \quad \textbf{(Linear Property):} \quad \langle au_1 + bu_2, v \rangle = a\langle u_1, v \rangle + b\langle u_2, v \rangle.$$

$$[I_2] \quad \textbf{(Symmetric Property):} \quad \langle u, v \rangle = \langle v, u \rangle.$$

$$[I_3] \quad \textbf{(Positive Definite Property):} \quad \langle u, u \rangle \geq 0.; \text{ and } \langle u, u \rangle = 0 \text{ if and only if } u = 0.$$

The vector space V with an inner product is called a (*real*) *inner product space*.

Axiom $[I_1]$ states that an inner product function is linear in the first position. Using $[I_1]$ and the symmetry axiom $[I_2]$, we obtain

$$\langle u, cv_1 + dv_2 \rangle = \langle cv_1 + dv_2, u \rangle = c\langle v_1, u \rangle + d\langle v_2, u \rangle = c\langle u, v_1 \rangle + d\langle u, v_2 \rangle$$

That is, the inner product function is also linear in its second position. Combining these two properties and using induction yields the following general formula:

$$\left\langle \sum_i a_i u_i, \sum_j b_j v_j \right\rangle = \sum_i \sum_j a_i b_j \langle u_i, v_j \rangle$$

That is, an inner product of linear combinations of vectors is equal to a linear combination of the inner products of the vectors.

EXAMPLE 7.1 Let V be a real inner product space. Then, by linearity,

$$\begin{aligned} \langle 3u_1 - 4u_2, 2v_1 - 5v_2 + 6v_3 \rangle &= 6\langle u_1, v_1 \rangle - 15\langle u_1, v_2 \rangle + 18\langle u_1, v_3 \rangle \\ &\quad - 8\langle u_2, v_1 \rangle + 20\langle u_2, v_2 \rangle - 24\langle u_2, v_3 \rangle \end{aligned}$$

$$\begin{aligned} \langle 2u - 5v, 4u + 6v \rangle &= 8\langle u, u \rangle + 12\langle u, v \rangle - 20\langle v, u \rangle - 30\langle v, v \rangle \\ &= 8\langle u, u \rangle - 8\langle v, u \rangle - 30\langle v, v \rangle \end{aligned}$$

Observe that in the last equation we have used the symmetry property that $\langle u, v \rangle = \langle v, u \rangle$.

Remark: Axiom $[I_1]$ by itself implies $\langle 0, 0 \rangle = \langle 0v, 0 \rangle = 0$ $\langle v, 0 \rangle = 0$. Thus, $[I_1]$, $[I_2]$, $[I_3]$ are equivalent to $[I_1]$, $[I_2]$, and the following axiom:

$[I'_3]$ If $u \neq 0$, then $\langle u, u \rangle$ is positive.

That is, a function satisfying $[I_1]$, $[I_2]$, $[I'_3]$ is an inner product.

Norm of a Vector

By the third axiom $[I_3]$ of an inner product, $\langle u, u \rangle$ is nonnegative for any vector u . Thus, its positive square root exists. We use the notation

$$\|u\| = \sqrt{\langle u, u \rangle}$$

This nonnegative number is called the *norm* or *length* of u . The relation $\|u\|^2 = \langle u, u \rangle$ will be used frequently.

Remark: If $\|u\| = 1$ or, equivalently, if $\langle u, u \rangle = 1$, then u is called a *unit vector* and it is said to be *normalized*. Every nonzero vector v in V can be multiplied by the reciprocal of its length to obtain the unit vector

$$\hat{v} = \frac{1}{\|v\|} v$$

which is a positive multiple of v . This process is called *normalizing* v .

7.3 Examples of Inner Product Spaces

This section lists the main examples of inner product spaces used in this text.

Euclidean n -Space $\mathbf{R}^n$

Consider the vector space $\mathbf{R}^n$. The *dot product* or *scalar product* in $\mathbf{R}^n$ is defined by

$$u \cdot v = a_1 b_1 + a_2 b_2 + \cdots + a_n b_n$$

where $u = (a_i)$ and $v = (b_i)$. This function defines an inner product on $\mathbf{R}^n$. The norm $\|u\|$ of the vector $u = (a_i)$ in this space is as follows:

$$\|u\| = \sqrt{u \cdot u} = \sqrt{a_1^2 + a_2^2 + \cdots + a_n^2}$$

On the other hand, by the Pythagorean theorem, the distance from the origin O in $\mathbf{R}^3$ to a point $P(a, b, c)$ is given by $\sqrt{a^2 + b^2 + c^2}$. This is precisely the same as the above-defined norm of the vector $v = (a, b, c)$ in $\mathbf{R}^3$. Because the Pythagorean theorem is a consequence of the axioms of Euclidean geometry, the vector space $\mathbf{R}^n$ with the above inner product and norm is called *Euclidean n -space*. Although there are many ways to define an inner product on $\mathbf{R}^n$, we shall assume this inner product unless otherwise stated or implied. It is called the *usual (or standard) inner product* on $\mathbf{R}^n$.

Remark: Frequently the vectors in $\mathbf{R}^n$ will be represented by column vectors—that is, by $n \times 1$ column matrices. In such a case, the formula

$$\langle u, v \rangle = u^T v$$

defines the usual inner product on $\mathbf{R}^n$.

EXAMPLE 7.2 Let $u = (1, 3, -4, 2)$, $v = (4, -2, 2, 1)$, $w = (5, -1, -2, 6)$ in $\mathbf{R}^4$.

(a) Show $\langle 3u - 2v, w \rangle = 3\langle u, w \rangle - 2\langle v, w \rangle$.

By definition,

$$\langle u, w \rangle = 5 - 3 + 8 + 12 = 22 \quad \text{and} \quad \langle v, w \rangle = 20 + 2 - 4 + 6 = 24$$

Note that $3u - 2v = (-5, 13, -16, 4)$. Thus,

$$\langle 3u - 2v, w \rangle = -25 - 13 + 32 + 24 = 18$$

As expected, $3\langle u, w \rangle - 2\langle v, w \rangle = 3(22) - 2(24) = 18 = \langle 3u - 2v, w \rangle$.

(b) Normalize u and v .

By definition,

$$\|u\| = \sqrt{1 + 9 + 16 + 4} = \sqrt{30} \quad \text{and} \quad \|v\| = \sqrt{16 + 4 + 4 + 1} = 5$$

We normalize u and v to obtain the following unit vectors in the directions of u and v , respectively:

$$\hat{u} = \frac{1}{\|u\|} u = \left(\frac{1}{\sqrt{30}}, \frac{3}{\sqrt{30}}, \frac{-4}{\sqrt{30}}, \frac{2}{\sqrt{30}} \right) \quad \text{and} \quad \hat{v} = \frac{1}{\|v\|} v = \left(\frac{4}{5}, \frac{-2}{5}, \frac{2}{5}, \frac{1}{5} \right)$$

Function Space $C[a, b]$ and Polynomial Space $P(t)$

The notation $C[a, b]$ is used to denote the vector space of all continuous functions on the closed interval $[a, b]$ —that is, where $a \leq t \leq b$. The following defines an inner product on $C[a, b]$, where $f(t)$ and $g(t)$ are functions in $C[a, b]$:

$$\langle f, g \rangle = \int_a^b f(t)g(t) dt$$

It is called the *usual inner product* on $C[a, b]$.

The vector space $P(t)$ of all polynomials is a subspace of $C[a, b]$ for any interval $[a, b]$, and hence, the above is also an inner product on $P(t)$.

EXAMPLE 7.3

Consider $f(t) = 3t - 5$ and $g(t) = t^2$ in the polynomial space $P(t)$ with inner product

$$\langle f, g \rangle = \int_0^1 f(t)g(t) dt.$$

(a) Find $\langle f, g \rangle$.

We have $f(t)g(t) = 3t^3 - 5t^2$. Hence,

$$\langle f, g \rangle = \int_0^1 (3t^3 - 5t^2) dt = \left. \frac{3}{4}t^4 - \frac{5}{3}t^3 \right|_0^1 = \frac{3}{4} - \frac{5}{3} = -\frac{11}{12}$$

(b) Find $\|f\|$ and $\|g\|$.

We have $[f(t)]^2 = f(t)f(t) = 9t^2 - 30t + 25$ and $[g(t)]^2 = t^4$. Then

$$\|f\|^2 = \langle f, f \rangle = \int_0^1 (9t^2 - 30t + 25) dt = \left. 3t^3 - 15t^2 + 25t \right|_0^1 = 13$$

$$\|g\|^2 = \langle g, g \rangle = \int_0^1 t^4 dt = \left. \frac{1}{5}t^5 \right|_0^1 = \frac{1}{5}$$

Therefore, $\|f\| = \sqrt{13}$ and $\|g\| = \sqrt{\frac{1}{5}} = \frac{1}{5}\sqrt{5}$.

Matrix Space $\mathbf{M} = \mathbf{M}_{m,n}$

Let $\mathbf{M} = \mathbf{M}_{m,n}$, the vector space of all real $m \times n$ matrices. An inner product is defined on $\mathbf{M}$ by

$$\langle A, B \rangle = \text{tr}(B^T A)$$

where, as usual, $\text{tr}(\)$ is the trace—the sum of the diagonal elements. If $A = [a_{ij}]$ and $B = [b_{ij}]$, then

$$\langle A, B \rangle = \text{tr}(B^T A) = \sum_{i=1}^m \sum_{j=1}^n a_{ij} b_{ij} \quad \text{and} \quad \|A\|^2 = \langle A, A \rangle = \sum_{i=1}^m \sum_{j=1}^n a_{ij}^2$$

That is, $\langle A, B \rangle$ is the sum of the products of the corresponding entries in A and B and, in particular, $\langle A, A \rangle$ is the sum of the squares of the entries of A .

Hilbert Space

Let V be the vector space of all infinite sequences of real numbers $(a_1, a_2, a_3, \dots)$ satisfying

$$\sum_{i=1}^{\infty} a_i^2 = a_1^2 + a_2^2 + \dots < \infty$$

that is, the sum converges. Addition and scalar multiplication are defined in V componentwise; that is, if

$$u = (a_1, a_2, \dots) \quad \text{and} \quad v = (b_1, b_2, \dots)$$

then

$$u + v = (a_1 + b_1, a_2 + b_2, \dots) \quad \text{and} \quad ku = (ka_1, ka_2, \dots)$$

An inner product is defined in V by

$$\langle u, v \rangle = a_1 b_1 + a_2 b_2 + \dots$$

The above sum converges absolutely for any pair of points in V . Hence, the inner product is well defined. This inner product space is called l_2 -space or *Hilbert space*.

7.4 Cauchy-Schwarz Inequality, Applications

The following formula (proved in [Problem 7.8](#)) is called the Cauchy-Schwarz inequality or Schwarz inequality. It is used in many branches of mathematics.

THEOREM 7.1: (Cauchy-Schwarz) For any vectors u and v in an inner product space V ,

$$\langle u, v \rangle^2 \leq \langle u, u \rangle \langle v, v \rangle \quad \text{or} \quad |\langle u, v \rangle| \leq \|u\| \|v\|$$

Next we examine this inequality in specific cases.

EXAMPLE 7.4

(a) Consider any real numbers $a_1, \dots, a_n, b_1, \dots, b_n$. Then, by the Cauchy-Schwarz inequality,

$$(a_1b_1 + a_2b_2 + \cdots + a_nb_n)^2 \leq (a_1^2 + \cdots + a_n^2)(b_1^2 + \cdots + b_n^2)$$

That is, $(u \cdot v)^2 \leq \|u\|^2 \|v\|^2$, where $u = (a_j)$ and $v = (b_j)$.

(b) Let f and g be continuous functions on the unit interval $[0, 1]$. Then, by the Cauchy-Schwarz inequality,

$$\left[\int_0^1 f(t)g(t) dt \right]^2 \leq \int_0^1 f^2(t) dt \int_0^1 g^2(t) dt$$

That is, $(\langle f, g \rangle)^2 \leq \|f\|^2 \|g\|^2$. Here V is the inner product space $C[0, 1]$.

The next theorem (proved in [Problem 7.9](#)) gives the basic properties of a norm. The proof of the third property requires the Cauchy-Schwarz inequality.

THEOREM 7.2: Let V be an inner product space. Then the norm in V satisfies the following properties:

$$[N_1] \quad \|v\| \geq 0; \text{ and } \|v\| = 0 \text{ if and only if } v = 0.$$

$$[N_2] \quad \|kv\| = |k| \|v\|.$$

$$[N_3] \quad \|u + v\| \leq \|u\| + \|v\|.$$

The property $[N_3]$ is called the *triangle inequality*, because if we view $u + v$ as the side of the triangle formed with sides u and v (as shown in [Fig. 7-1](#)), then $[N_3]$ states

that the length of one side of a triangle cannot be greater than the sum of the lengths of the other two sides.

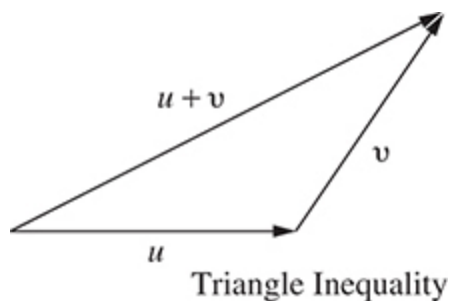


Figure 7-1

Angle Between Vectors

For any nonzero vectors u and v in an inner product space V , the *angle between u and v* is defined to be the angle θ such that $0 \leq \theta \leq \pi$ and

$$\cos \theta = \frac{\langle u, v \rangle}{\|u\| \|v\|}$$

By the Cauchy-Schwartz inequality, $-1 \leq \cos \theta \leq 1$, and so the angle exists and is unique.

EXAMPLE 7.5

(a) Consider vectors $u = (2, 3, 5)$ and $v = (1, -4, 3)$ in $\mathbf{R}^3$.
Then

$$\langle u, v \rangle = 2 - 12 + 15 = 5, \quad \|u\| = \sqrt{4 + 9 + 25} = \sqrt{38}, \quad \|v\| = \sqrt{1 + 16 + 9} = \sqrt{26}$$

Then the angle θ between u and v is given by

$$\cos \theta = \frac{5}{\sqrt{38}\sqrt{26}}$$

Note that θ is an acute angle, because $\cos \theta$ is positive.

(b) Let $f(t) = 3t - 5$ and $g(t) = t^2$ in the polynomial space $\mathbf{P}(t)$ with inner product $\langle f, g \rangle = \int_0^1 f(t)g(t) dt$. By [Example 7.3](#),

$$\langle f, g \rangle = -\frac{11}{12}, \quad \|f\| = \sqrt{13}, \quad \|g\| = \frac{1}{5}\sqrt{5}$$

Then the “angle” θ between f and g is given by

$$\cos \theta = \frac{-\frac{11}{12}}{(\sqrt{13})(\frac{1}{5}\sqrt{5})} = -\frac{55}{12\sqrt{13}\sqrt{5}}$$

Note that θ is an obtuse angle, because $\cos \theta$ is negative.

7.5 Orthogonality

Let V be an inner product space. The vectors $u, v \in V$ are said to be *orthogonal* and u is said to be *orthogonal* to v if

$$\langle u, v \rangle = 0$$

The relation is clearly symmetric—if u is orthogonal to v , then $\langle v, u \rangle = 0$, and so v is orthogonal to u . We note that $0 \in V$ is orthogonal to every $v \in V$, because

$$\langle 0, v \rangle = \langle 0v, v \rangle = 0\langle v, v \rangle = 0$$

Conversely, if u is orthogonal to every $v \in V$, then $\langle u, u \rangle = 0$ and hence $u = 0$ by [I₃]. Observe that u and v are orthogonal if and only if $\cos \theta = 0$, where θ is the angle

between u and v . Also, this is true if and only if u and v are “perpendicular”—that is, $\theta = \pi/2$ (or $\theta = 90^\circ$).

EXAMPLE 7.6

(a) Consider the vectors $u = (1, 1, 1)$, $v = (1, 2, -3)$, $w = (1, -4, 3)$ in $\mathbf{R}^3$. Then

$$\langle u, v \rangle = 1 + 2 - 3 = 0, \quad \langle u, w \rangle = 1 - 4 + 3 = 0, \quad \langle v, w \rangle = 1 - 8 - 9 = -16$$

Thus, u is orthogonal to v and w , but v and w are not orthogonal.

(b) Consider the functions $\sin t$ and $\cos t$ in the vector space $C[-\pi, \pi]$ of continuous functions on the closed interval $[-\pi, \pi]$. Then

$$\langle \sin t, \cos t \rangle = \int_{-\pi}^{\pi} \sin t \cos t \, dt = \frac{1}{2} \sin^2 t \Big|_{-\pi}^{\pi} = 0 - 0 = 0$$

Thus, $\sin t$ and $\cos t$ are orthogonal functions in the vector space $C[-\pi, \pi]$.

Remark: A vector $w = (x_1, x_2, \dots, x_n)$ is orthogonal to $u = (a_1, a_2, \dots, a_n)$ in $\mathbf{R}^n$ if

$$\langle u, w \rangle = a_1 x_1 + a_2 x_2 + \cdots + a_n x_n = 0$$

That is, w is orthogonal to u if w satisfies a homogeneous equation whose coefficients are the elements of u .

EXAMPLE 7.7 Find a nonzero vector w that is orthogonal to $u_1 = (1, 2, 1)$ and $u_2 = (2, 5, 4)$ in $\mathbf{R}^3$.

Let $w = (x, y, z)$. Then we want $\langle u_1, w \rangle = 0$ and $\langle u_2, w \rangle = 0$. This yields the homogeneous system

$$\begin{array}{rcl} x + 2y + z = 0 & & x + 2y + z = 0 \\ 2x + 5y + 4z = 0 & \text{or} & y + 2z = 0 \end{array}$$

Here z is the only free variable in the echelon system. Set $z = 1$ to obtain $y = -2$ and $x = 3$. Thus, $w = (3, -2, 1)$ is a desired nonzero vector orthogonal to u_1 and u_2 .

Any multiple of w will also be orthogonal to u_1 and u_2 . Normalizing w , we obtain the following unit vector orthogonal to u_1 and u_2 :

$$\hat{w} = \frac{w}{\|w\|} = \left(\frac{3}{\sqrt{14}}, -\frac{2}{\sqrt{14}}, \frac{1}{\sqrt{14}} \right)$$

Orthogonal Complements

Let S be a subset of an inner product space V . The orthogonal complement of S , denoted by $S^\perp$ (read “ S perp”) consists of those vectors in V that are orthogonal to every vector $u \in S$; that is,

$$S^\perp = \{v \in V : \langle v, u \rangle = 0 \text{ for every } u \in S\}$$

In particular, for a given vector u in V , we have

$$u^\perp = \{v \in V : \langle v, u \rangle = 0\}$$

that is, $u^\perp$ consists of all vectors in V that are orthogonal to the given vector u .

We show that $S^\perp$ is a subspace of V . Clearly $0 \in S^\perp$, because 0 is orthogonal to every vector in V . Now suppose $v, w \in S^\perp$. Then, for any scalars a and b and any vector $u \in S$, we have

$$\langle av + bw, u \rangle = a\langle v, u \rangle + b\langle w, u \rangle = a \cdot 0 + b \cdot 0 = 0$$

Thus, $av + bw \in S^\perp$, and therefore $S^\perp$ is a subspace of V .
We state this result formally.

PROPOSITION 7.3: Let S be a subset of a vector space V .
Then $S^\perp$ is a subspace of V .

Remark 1: Suppose u is a nonzero vector in $\mathbf{R}^3$. Then there is a geometrical description of $u^\perp$. Specifically, $u^\perp$ is the plane in $\mathbf{R}^3$ through the origin O and perpendicular to the vector u . This is shown in Fig. 7-2.

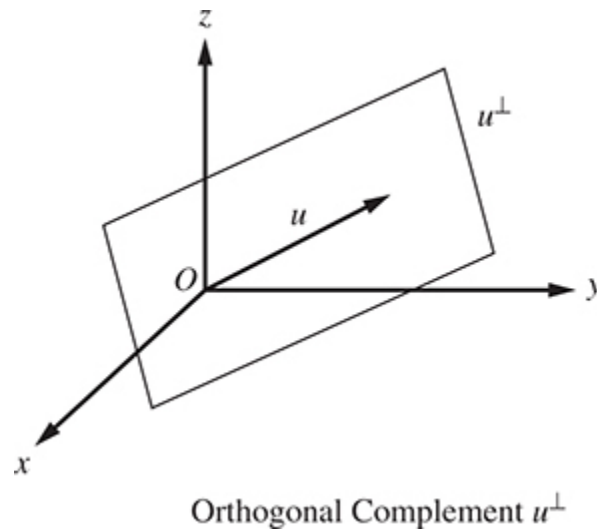


Figure 7-2

Remark 2: Let W be the solution space of an $m \times n$ homogeneous system $AX = 0$, where $A = [a_{ij}]$ and $X = [x_i]$. Recall that W may be viewed as the kernel of the linear mapping $A: \mathbf{R}^n \rightarrow \mathbf{R}^m$. Now we can give another interpretation of W using the notion of orthogonality. Specifically, each solution vector $w = (x_1, x_2, \dots, x_n)$ is orthogonal to each row of A ; hence, W is the orthogonal complement of the row space of A .

EXAMPLE 7.8 Find a basis for the subspace $u^\perp$ of $\mathbf{R}^3$, where $u = (1, 3, -4)$.

Note that $u^\perp$ consists of all vectors $w = (x, y, z)$ such that $\langle u, w \rangle = 0$, or $x + 3y - 4z = 0$. The free variables are y and z .

- (1) Set $y = 1, z = 0$ to obtain the solution $w_1 = (-3, 1, 0)$.
- (2) Set $y = 0, z = 1$ to obtain the solution $w_2 = (4, 0, 1)$.

The vectors w_1 and w_2 form a basis for the solution space of the equation, and hence a basis for $u^\perp$.

Suppose W is a subspace of V . Then both W and $W^\perp$ are subspaces of V . The next theorem, whose proof ([Problem 7.28](#)) requires results of later sections, is a basic result in linear algebra.

THEOREM 7.4: Let W be a subspace of V . Then V is the direct sum of W and $W^\perp$; that is, $V = W \oplus W^\perp$.

7.6 Orthogonal Sets and Bases

Consider a set $S = \{u_1, u_2, \dots, u_r\}$ of nonzero vectors in an inner product space V . S is called *orthogonal* if each pair of vectors in S are orthogonal, and S is called *orthonormal* if S is orthogonal and each vector in S has unit length. That is,

- (i) **Orthogonal:** $\langle u_i, u_j \rangle = 0$ for $i \neq j$
- (ii) **Orthonormal:** $\langle u_i, u_j \rangle = \begin{cases} 0 & \text{for } i \neq j \\ 1 & \text{for } i = j \end{cases}$

Normalizing an orthogonal set S refers to the process of multiplying each vector in S by the reciprocal of its length in order to transform S into an orthonormal set of vectors.

The following theorems apply.

THEOREM 7.5: Suppose S is an orthogonal set of nonzero vectors. Then S is linearly independent.

THEOREM 7.6: (Pythagoras) Suppose $\{u_1, u_2, \dots, u_r\}$ is an orthogonal set of vectors. Then

$$\|u_1 + u_2 + \cdots + u_r\|^2 = \|u_1\|^2 + \|u_2\|^2 + \cdots + \|u_r\|^2$$

These theorems are proved in Problems 7.15 and 7.16, respectively. Here we prove the Pythagorean theorem in the special and familiar case for two vectors. Specifically, suppose $\langle u, v \rangle = 0$. Then

$$\|u + v\|^2 = \langle u + v, u + v \rangle = \langle u, u \rangle + 2\langle u, v \rangle + \langle v, v \rangle = \langle u, u \rangle + \langle v, v \rangle = \|u\|^2 + \|v\|^2$$

which gives our result.

EXAMPLE 7.9

(a) Let $E = \{e_1, e_2, e_3\} = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ be the usual basis of Euclidean space $\mathbf{R}^3$. It is clear that

$$\langle e_1, e_2 \rangle = \langle e_1, e_3 \rangle = \langle e_2, e_3 \rangle = 0 \quad \text{and} \quad \langle e_1, e_1 \rangle = \langle e_2, e_2 \rangle = \langle e_3, e_3 \rangle = 1$$

Namely, E is an orthonormal basis of $\mathbf{R}^3$. More generally, the usual basis of $\mathbf{R}^n$ is orthonormal for every n .

(b) Let $V = C[-\pi, \pi]$ be the vector space of continuous functions on the interval $-\pi \leq t \leq \pi$ with inner product defined by $\langle f, g \rangle = \int_{-\pi}^{\pi} f(t)g(t) dt$. Then the following is a classical example of an orthogonal set in V :

$$\{1, \cos t, \cos 2t, \cos 3t, \dots, \sin t, \sin 2t, \sin 3t, \dots\}$$

This orthogonal set plays a fundamental role in the theory of Fourier series.

Orthogonal Basis and Linear Combinations, Fourier Coefficients

Let S consist of the following three vectors in $\mathbf{R}^3$:

$$u_1 = (1, 2, 1), \quad u_2 = (2, 1, -4), \quad u_3 = (3, -2, 1)$$

The reader can verify that the vectors are orthogonal; hence, they are linearly independent. Thus, S is an orthogonal basis of $\mathbf{R}^3$.

Suppose we want to write $v = (7, 1, 9)$ as a linear combination of u_1, u_2, u_3 . First we set v as a linear combination of u_1, u_2, u_3 using unknowns x_1, x_2, x_3 as follows:

$$v = x_1 u_1 + x_2 u_2 + x_3 u_3 \quad \text{or} \quad (7, 1, 9) = x_1(1, 2, 1) + x_2(2, 1, -4) + x_3(3, -2, 1) \quad (*)$$

We can proceed in two ways.

METHOD 1: Expand (*) (as in [Chapter 3](#)) to obtain the system

$$x_1 + 2x_2 + 3x_3 = 7, \quad 2x_1 + x_2 - 2x_3 = 1, \quad x_1 - 4x_2 + x_3 = 9$$

Solve the system by Gaussian elimination to obtain $x_1 = 3, x_2 = -1, x_3 = 2$. Thus, $v = 3u_1 - u_2 + 2u_3$.

METHOD 2: (This method uses the fact that the basis vectors are orthogonal, and the arithmetic is much simpler.) If we take the inner product of each side of (*) with respect to u_i , we get

$$\langle v, u_i \rangle = \langle x_1 u_1 + x_2 u_2 + x_3 u_3, u_i \rangle \quad \text{or} \quad \langle v, u_i \rangle = x_i \langle u_i, u_i \rangle \quad \text{or} \quad x_i = \frac{\langle v, u_i \rangle}{\langle u_i, u_i \rangle}$$

Here two terms drop out, because u_1, u_2, u_3 are orthogonal. Accordingly,

$$x_1 = \frac{\langle v, u_1 \rangle}{\langle u_1, u_1 \rangle} = \frac{7 + 2 + 9}{1 + 4 + 1} = \frac{18}{6} = 3, \quad x_2 = \frac{\langle v, u_2 \rangle}{\langle u_2, u_2 \rangle} = \frac{14 + 1 - 36}{4 + 1 + 16} = \frac{-21}{21} = -1$$

$$x_3 = \frac{\langle v, u_3 \rangle}{\langle u_3, u_3 \rangle} = \frac{21 - 2 + 9}{9 + 4 + 1} = \frac{28}{14} = 2$$

Thus, again, we get $v = 3u_1 - u_2 + 2u_3$.

The procedure in Method 2 is true in general. Namely, we have the following theorem (proved in [Problem 7.17](#)).

THEOREM 7.7: Let $\{u_1, u_2, \dots, u_n\}$ be an orthogonal basis of V . Then, for any $v \in V$,

$$v = \frac{\langle v, u_1 \rangle}{\langle u_1, u_1 \rangle} u_1 + \frac{\langle v, u_2 \rangle}{\langle u_2, u_2 \rangle} u_2 + \dots + \frac{\langle v, u_n \rangle}{\langle u_n, u_n \rangle} u_n$$

Remark: The scalar $k_i \equiv \frac{\langle v, u_i \rangle}{\langle u_i, u_i \rangle}$ is called the *Fourier coefficient* of v with respect to u_i , because it is analogous to a coefficient in the Fourier series of a function. This scalar also has a geometric interpretation, which is discussed below.

Projections

Let V be an inner product space. Suppose w is a given nonzero vector in V , and suppose v is another vector. We seek the “projection of v along w ,” which, as indicated in [Fig. 7-3\(a\)](#), will be the multiple cw of w such that $v' = v - cw$ is orthogonal to w . This means

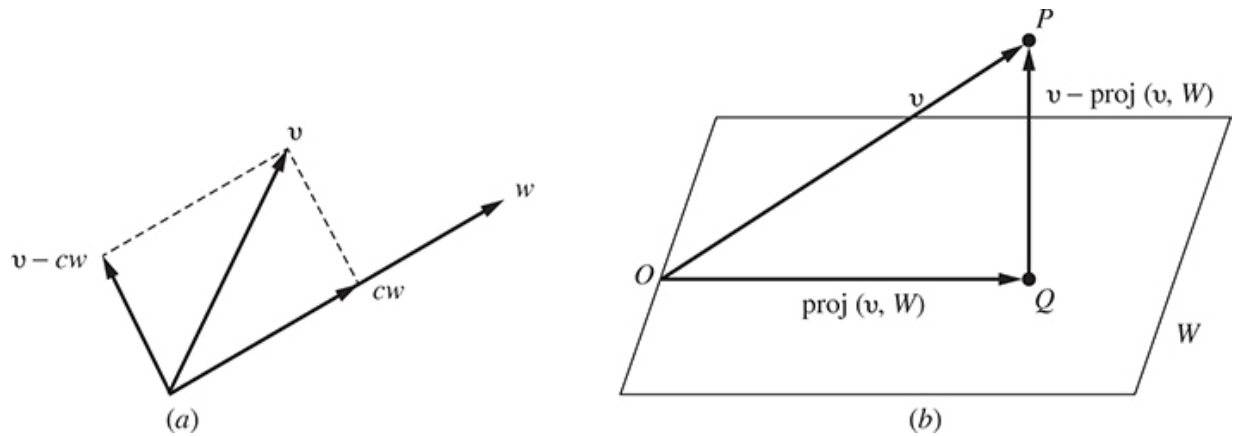


Figure 7-3

$$\langle v - cw, w \rangle = 0 \quad \text{or} \quad \langle v, w \rangle - c\langle w, w \rangle = 0 \quad \text{or} \quad c = \frac{\langle v, w \rangle}{\langle w, w \rangle}$$

Accordingly, the *projection of v along w* is denoted and defined by

$$\text{proj}(v, w) = cw = \frac{\langle v, w \rangle}{\langle w, w \rangle} w$$

Such a scalar c is unique, and it is called the *Fourier coefficient* of v with respect to w or the *component* of v along w .

The above notion is generalized as follows (see [Problem 7.25](#)).

THEOREM 7.8: Suppose $w_1, w_2, \dots, w_r$ form an orthogonal set of nonzero vectors in V . Let v be any vector in V . Define

$$v' = v - (c_1 w_1 + c_2 w_2 + \cdots + c_r w_r)$$

where

$$c_1 = \frac{\langle v, w_1 \rangle}{\langle w_1, w_1 \rangle}, \quad c_2 = \frac{\langle v, w_2 \rangle}{\langle w_2, w_2 \rangle}, \quad \dots, \quad c_r = \frac{\langle v, w_r \rangle}{\langle w_r, w_r \rangle}$$

Then v is orthogonal to $w_1, w_2, \dots, w_r$.

Note that each c_i in the above theorem is the component (Fourier coefficient) of v along the given w_i .

Remark: The notion of the projection of a vector $v \in V$ along a subspace W of V is defined as follows. By [Theorem 7.4](#), $V = W \oplus W^\perp$. Hence, v may be expressed uniquely in the form

$$v = w + w', \quad \text{where} \quad w \in W \quad \text{and} \quad w' \in W^\perp$$

We define w to be the *projection of v along W* , and denote it by $\text{proj}(v, W)$, as pictured in [Fig. 7-3\(b\)](#). In particular, if $W = \text{span}(w_1, w_2, \dots, w_r)$, where the w_i form an orthogonal set, then

$$\text{proj}(v, W) = c_1 w_1 + c_2 w_2 + \dots + c_r w_r$$

Here c_i is the component of v along w_i , as above.

7.7 Gram-Schmidt Orthogonalization Process

Suppose $\{v_1, v_2, \dots, v_n\}$ is a basis of an inner product space V . One can use this basis to construct an orthogonal basis $\{w_1, w_2, \dots, w_n\}$ of V as follows. Set

$$w_1 = v_1$$

$$w_2 = v_2 - \frac{\langle v_2, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1$$

$$w_3 = v_3 - \frac{\langle v_3, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1 - \frac{\langle v_3, w_2 \rangle}{\langle w_2, w_2 \rangle} w_2$$

.....

$$w_n = v_n - \frac{\langle v_n, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1 - \frac{\langle v_n, w_2 \rangle}{\langle w_2, w_2 \rangle} w_2 - \cdots - \frac{\langle v_n, w_{n-1} \rangle}{\langle w_{n-1}, w_{n-1} \rangle} w_{n-1}$$

In other words, for $k = 2, 3, \dots, n$, we define

$$w_k = v_k - c_{k1}w_1 - c_{k2}w_2 - \cdots - c_{k,k-1}w_{k-1}$$

where $c_{ki} = \langle v_k, w_i \rangle / \langle w_i, w_i \rangle$ is the component of v_k along w_i . By [Theorem 7.8](#), each w_k is orthogonal to the preceding w 's. Thus, $w_1, w_2, \dots, w_n$ form an orthogonal basis for V as claimed. Normalizing each w_i will then yield an orthonormal basis for V .

The above construction is known as the *Gram-Schmidt orthogonalization process*. The following remarks are in order.

Remark 1: Each vector w_k is a linear combination of v_k and the preceding w 's. Hence, one can easily show, by induction, that each w_k is a linear combination of $v_1, v_2, \dots, v_n$.

Remark 2: Because taking multiples of vectors does not affect orthogonality, it may be simpler in hand calculations to clear fractions in any new w_k , by multiplying w_k by an appropriate scalar, before obtaining the next w_{k+1} .

Remark 3: Suppose $u_1, u_2, \dots, u_r$ are linearly independent, and so they form a basis for $U = \text{span}(u_i)$. Applying the Gram-Schmidt orthogonalization process to the u 's yields an orthogonal basis for U .

The following theorems (proved in Problems 7.26 and 7.27) use the above algorithm and remarks.

THEOREM 7.9: Let $\{v_1, v_2, \dots, v_n\}$ be any basis of an inner product space V . Then there exists an orthonormal basis $\{u_1, u_2, \dots, u_n\}$ of V such that the change-of-basis matrix from $\{v_i\}$ to $\{u_i\}$ is triangular; that is, for $k = 1, \dots, n$,

$$u_k = a_{k1}v_1 + a_{k2}v_2 + \cdots + a_{kk}v_k$$

THEOREM 7.10: Suppose $S = \{w_1, w_2, \dots, w_r\}$ is an orthogonal basis for a subspace W of a vector space V . Then one may extend S to an orthogonal basis for V ; that is, one may find vectors $w_{r+1}, \dots, w_n$ such that $\{w_1, w_2, \dots, w_n\}$ is an orthogonal basis for V .

EXAMPLE 7.10 Apply the Gram-Schmidt orthogonalization process to find an orthogonal basis and then an orthonormal basis for the subspace U of $\mathbf{R}^4$ spanned by

$$v_1 = (1, 1, 1, 1), \quad v_2 = (1, 2, 4, 5), \quad v_3 = (1, -3, -4, -2)$$

(1) First set $w_1 = v_1 = (1, 1, 1, 1)$.

(2) Compute

$$v_2 - \frac{\langle v_2, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1 = v_2 - \frac{12}{4} w_1 = (-2, -1, 1, 2)$$

Set $w_2 = (-2, -1, 1, 2)$.

(3) Compute

$$v_3 - \frac{\langle v_3, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1 - \frac{\langle v_3, w_2 \rangle}{\langle w_2, w_2 \rangle} w_2 = v_3 - \frac{(-8)}{4} w_1 - \frac{(-7)}{10} w_2 = \left(\frac{8}{5}, -\frac{17}{10}, -\frac{13}{10}, \frac{7}{5}\right)$$

Clear fractions to obtain $w_3 = (-6, -17, -13, 14)$.

Thus, w_1, w_2, w_3 form an orthogonal basis for U .

Normalize these vectors to obtain an orthonormal basis $\{u_1, u_2, u_3\}$ of U . We have $\|w_1\|^2 = 4$, $\|w_2\|^2 = 10$, $\|w_3\|^2 = 910$, so

$$u_1 = \frac{1}{2}(1, 1, 1, 1), \quad u_2 = \frac{1}{\sqrt{10}}(-2, -1, 1, 2), \quad u_3 = \frac{1}{\sqrt{910}}(16, -17, -13, 14)$$

EXAMPLE 7.11 Let V be the vector space of polynomials $f(t)$ with inner product $\langle f, g \rangle = \int_{-1}^1 f(t)g(t) dt$. Apply the Gram-Schmidt orthogonalization process to $\{1, t, t^2, t^3\}$ to find an orthogonal basis $\{f_0, f_1, f_2, f_3\}$ with integer coefficients for $\mathbf{P}_3(t)$.

Here we use the fact that, for $r + s = n$,

$$\langle t^r, t^s \rangle = \int_{-1}^1 t^n dt = \frac{t^{n+1}}{n+1} \Big|_{-1}^1 = \begin{cases} 2/(n+1) & \text{when } n \text{ is even} \\ 0 & \text{when } n \text{ is odd} \end{cases}$$

(1) First set $f_0 = 1$.

(2) Compute $t - \frac{\langle t, 1 \rangle}{\langle 1, 1 \rangle} (1) = t - 0 = t$. Set $f_1 = t$.

(3) Compute

$$t^2 - \frac{\langle t^2, 1 \rangle}{\langle 1, 1 \rangle} (1) - \frac{\langle t^2, t \rangle}{\langle t, t \rangle} (t) = t^2 - \frac{\frac{2}{3}}{2} (1) + \frac{0}{2} (t) = t^2 - \frac{1}{3}$$

Multiply by 3 to obtain $f_2 = 3t^2 - 1$.

(4) Compute

$$\begin{aligned} t^3 - \frac{\langle t^3, 1 \rangle}{\langle 1, 1 \rangle} (1) - \frac{\langle t^3, t \rangle}{\langle t, t \rangle} (t) - \frac{\langle t^3, 3t^2 - 1 \rangle}{\langle 3t^2 - 1, 3t^2 - 1 \rangle} (3t^2 - 1) \\ = t^3 - 0(1) - \frac{\frac{2}{3}}{\frac{2}{3}}(t) - 0(3t^2 - 1) = t^3 - \frac{3}{5}t \end{aligned}$$

Multiply by 5 to obtain $f_3 = 5t^3 - 3t$.

Thus, $\{1, t, 3t^2 - 1, 5t^3 - 3t\}$ is the required orthogonal basis.

Remark: Normalizing the polynomials in [Example 7.11](#) so that $p(1) = 1$ yields the polynomials

$$1, t, \frac{1}{2}(3t^2 - 1), \frac{1}{2}(5t^3 - 3t)$$

These are the first four *Legendre polynomials*, which appear in the study of differential equations.

7.8 Orthogonal and Positive Definite Matrices

This section discusses two types of matrices that are closely related to real inner product spaces V . Here vectors in $\mathbf{R}^n$ will be represented by column vectors. Thus, $\langle u, v \rangle = u^T v$ denotes the inner product in Euclidean space $\mathbf{R}^n$.

Orthogonal Matrices

A real matrix P is *orthogonal* if P is nonsingular and $P^{-1} = P^T$, or, in other words, if $PP^T = P^T P = I$. First we recall

(Theorem 2.6) an important characterization of such matrices.

THEOREM 7.11: Let P be a real matrix. Then the following are equivalent: (a) P is orthogonal; (b) the rows of P form an orthonormal set; (c) the columns of P form an orthonormal set.

(This theorem is true only using the usual inner product on $\mathbf{R}^n$. It is not true if $\mathbf{R}^n$ is given any other inner product.)

EXAMPLE 7.12

(a) Let $P = \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \\ 0 & 1/\sqrt{2} & -1/\sqrt{2} \\ 2/\sqrt{6} & -1/\sqrt{6} & -1/\sqrt{6} \end{bmatrix}$. The rows of P are orthogonal to each other and are unit vectors. Thus P is an orthogonal matrix.

(b) Let P be a 2×2 orthogonal matrix. Then, for some real number θ , we have

$$P = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \quad \text{or} \quad P = \begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{bmatrix}$$

The following two theorems (proved in Problems 7.37 and 7.38) show important relationships between orthogonal matrices and orthonormal bases of a real inner product space V .

THEOREM 7.12: Suppose $E = \{e_i\}$ and $E' = \{e'_i\}$ are orthonormal bases of V . Let P be the change-of-basis matrix from the basis E to the basis E' . Then P is orthogonal.

THEOREM 7.13: Let $\{e_1, \dots, e_n\}$ be an orthonormal basis of an inner product space V . Let $P = [a_{ij}]$ be

an orthogonal matrix. Then the following n vectors form an orthonormal basis for V :

$$e'_i = a_{1i}e_1 + a_{2i}e_2 + \cdots + a_{ni}e_n, \quad i = 1, 2, \dots, n$$

Positive Definite Matrices

Let A be a real symmetric matrix; that is, $A^T = A$. Then A is said to be *positive definite* if, for every nonzero vector u in $\mathbf{R}^n$,

$$\langle u, Au \rangle = u^T Au > 0$$

Algorithms to decide whether or not a matrix A is positive definite will be given in [Chapter 13](#). However, for 2×2 matrices, we have simple criteria that we state formally in the following theorem (proved in [Problem 7.43](#)).

THEOREM 7.14: A 2×2 real symmetric matrix

$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a & b \\ b & d \end{bmatrix}$ is positive definite if and only if the diagonal entries a and d are positive and the determinant $|A| = ad - bc = ad - b^2$ is positive.

EXAMPLE 7.13 Consider the following symmetric matrices:

$$A = \begin{bmatrix} 1 & 3 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & -2 \\ -2 & -3 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & -2 \\ -2 & 5 \end{bmatrix}$$

A is not positive definite, because $|A| = 4 - 9 = -5$ is negative. B is not positive definite, because the diagonal entry -3 is negative. However, C is positive definite, because the diagonal entries 1 and 5 are positive, and the determinant $|C| = 5 - 4 = 1$ is also positive.

The following theorem (proved in [Problem 7.44](#)) holds.

THEOREM 7.15: Let A be a real positive definite matrix. Then the function $\langle u, v \rangle = u^T A v$ is an inner product on $\mathbf{R}^n$.

Matrix Representation of an Inner Product (Optional)

Theorem 7.15 says that every positive definite matrix A determines an inner product on $\mathbf{R}^n$. This subsection may be viewed as giving the converse of this result.

Let V be a real inner product space with basis $S = \{u_1, u_2, \dots, u_n\}$. The matrix

$$A = [a_{ij}], \quad \text{where} \quad a_{ij} = \langle u_i, u_j \rangle$$

is called the *matrix representation of the inner product on V relative to the basis S* .

Observe that A is symmetric, because the inner product is symmetric; that is, $\langle u_i, u_j \rangle = \langle u_j, u_i \rangle$. Also, A depends on both the inner product on V and the basis S for V . Moreover, if S is an orthogonal basis, then A is diagonal, and if S is an orthonormal basis, then A is the identity matrix.

EXAMPLE 7.14 The vectors $u_1 = (1, 1, 0)$, $u_2 = (1, 2, 3)$, $u_3 = (1, 3, 5)$ form a basis S for Euclidean space $\mathbf{R}^3$. Find the matrix A that represents the inner product in $\mathbf{R}^3$ relative to this basis S .

First compute each $\langle u_i, u_j \rangle$ to obtain

$$\begin{aligned} \langle u_1, u_1 \rangle &= 1 + 1 + 0 = 2, & \langle u_1, u_2 \rangle &= 1 + 2 + 0 = 3, & \langle u_1, u_3 \rangle &= 1 + 3 + 0 = 4 \\ \langle u_2, u_2 \rangle &= 1 + 4 + 9 = 14, & \langle u_2, u_3 \rangle &= 1 + 6 + 15 = 22, & \langle u_3, u_3 \rangle &= 1 + 9 + 25 = 35 \end{aligned}$$

Then $A = \begin{bmatrix} 2 & 3 & 4 \\ 3 & 14 & 22 \\ 4 & 22 & 35 \end{bmatrix}$. As expected, A is symmetric.

The following theorems (proved in Problems 7.45 and 7.46, respectively) hold.

THEOREM 7.16: Let A be the matrix representation of an inner product relative to basis S for V . Then, for any vectors $u, v \in V$, we have

$$\langle u, v \rangle = [u]^T A [v]$$

where $[u]$ and $[v]$ denote the (column) coordinate vectors relative to the basis S .

THEOREM 7.17: Let A be the matrix representation of any inner product on V . Then A is a positive definite matrix.

7.9 Complex Inner Product Spaces

This section considers vector spaces over the complex field $\mathbf{C}$. First we recall some properties of the complex numbers (Section 1.7), especially the relations between a complex number $z = a + bi$, where $a, b \in \mathbf{R}$, and its complex conjugate $\bar{z} = a - bi$:

$$z\bar{z} = a^2 + b^2, \quad |z| = \sqrt{a^2 + b^2}, \quad \overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2, \quad \overline{z_1 z_2} = \bar{z}_1 \bar{z}_2, \quad \bar{\bar{z}} = z$$

Also, z is real if and only if $\bar{z} = z$.

The following definition applies.

DEFINITION: Let V be a vector space over $\mathbf{C}$. Suppose to each pair of vectors, $u, v \in V$ there is assigned a complex number, denoted by $\langle u, v \rangle$. This function is called a (*complex*) *inner product* on V if it satisfies the following axioms:

- $[I_1^*]$ (**Linear Property**) $\langle au_1 + bu_2, v \rangle = a\langle u_1, v \rangle + b\langle u_2, v \rangle$
 $[I_2^*]$ (**Conjugate Symmetric Property**) $\langle u, v \rangle = \overline{\langle v, u \rangle}$
 $[I_3^*]$ (**Positive Definite Property**) $\langle u, u \rangle \geq 0$; and $\langle u, u \rangle = 0$ if and only if $u = 0$.

The vector space V over C with an inner product is called a (*complex*) *inner product space*. Observe that a complex inner product differs from the real case only in the second axiom $[I_2^*]$. Axiom $[I_1^*]$ (Linear Property) is equivalent to the two conditions:

$$(a) \quad \langle u_1 + u_2, v \rangle = \langle u_1, v \rangle + \langle u_2, v \rangle, \quad (b) \quad \langle ku, v \rangle = k\langle u, v \rangle$$

On the other hand, applying $[I_1^*]$ and $[I_2^*]$, we obtain

$$\langle u, kv \rangle = \overline{\langle kv, u \rangle} = \overline{k\langle v, u \rangle} = \bar{k}\overline{\langle v, u \rangle} = \bar{k}\langle u, v \rangle$$

That is, we must take the conjugate of a complex number when it is taken out of the second position of a complex inner product. In fact ([Problem 7.47](#)), the inner product is *conjugate linear* in the second position; that is,

$$\langle u, av_1 + bv_2 \rangle = \bar{a}\langle u, v_1 \rangle + \bar{b}\langle u, v_2 \rangle$$

Combining linear in the first position and conjugate linear in the second position, we obtain, by induction,

$$\left\langle \sum_i a_i u_i, \sum_j b_j v_j \right\rangle = \sum_{i,j} a_i \bar{b}_j \langle u_i, v_j \rangle$$

The following remarks are in order.

Remark 1: Axiom $[I_1^*]$ by itself implies that $\langle 0, 0 \rangle = \langle 0v, 0 \rangle = 0\langle v, 0 \rangle = 0$. Accordingly, $[I_1^*]$, $[I_2^*]$, and $[I_3^*]$ are equivalent to $[I_1^*]$, $[I_2^*]$, and the following axiom:

$$[I_3^{*'}] \text{ If } u \neq 0, \text{ then } \langle u, u \rangle > 0.$$

That is, a function satisfying $[I_1^*]$, $[I_2^*]$, and $[I_3^{*'}]$ is a (complex) inner product on V .

Remark 2: By $[I_2^*]$, $\langle u, u \rangle = \overline{\langle u, u \rangle}$. Thus, $\langle u, u \rangle$ must be real. By $[I_3^*]$, $\langle u, u \rangle$ must be nonnegative, and hence, its positive real square root exists. As with real inner product spaces, we define $\|u\| = \sqrt{\langle u, u \rangle}$ to be the norm or length of u .

Remark 3: In addition to the norm, we define the notions of orthogonality, orthogonal complement, and orthogonal and orthonormal sets as before. In fact, the definitions of distance and Fourier coefficient and projections are the same as in the real case.

EXAMPLE 7.15 (Complex Euclidean Space $\mathbf{C}^n$). Let $V = \mathbf{C}^n$, and let $u = (z_j)$ and $v = (w_j)$ be vectors in $\mathbf{C}^n$. Then

$$\langle u, v \rangle = \sum_k z_k \overline{w_k} = z_1 \overline{w_1} + z_2 \overline{w_2} + \cdots + z_n \overline{w_n}$$

is an inner product on V , called the *usual* or *standard inner product* on $\mathbf{C}^n$. V with this inner product is called Complex Euclidean Space. We assume this inner product on $\mathbf{C}^n$ unless otherwise stated or implied. Assuming u and v are column vectors, the above inner product may be defined by

$$\langle u, v \rangle = u^T \bar{v}$$

where, as with matrices, $\bar{v}$ means the conjugate of each element of v . If u and v are real, we have $\bar{w}_i = w_i$. In this case, the inner product reduced to the analogous one on $\mathbf{R}^n$.

EXAMPLE 7.16

- (a) Let V be the vector space of complex continuous functions on the (real) interval $a \leq t \leq b$. Then the following is the *usual inner product* on V :

$$\langle f, g \rangle = \int_a^b f(t) \overline{g(t)} dt$$

- (b) Let U be the vector space of $m \times n$ matrices over $\mathbf{C}$. Suppose $A = (z_{ij})$ and $B = (w_{ij})$ are elements of U . Then the following is the usual inner product on U :

$$\langle A, B \rangle = \text{tr}(B^H A) = \sum_{i=1}^m \sum_{j=1}^n \bar{w}_{ij} z_{ij}$$

As usual, $B^H = \bar{B}^T$; that is, B^H is the conjugate transpose of B .

The following is a list of theorems for complex inner product spaces that are analogous to those for the real case. Here a Hermitian matrix A (i.e., one where $A^H = \bar{A}^T = A$) plays the same role that a symmetric matrix A (i.e., one where $A^T = A$) plays in the real case. ([Theorem 7.18](#) is proved in [Problem 7.50](#).)

THEOREM 7.18: (Cauchy-Schwarz) Let V be a complex inner product space. Then

$$|\langle u, v \rangle| \leq \|u\| \|v\|$$

THEOREM 7.19: Let W be a subspace of a complex inner product space V . Then $V = W \oplus W^\perp$.

THEOREM 7.20: Suppose $\{u_1, u_2, \dots, u_n\}$ is a basis for a complex inner product space V . Then, for any $v \in V$,

$$v = \frac{\langle v, u_1 \rangle}{\langle u_1, u_1 \rangle} u_1 + \frac{\langle v, u_2 \rangle}{\langle u_2, u_2 \rangle} u_2 + \cdots + \frac{\langle v, u_n \rangle}{\langle u_n, u_n \rangle} u_n$$

THEOREM 7.21: Suppose $\{u_1, u_2, \dots, u_n\}$ is a basis for a complex inner product space V . Let $A = [a_{ij}]$ be the complex matrix defined by $a_{ij} = \langle u_i, u_j \rangle$. Then, for any $u, v \in V$,

$$\langle u, v \rangle = [u]^T A [\overline{v}]$$

where $[u]$ and $[v]$ are the coordinate column vectors in the given basis $\{u_i\}$. (*Remark:* This matrix A is said to represent the inner product on V .)

THEOREM 7.22: Let A be a Hermitian matrix (i.e., $A^H = \bar{A}^T = A$) such that $X^T A \bar{X}$ is real and positive for every nonzero vector $X \in \mathbf{C}^n$. Then $\langle u, v \rangle = u^T A \bar{v}$ is an inner product on $\mathbf{C}^n$.

THEOREM 7.23: Let A be the matrix that represents an inner product on V . Then A is Hermitian, and $X^T A X$ is real and positive for any nonzero vector in $\mathbf{C}^n$.

7.10 Normed Vector Spaces (Optional)

We begin with a definition.

DEFINITION: Let V be a real or complex vector space. Suppose to each $v \in V$ there is assigned a real number, denoted by $\|v\|$. This function $\|\cdot\|$ is

called a *norm* on V if it satisfies the following axioms:

[N₁] $\|v\| \geq 0$; and $\|v\| = 0$ if and only if $v = 0$.

[N₂] $\|kv\| = |k|\|v\|$.

[N₃] $\|u + v\| \leq \|u\| + \|v\|$.

A vector space V with a norm is called a *normed vector space*.

Suppose V is a normed vector space. The *distance* between two vectors u and v in V is denoted and defined by

$$d(u, v) = \|u - v\|$$

The following theorem (proved in [Problem 7.56](#)) is the main reason why $d(u, v)$ is called the distance between u and v .

THEOREM 7.24: Let V be a normed vector space. Then the function $d(u, v) = \|u - v\|$ satisfies the following three axioms of a metric space:

[M₁] $d(u, v) \geq 0$; and $d(u, v) = 0$ if and only if $u = v$.

[M₂] $d(u, v) = d(v, u)$.

[M₃] $d(u, v) \leq d(u, w) + d(w, v)$.

Normed Vector Spaces and Inner Product Spaces

Suppose V is an inner product space. Recall that the norm of a vector v in V is defined by

$$\|v\| = \sqrt{\langle v, v \rangle}$$

One can prove ([Theorem 7.2](#)) that this norm satisfies [N₁], [N₂], and [N₃]. Thus, every inner product space V is a

normed vector space. On the other hand, there may be norms on a vector space V that do not come from an inner product on V , as shown below.

Norms on $\mathbf{R}^n$ and $\mathbf{C}^n$

The following define three important norms on $\mathbf{R}^n$ and $\mathbf{C}^n$:

$$\begin{aligned}\|(a_1, \dots, a_n)\|_\infty &= \max(|a_i|) \\ \|(a_1, \dots, a_n)\|_1 &= |a_1| + |a_2| + \dots + |a_n| \\ \|(a_1, \dots, a_n)\|_2 &= \sqrt{|a_1|^2 + |a_2|^2 + \dots + |a_n|^2}\end{aligned}$$

(Note that subscripts are used to distinguish between the three norms.) The norms $\|\cdot\|_\infty$, $\|\cdot\|_1$, and $\|\cdot\|_2$ are called the *infinity-norm*, *one-norm*, and *two-norm*, respectively.

Observe that $\|\cdot\|_2$ is the norm on $\mathbf{R}^n$ (respectively, $\mathbf{C}^n$) induced by the usual inner product on $\mathbf{R}^n$ (respectively, $\mathbf{C}^n$). We will let d_∞ , d_1 , d_2 denote the corresponding distance functions.

EXAMPLE 7.17 Consider vectors $u = (1, -5, 3)$ and $v = (4, 2, -3)$ in $\mathbf{R}^3$.

(a) The infinity norm chooses the maximum of the absolute values of the components. Hence,

$$\|u\|_\infty = 5 \quad \text{and} \quad \|v\|_\infty = 4$$

(b) The one-norm adds the absolute values of the components. Thus,

$$\|u\|_1 = 1 + 5 + 3 = 9 \quad \text{and} \quad \|v\|_1 = 4 + 2 + 3 = 9$$

(c) The two-norm is equal to the square root of the sum of the squares of the components (i.e., the norm induced by

the usual inner product on $\mathbf{R}^3$). Thus,

$$\|u\|_2 = \sqrt{1 + 25 + 9} = \sqrt{35} \quad \text{and} \quad \|v\|_2 = \sqrt{16 + 4 + 9} = \sqrt{29}$$

(d) Because $u - v = (1 - 4, -5 - 2, 3 + 3) = (-3, -7, 6)$, we have

$$d_\infty(u, v) = 7, \quad d_1(u, v) = 3 + 7 + 6 = 16, \quad d_2(u, v) = \sqrt{9 + 49 + 36} = \sqrt{94}$$

EXAMPLE 7.18 Consider the Cartesian plane $\mathbf{R}^2$ shown in Fig. 7-4.

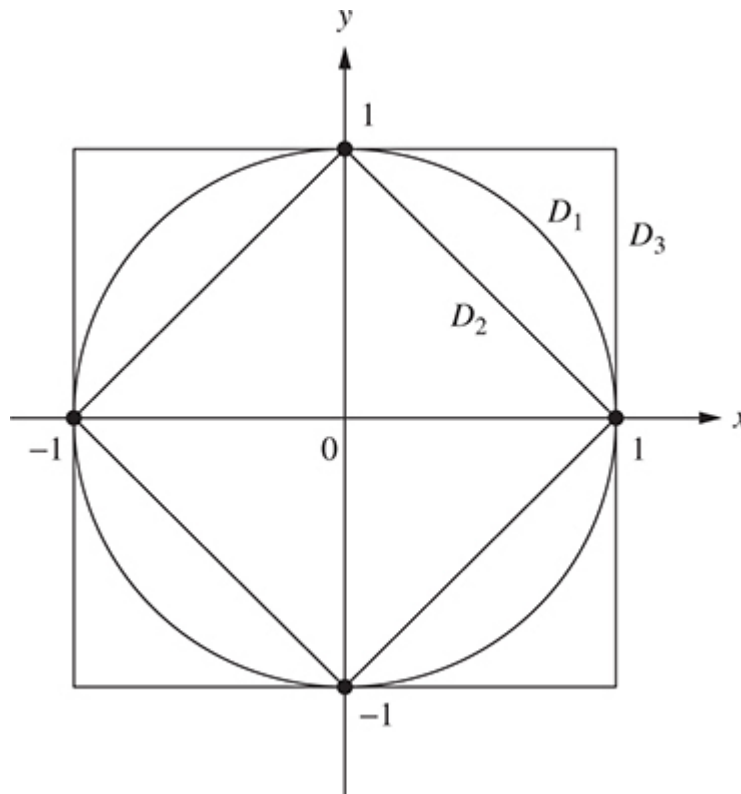


Figure 7-4

(a) Let D_1 be the set of points $u = (x, y)$ in $\mathbf{R}^2$ such that $\|u\|_2 = 1$. Then D_1 consists of the points (x, y) such that $\|u\|_2^2 = x^2 + y^2 = 1$. Thus, D_1 is the unit circle, as shown in Fig. 7-4.

- (b) Let D_2 be the set of points $u = (x, y)$ in $\mathbf{R}^2$ such that $\|u\|_1 = 1$. Then D_1 consists of the points (x, y) such that $\|u\|_1 = |x| + |y| = 1$. Thus, D_2 is the diamond inside the unit circle, as shown in [Fig. 7-4](#).
- (c) Let D_3 be the set of points $u = (x, y)$ in $\mathbf{R}^2$ such that $\|u\|_\infty = 1$. Then D_3 consists of the points (x, y) such that $\|u\|_\infty = \max(|x|, |y|) = 1$. Thus, D_3 is the square circumscribing the unit circle, as shown in [Fig. 7-4](#).

Norms on $C[a, b]$

Consider the vector space $V = C[a, b]$ of real continuous functions on the interval $a \leq t \leq b$. Recall that the following defines an inner product on V :

$$\langle f, g \rangle = \int_a^b f(t)g(t) dt$$

Accordingly, the above inner product defines the following norm on $V = C[a, b]$ (which is analogous to the $\|\cdot\|_2$ norm on $\mathbf{R}^n$):

$$\|f\|_2 = \sqrt{\int_a^b [f(t)]^2 dt}$$

The following define the other norms on $V = C[a, b]$:

$$\|f\|_1 = \int_a^b |f(t)| dt \quad \text{and} \quad \|f\|_\infty = \max(|f(t)|)$$

There are geometrical descriptions of these two norms and their corresponding distance functions, which are described below.

The first norm is pictured in [Fig. 7-5](#). Here

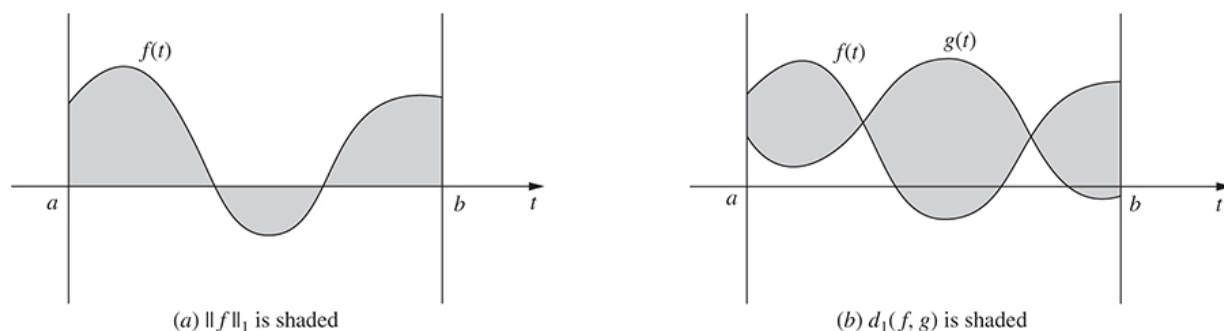


Figure 7-5

$\|f\|_1 = \text{area between the function } |f| \text{ and the } t\text{-axis}$

$d_1(f, g) = \text{area between the functions } f \text{ and } g$

This norm is analogous to the norm $\|\cdot\|_1$ on $\mathbf{R}^n$.

The second norm is pictured in [Fig. 7-6](#). Here

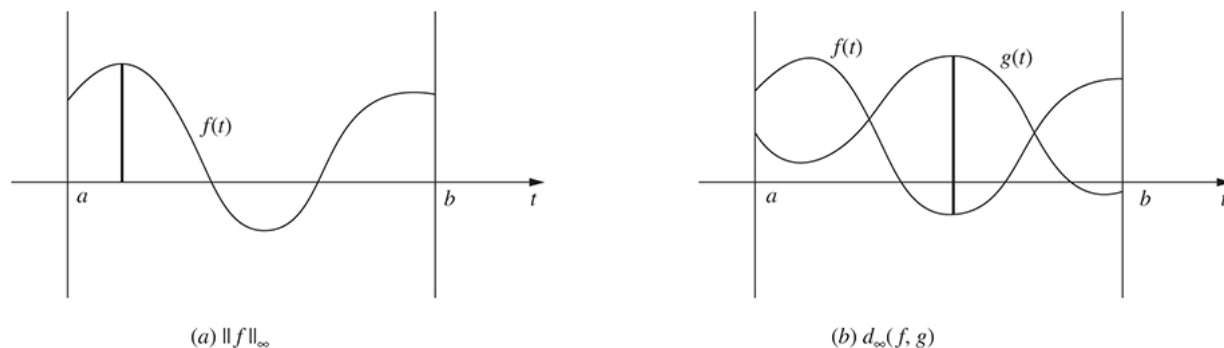


Figure 7-6

$\|f\|_\infty = \text{maximum distance between } f \text{ and the } t\text{-axis}$

$d_\infty(f, g) = \text{maximum distance between } f \text{ and } g$

This norm is analogous to the norms $\|\cdot\|_\infty$ on $\mathbf{R}^n$.

Inner Products

7.1. Expand:

(a) $\langle 5u_1 + 8u_2, 6v_1 - 7v_2 \rangle,$

(b) $\langle 3u + 5v, 4u + 6v \rangle,$

(c) $\|2u - 3v\|^2$

Use linearity in both positions and, when possible, symmetry, $\langle u, v \rangle = \langle v, u \rangle$.

(a) Take the inner product of each term on the left with each term on the right:

$$\begin{aligned}\langle 5u_1 + 8u_2, 6v_1 - 7v_2 \rangle &= \langle 5u_1, 6v_1 \rangle + \langle 5u_1, -7v_2 \rangle + \langle 8u_2, 6v_1 \rangle + \langle 8u_2, -7v_2 \rangle \\ &= 30\langle u_1, v_1 \rangle - 35\langle u_1, v_2 \rangle + 48\langle u_2, v_1 \rangle - 56\langle u_2, v_2 \rangle\end{aligned}$$

[*Remark:* Observe the similarity between the above expansion and the expansion $(5a-8b)(6c-7d)$ in ordinary algebra.]

(b) $\begin{aligned}\langle 3u + 5v, 4u - 6v \rangle &= 12\langle u, u \rangle - 18\langle u, v \rangle + 20\langle v, u \rangle - 30\langle v, v \rangle \\ &= 12\langle u, u \rangle + 2\langle u, v \rangle - 30\langle v, v \rangle\end{aligned}$

(c) $\begin{aligned}\|2u - 3v\|^2 &= \langle 2u - 3v, 2u - 3v \rangle = 4\langle u, u \rangle - 6\langle u, v \rangle - 6\langle v, u \rangle + 9\langle v, v \rangle \\ &= 4\|u\|^2 - 12\langle u, v \rangle + 9\|v\|^2\end{aligned}$

7.2. Consider vectors $u = (1, 2, 4)$, $v = (2, 3, 5)$, $w = (4, 2, 3)$ in $\mathbf{R}^3$. Find

(a) $u \cdot v$, (b) $u \cdot w$, (c) $v \cdot w$, (d) $(u + v) \cdot w$, (e) $\|u\|$, (f) $\|v\|$.

(a) Multiply corresponding components and add to get $u \cdot v = 2 + 6 + 20 = 28$.

(b) $u \cdot w = 4 + 4 - 12 = -4$.

(c) $v \cdot w = 8 - 6 - 15 = -13$.

(d) First find $u + v = (3, -1, 9)$. Then $(u + v) \cdot w = 12 - 2 - 27 = -17$. Alternatively, using $[I_1]$, $(u + v) \cdot w = u \cdot w + v \cdot w = -4 - 13 = -17$.

(e) First find $\|u\|^2$ by squaring the components of u and adding:

$$\|u\|^2 = 1^2 + 2^2 + 4^2 = 1 + 4 + 16 = 21, \quad \text{and so} \quad \|u\| = \sqrt{21}$$

(f) $\|v\|^2 = 4 + 9 + 25 = 38$, and so $\|v\| = \sqrt{38}$.

7.3. Verify that the following defines an inner product in $\mathbf{R}^2$:

$$\langle u, v \rangle = x_1 y_1 - x_1 y_2 - x_2 y_1 + 3x_2 y_2, \quad \text{where} \quad u = (x_1, x_2), \quad v = (y_1, y_2)$$

We argue via matrices. We can write $\langle u, v \rangle$ in matrix notation as follows:

$$\langle u, v \rangle = u^T A v = [x_1, x_2] \begin{bmatrix} 1 & -1 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

Because A is real and symmetric, we need only show that A is positive definite. The diagonal elements 1 and 3 are positive, and the determinant $\|A\| = 3 - 1 = 2$ is positive. Thus, by [Theorem 7.14](#), A is positive definite. Accordingly, by [Theorem 7.15](#), $\langle u, v \rangle$ is an inner product.

7.4. Consider the vectors $u = (1, 5)$ and $v = (3, 4)$ in $\mathbf{R}^2$. Find

(a) $\langle u, v \rangle$ with respect to the usual inner product in $\mathbf{R}^2$.

(b) $\langle u, v \rangle$ with respect to the inner product in $\mathbf{R}^2$ in [Problem 7.3](#).

(c) $\|v\|$ using the usual inner product in $\mathbf{R}^2$.

(d) $\|v\|$ using the inner product in $\mathbf{R}^2$ in [Problem 7.3](#).

(a) $\langle u, v \rangle = 3 + 20 = 23$.

(b) $\langle u, v \rangle = 1 \cdot 3 - 1 \cdot 4 - 5 \cdot 3 + 3 \cdot 5 \cdot 4 = 3 - 4 - 15 + 60 = 44$.

(c) $\|v\|^2 = \langle v, v \rangle = \langle (3, 4), (3, 4) \rangle = 9 + 16 = 25$; hence, $\|v\| = 5$.

(d) $\|v\|^2 = \langle v, v \rangle = \langle (3, 4), (3, 4) \rangle = 9 - 12 - 12 + 48 = 33$; hence, $\|v\| = \sqrt{33}$.

7.5. Consider the following polynomials in $\mathbf{P}(t)$ with the inner product $\langle f, g \rangle = \int_0^1 f(t)g(t) dt$:

$$f(t) = t + 2, \quad g(t) = 3t - 2, \quad h(t) = t^2 - 2t - 3$$

(a) Find $\langle f, g \rangle$ and $\langle f, h \rangle$.

(b) Find $\|f\|$ and $\|g\|$.

(c) Normalize f and g .

(a) Integrate as follows:

$$\langle f, g \rangle = \int_0^1 (t+2)(3t-2) dt = \int_0^1 (3t^2 + 4t - 4) dt = \left(t^3 + 2t^2 - 4t \right) \Big|_0^1 = -1$$

$$\langle f, h \rangle = \int_0^1 (t+2)(t^2 - 2t - 3) dt = \left(\frac{t^4}{4} - \frac{7t^2}{2} - 6t \right) \Big|_0^1 = -\frac{37}{4}$$

(b)

$$\langle f, f \rangle = \int_0^1 (t+2)(t+2) dt = \frac{19}{3}; \quad \text{hence,} \quad \|f\| = \sqrt{\frac{19}{3}} = \frac{1}{3}\sqrt{57}$$

$$\langle g, g \rangle = \int_0^1 (3t-2)(3t-2) dt = 1; \quad \text{hence,} \quad \|g\| = \sqrt{1} = 1$$

(c) Because $\|f\| = \frac{1}{3}\sqrt{57}$ and g is already a unit vector, we have

$$\hat{f} = \frac{1}{\|f\|} f = \frac{3}{\sqrt{57}}(t+2) \quad \text{and} \quad \hat{g} = g = 3t-2$$

7.6. Find $\cos \theta$ where θ is the angle between:

(a) $u = (1, 3, -5, 4)$ and $v = (2, -3, 4, 1)$ in $\mathbf{R}^4$,

(b) $A = \begin{bmatrix} 9 & 8 & 7 \\ 6 & 5 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$, where $\langle A, B \rangle = \text{tr}(B^T A)$.

$$\text{Use } \cos \theta = \frac{\langle u, v \rangle}{\|u\| \|v\|}$$

(a) Compute:

$$\langle u, v \rangle = 2 - 9 - 20 + 4 = -23, \quad \|u\|^2 = 1 + 9 + 25 + 16 = 51, \quad \|v\|^2 = 4 + 9 + 16 + 1 = 30$$

Thus,

$$\cos \theta = \frac{-23}{\sqrt{51}\sqrt{30}} = \frac{-23}{3\sqrt{170}}$$

(b) Use $\langle A, B \rangle = \text{tr}(B^T A) = \sum_{i=1}^m \sum_{j=1}^n a_{ij} b_{ij}$, the sum of the products of corresponding entries.

$$\langle A, B \rangle = 9 + 16 + 21 + 24 + 25 + 24 = 119$$

Use $\|A\|^2 = \langle A, A \rangle = \sum_{i=1}^m \sum_{j=1}^n a_{ij}^2$, the sum of the squares of all the elements of A .

$$\|A\|^2 = \langle A, A \rangle = 9^2 + 8^2 + 7^2 + 6^2 + 5^2 + 4^2 = 271, \quad \text{and so} \quad \|A\| = \sqrt{271}$$

$$\|B\|^2 = \langle B, B \rangle = 1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2 = 91, \quad \text{and so} \quad \|B\| = \sqrt{91}$$

Thus,

$$\cos \theta = \frac{119}{\sqrt{271}\sqrt{91}}$$

7.7. Verify each of the following:

- (a) Parallelogram Law ([Fig. 7-7](#)): $\|u + v\|^2 + \|u - v\|^2 = 2\|u\|^2 + 2\|v\|^2$.

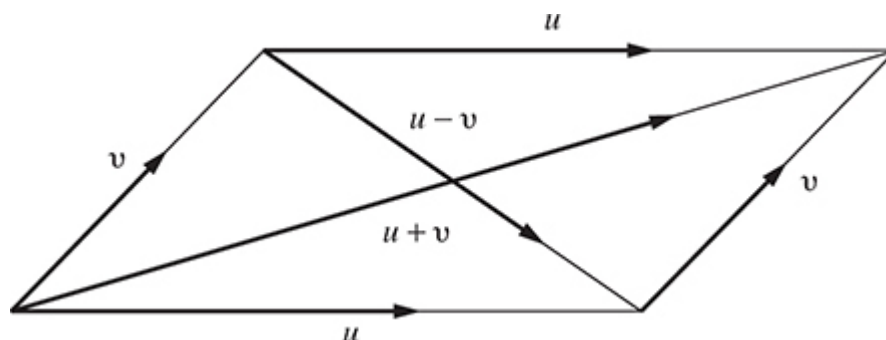


Figure 7-7

- (b) Polar form for $\langle u, v \rangle$ (which shows the inner product can be obtained from the norm function):

$$\langle u, v \rangle = \frac{1}{4}(\|u + v\|^2 - \|u - v\|^2).$$

Expand as follows to obtain

$$\|u + v\|^2 = \langle u + v, u + v \rangle = \|u\|^2 + 2\langle u, v \rangle + \|v\|^2 \quad (1)$$

$$\|u - v\|^2 = \langle u - v, u - v \rangle = \|u\|^2 - 2\langle u, v \rangle + \|v\|^2 \quad (2)$$

Add (1) and (2) to get the Parallelogram Law (a).

Subtract (2) from (1) to obtain

$$\|u + v\|^2 - \|u - v\|^2 = 4\langle u, v \rangle$$

Divide by 4 to obtain the (real) polar form (b).

7.8. Prove [Theorem 7.1](#) (Cauchy-Schwarz): For u and v in a real inner product space V , $\langle u, u \rangle^2 \leq \langle u, u \rangle \langle v, v \rangle$ or $|\langle u, v \rangle| \leq \|u\| \|v\|$.

For any real number t ,

$$\langle tu + v, tu + v \rangle = t^2 \langle u, u \rangle + 2t \langle u, v \rangle + \langle v, v \rangle = t^2 \|u\|^2 + 2t \langle u, v \rangle + \|v\|^2$$

Let $a = \|u\|^2$, $b = 2\langle u, v \rangle$, $c = \|v\|^2$. Because $\|tu + v\|^2 \geq 0$, we have

$$at^2 + bt + c \geq 0$$

for every value of t . This means that the quadratic polynomial cannot have two real roots, which implies that $b^2 - 4ac \leq 0$ or $b^2 \leq 4ac$. Thus,

$$4\langle u, v \rangle^2 \leq 4\|u\|^2 \|v\|^2$$

Dividing by 4 gives our result.

7.9. Prove [Theorem 7.2](#): The norm in an inner product space V satisfies

(a) $[N_1]$ $\|v\| \geq 0$; and $\|v\| = 0$ if and only if $v = 0$.

(b) $[N_2]$ $\|kv\| = |k| \|v\|$.

(c) $[N_3]$ $\|u + v\| \leq \|u\| + \|v\|$.

(a) If $v \neq 0$, then $\langle v, v \rangle > 0$, and hence, $\|v\| = \sqrt{\langle v, v \rangle} > 0$.
If $v = 0$, then $\langle 0, 0 \rangle = 0$. Consequently, $\|0\| = \sqrt{0} = 0$.
Thus, $[N_1]$ is true.

(b) We have $\|kv\|^2 = \langle kv, kv \rangle = k^2 \langle v, v \rangle = k^2 \|v\|^2$.
Taking the square root of both sides gives $[N_2]$.

(c) Using the Cauchy-Schwarz inequality, we obtain

$$\begin{aligned}\|u + v\|^2 &= \langle u + v, u + v \rangle = \langle u, u \rangle + \langle u, v \rangle + \langle u, v \rangle + \langle v, v \rangle \\ &\leq \|u\|^2 + 2\|u\|\|v\| + \|v\|^2 = (\|u\| + \|v\|)^2\end{aligned}$$

Taking the square root of both sides yields [N₃].

Orthogonality, Orthonormal Complements, Orthogonal Sets

- 7.10.** Find k so that $u = (1, 2, k, 3)$ and $v = (3, k, 7, -5)$ in $\mathbf{R}^4$ are orthogonal.

First find

$$\langle u, v \rangle = (1, 2, k, 3) \cdot (3, k, 7, -5) = 3 + 2k + 7k - 15 = 9k - 12$$

Then set $\langle u, v \rangle = 9k - 12 = 0$ to obtain $k = \frac{4}{3}$.

- 7.11.** Let W be the subspace of $\mathbf{R}^5$ spanned by $u = (1, 2, 3, -1, 2)$ and $v = (2, 4, 7, 2, -1)$. Find a basis of the orthogonal complement $W^\perp$ of W .

We seek all vectors $w = (x, y, z, s, t)$ such that

$$\langle w, u \rangle = x + 2y + 3z - s + 2t = 0$$

$$\langle w, v \rangle = 2x + 4y + 7z + 2s - t = 0$$

Eliminating x from the second equation, we find the equivalent system

$$\begin{aligned}x + 2y + 3z - s + 2t &= 0 \\ z + 4s - 5t &= 0\end{aligned}$$

The free variables are y , s , and t . Therefore,

- (1) Set $y = -1$, $s = 0$, $t = 0$ to obtain the solution $w_1 = (2, -1, 0, 0, 0)$.
- (2) Set $y = 0$, $s = 1$, $t = 0$ to find the solution $w_2 = (13, 0, -4, 1, 0)$.

- (3) Set $y = 0$, $s = 0$, $t = 1$ to obtain the solution $w_3 = (-17, 0, 5, 0, 1)$.

The set $\{w_1, w_2, w_3\}$ is a basis of $W^\perp$.

7.12. Let $w = (1, 2, 3, 1)$ be a vector in $\mathbf{R}^4$. Find an orthogonal basis for $w^\perp$.

Find a nonzero solution of $x + 2y + 3z + t = 0$, say $v_1 = (0, 0, 1, -3)$. Now find a nonzero solution of the system

$$x + 2y + 3z + t = 0, \quad z - 3t = 0$$

say $v_2 = (0, -5, 3, 1)$. Last, find a nonzero solution of the system

$$x + 2y + 3z + t = 0, \quad -5y + 3z + t = 0, \quad z - 3t = 0$$

say $v_3 = (-14, 2, 3, 1)$. Thus, v_1, v_2, v_3 form an orthogonal basis for $w^\perp$.

7.13. Let S consist of the following vectors in $\mathbf{R}^4$:

$$u_1 = (1, 1, 0, -1), \quad u_2 = (1, 2, 1, 3), \quad u_3 = (1, 1, -9, 2), \quad u_4 = (16, -13, 1, 3)$$

(a) Show that S is orthogonal and a basis of $\mathbf{R}^4$.

(b) Find the coordinates of an arbitrary vector $v = (a, b, c, d)$ in $\mathbf{R}^4$ relative to the basis S .

(a) Compute

$$\begin{aligned} u_1 \cdot u_2 &= 1 + 2 + 0 - 3 = 0, & u_1 \cdot u_3 &= 1 + 1 + 0 - 2 = 0, & u_1 \cdot u_4 &= 16 - 13 + 0 - 3 = 0 \\ u_2 \cdot u_3 &= 1 + 2 - 9 + 6 = 0, & u_2 \cdot u_4 &= 16 - 26 + 1 + 9 = 0, & u_3 \cdot u_4 &= 16 - 13 - 9 + 6 = 0 \end{aligned}$$

Thus, S is orthogonal, and S is linearly independent. Accordingly, S is a basis for $\mathbf{R}^4$ because any four linearly independent vectors form a basis of $\mathbf{R}^4$.

(b) Because S is orthogonal, we need only find the Fourier coefficients of v with respect to the basis vectors, as in [Theorem 7.7](#). Thus,

$$\begin{aligned} k_1 &= \frac{\langle v, u_1 \rangle}{\langle u_1, u_1 \rangle} = \frac{a + b - d}{3}, & k_3 &= \frac{\langle v, u_3 \rangle}{\langle u_3, u_3 \rangle} = \frac{a + b - 9c + 2d}{87} \\ k_2 &= \frac{\langle v, u_2 \rangle}{\langle u_2, u_2 \rangle} = \frac{a + 2b + c + 3d}{15}, & k_4 &= \frac{\langle v, u_4 \rangle}{\langle u_4, u_4 \rangle} = \frac{16a - 13b + c + 3d}{435} \end{aligned}$$

are the coordinates of v with respect to the basis S .

7.14. Suppose S, S_1, S_2 are the subsets of V . Prove the following (where $S^{\perp\perp}$ means $(S^\perp)^\perp$):

(a) $S \subseteq S^{\perp\perp}$.

(b) If $S_1 \subseteq S_2$, then $S_2^\perp \subseteq S_1^\perp$.

(c) $S^\perp = \text{span}(S)^\perp$.

(a) Let $w \in S$. Then $\langle w, v \rangle = 0$ for every $v \in S^\perp$; hence, $w \in S^{\perp\perp}$. Accordingly, $S \subseteq S^{\perp\perp}$.

(b) Let $w \in S_2^\perp$. Then $\langle w, v \rangle = 0$ for every $v \in S_2$. Because $S_1 \subseteq S_2$, $\langle w, v \rangle = 0$ for every $v \in S_1$. Thus, $w \in S_1^\perp$, and hence, $S_2^\perp \subseteq S_1^\perp$.

(c) Because $S \subseteq \text{span}(S)$, part (b) gives us $\text{span}(S)^\perp \subseteq S^\perp$. Suppose $u \in S^\perp$ and $v \in \text{span}(S)$. Then there exist $w_1, w_2, \dots, w_k$ in S such that $v = a_1 w_1 + a_2 w_2 + \dots + a_k w_k$. Then, using $u \in S^\perp$, we have

$$\begin{aligned} \langle u, v \rangle &= \langle u, a_1 w_1 + a_2 w_2 + \dots + a_k w_k \rangle = a_1 \langle u, w_1 \rangle + a_2 \langle u, w_2 \rangle + \dots + a_k \langle u, w_k \rangle \\ &= a_1(0) + a_2(0) + \dots + a_k(0) = 0 \end{aligned}$$

Thus, $u \in \text{span}(S)^\perp$. Accordingly, $S^\perp \subseteq \text{span}(S)^\perp$. Both inclusions give $S^\perp = \text{span}(S)^\perp$.

7.15. Prove [Theorem 7.5](#): Suppose S is an orthogonal set of nonzero vectors. Then S is linearly independent.

Suppose $S = \{u_1, u_2, \dots, u_r\}$ and suppose

$$a_1u_1 + a_2u_2 + \cdots + a_ru_r = 0 \quad (1)$$

Taking the inner product of (1) with u_1 , we get

$$\begin{aligned} 0 &= \langle 0, u_1 \rangle = \langle a_1u_1 + a_2u_2 + \cdots + a_ru_r, u_1 \rangle \\ &= a_1\langle u_1, u_1 \rangle + a_2\langle u_2, u_1 \rangle + \cdots + a_r\langle u_r, u_1 \rangle \\ &= a_1\langle u_1, u_1 \rangle + a_2 \cdot 0 + \cdots + a_r \cdot 0 = a_1\langle u_1, u_1 \rangle \end{aligned}$$

Because $u_1 \neq 0$, we have $\langle u_1, u_1 \rangle \neq 0$. Thus, $a_1 = 0$. Similarly, for $i = 2, \dots, r$, taking the inner product of (1) with u_i ,

$$\begin{aligned} 0 &= \langle 0, u_i \rangle = \langle a_1u_1 + \cdots + a_ru_r, u_i \rangle \\ &= a_1\langle u_1, u_i \rangle + \cdots + a_i\langle u_i, u_i \rangle + \cdots + a_r\langle u_r, u_i \rangle = a_i\langle u_i, u_i \rangle \end{aligned}$$

But $\langle u_i, u_i \rangle \neq 0$, and hence, every $a_i = 0$. Thus, S is linearly independent.

7.16. Prove [Theorem 7.6](#) (Pythagoras): Suppose $\{u_1, u_2, \dots, u_r\}$ is an orthogonal set of vectors. Then

$$\|u_1 + u_2 + \cdots + u_r\|^2 = \|u_1\|^2 + \|u_2\|^2 + \cdots + \|u_r\|^2$$

Expanding the inner product, we have

$$\begin{aligned} \|u_1 + u_2 + \cdots + u_r\|^2 &= \langle u_1 + u_2 + \cdots + u_r, u_1 + u_2 + \cdots + u_r \rangle \\ &= \langle u_1, u_1 \rangle + \langle u_2, u_2 \rangle + \cdots + \langle u_r, u_r \rangle + \sum_{i \neq j} \langle u_i, u_j \rangle \end{aligned}$$

The theorem follows from the fact that $\langle u_i, u_i \rangle = \|u_i\|^2$ and $\langle u_i, u_j \rangle = 0$ for $i \neq j$.

7.17. Prove [Theorem 7.7](#): Let $\{u_1, u_2, \dots, u_n\}$ be an orthogonal basis of V . Then for any $v \in V$,

$$v = \frac{\langle v, u_1 \rangle}{\langle u_1, u_1 \rangle} u_1 + \frac{\langle v, u_2 \rangle}{\langle u_2, u_2 \rangle} u_2 + \dots + \frac{\langle v, u_n \rangle}{\langle u_n, u_n \rangle} u_n$$

Suppose $v = k_1 u_1 + k_2 u_2 + \dots + k_n u_n$. Taking the inner product of both sides with u_1 yields

$$\begin{aligned} \langle v, u_1 \rangle &= \langle k_1 u_1 + k_2 u_2 + \dots + k_n u_n, u_1 \rangle \\ &= k_1 \langle u_1, u_1 \rangle + k_2 \langle u_2, u_1 \rangle + \dots + k_n \langle u_n, u_1 \rangle \\ &= k_1 \langle u_1, u_1 \rangle + k_2 \cdot 0 + \dots + k_n \cdot 0 = k_1 \langle u_1, u_1 \rangle \end{aligned}$$

Thus, $k_1 = \frac{\langle v, u_1 \rangle}{\langle u_1, u_1 \rangle}$. Similarly, for $i = 2, \dots, n$,

$$\begin{aligned} \langle v, u_i \rangle &= \langle k_1 u_1 + k_2 u_2 + \dots + k_n u_n, u_i \rangle \\ &= k_1 \langle u_1, u_i \rangle + k_2 \langle u_2, u_i \rangle + \dots + k_n \langle u_n, u_i \rangle \\ &= k_1 \cdot 0 + \dots + k_i \langle u_i, u_i \rangle + \dots + k_n \cdot 0 = k_i \langle u_i, u_i \rangle \end{aligned}$$

Thus, $k_i = \frac{\langle v, u_i \rangle}{\langle u_i, u_i \rangle}$. Substituting for k_i in the equation $v = k_1 u_1 + \dots + k_n u_n$, we obtain the desired result.

7.18. Suppose $E = \{e_1, e_2, \dots, e_n\}$ is an orthonormal basis of V . Prove

(a) For any $u \in V$, we have $u = \langle u, e_1 \rangle e_1 + \langle u, e_2 \rangle e_2 + \dots + \langle u, e_n \rangle e_n$.

(b) $\langle a_1 e_1 + \dots + a_n e_n, b_1 e_1 + \dots + b_n e_n \rangle = a_1 b_1 + a_2 b_2 + \dots + a_n b_n$.

(c) For any $u, v \in V$, we have $\langle u, v \rangle = \langle u, e_1 \rangle \langle v, e_1 \rangle + \dots + \langle u, e_n \rangle \langle v, e_n \rangle$.

(a) Suppose $u = k_1 e_1 + k_2 e_2 + \dots + k_n e_n$. Taking the inner product of u with e_1 ,

$$\begin{aligned}\langle u, e_1 \rangle &= \langle k_1 e_1 + k_2 e_2 + \dots + k_n e_n, e_1 \rangle \\ &= k_1 \langle e_1, e_1 \rangle + k_2 \langle e_2, e_1 \rangle + \dots + k_n \langle e_n, e_1 \rangle \\ &= k_1(1) + k_2(0) + \dots + k_n(0) = k_1\end{aligned}$$

Similarly, for $i = 2, \dots, n$,

$$\begin{aligned}\langle u, e_i \rangle &= \langle k_1 e_1 + \dots + k_i e_i + \dots + k_n e_n, e_i \rangle \\ &= k_1 \langle e_1, e_i \rangle + \dots + k_i \langle e_i, e_i \rangle + \dots + k_n \langle e_n, e_i \rangle \\ &= k_1(0) + \dots + k_i(1) + \dots + k_n(0) = k_i\end{aligned}$$

Substituting $\langle u, e_i \rangle$ for k_i in the equation $u = k_1 e_1 + \dots + k_n e_n$, we obtain the desired result.

(b) We have

$$\left\langle \sum_{i=1}^n a_i e_i, \sum_{j=1}^n b_j e_j \right\rangle = \sum_{i,j=1}^n a_i b_j \langle e_i, e_j \rangle = \sum_{i=1}^n a_i b_i \langle e_i, e_i \rangle + \sum_{i \neq j} a_i b_j \langle e_i, e_j \rangle$$

But $\langle e_i, e_j \rangle = 0$ for $i \neq j$, and $\langle e_i, e_j \rangle = 1$ for $i = j$.

Hence, as required,

$$\left\langle \sum_{i=1}^n a_i e_i, \sum_{j=1}^n b_j e_j \right\rangle = \sum_{i=1}^n a_i b_i = a_1 b_1 + a_2 b_2 + \dots + a_n b_n$$

(c) By part (a), we have

$$u = \langle u, e_1 \rangle e_1 + \dots + \langle u, e_n \rangle e_n \quad \text{and} \quad v = \langle v, e_1 \rangle e_1 + \dots + \langle v, e_n \rangle e_n$$

Thus, by part (b),

$$\langle u, v \rangle = \langle u, e_1 \rangle \langle v, e_1 \rangle + \langle u, e_2 \rangle \langle v, e_2 \rangle + \cdots + \langle u, e_n \rangle \langle v, e_n \rangle$$

Projections, Gram-Schmidt Algorithm, Applications

7.19. Suppose $w \neq 0$. Let v be any vector in V . Show that

$$c = \frac{\langle v, w \rangle}{\langle w, w \rangle} = \frac{\langle v, w \rangle}{\|w\|^2}$$

is the unique scalar such that $v' = v - cw$ is orthogonal to w .

In order for v' to be orthogonal to w we must have

$$\langle v - cw, w \rangle = 0 \quad \text{or} \quad \langle v, w \rangle - c\langle w, w \rangle = 0 \quad \text{or} \quad \langle v, w \rangle = c\langle w, w \rangle$$

Thus, $c = \frac{\langle v, w \rangle}{\langle w, w \rangle}$. Conversely, suppose $c = \frac{\langle v, w \rangle}{\langle w, w \rangle}$. Then

$$\langle v - cw, w \rangle = \langle v, w \rangle - c\langle w, w \rangle = \langle v, w \rangle - \frac{\langle v, w \rangle}{\langle w, w \rangle} \langle w, w \rangle = 0$$

7.20. Find the Fourier coefficient c and the projection of $v = (1, -2, 3, -4)$ along $w = (1, 2, 1, 2)$ in $\mathbf{R}^4$.

Compute $\langle v, w \rangle = 1 - 4 + 3 - 8 = -8$ and $\|w\|^2 = 1 + 4 + 1 + 4 = 10$. Then

$$c = -\frac{8}{10} = -\frac{4}{5} \quad \text{and} \quad \text{proj}(v, w) = cw = \left(-\frac{4}{5}, -\frac{8}{5}, -\frac{4}{5}, -\frac{8}{5}\right)$$

7.21. Consider the subspace U of $\mathbf{R}^4$ spanned by the vectors:

$$v_1 = (1, 1, 1, 1), \quad v_2 = (1, 1, 2, 4), \quad v_3 = (1, 2, -4, -3)$$

Find (a) an orthogonal basis of U ; (b) an orthonormal basis of U .

(a) Use the Gram-Schmidt algorithm. Begin by setting $w_1 = u = (1, 1, 1, 1)$. Next find

$$v_2 - \frac{\langle v_2, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1 = (1, 1, 2, 4) - \frac{8}{4}(1, 1, 1, 1) = (-1, -1, 0, 2)$$

Set $w_2 = (-1, -1, 0, 2)$. Then find

$$\begin{aligned} v_3 - \frac{\langle v_3, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1 - \frac{\langle v_3, w_2 \rangle}{\langle w_2, w_2 \rangle} w_2 &= (1, 2, -4, -3) - \frac{(-4)}{4}(1, 1, 1, 1) - \frac{(-9)}{6}(-1, -1, 0, 2) \\ &= \left(\frac{1}{2}, \frac{3}{2}, -3, 1\right) \end{aligned}$$

Clear fractions to obtain $w_3 = (1, 3, -6, 2)$. Then w_1, w_2, w_3 form an orthogonal basis of U .

(b) Normalize the orthogonal basis consisting of w_1, w_2, w_3 . Because $\|w_1\|^2 = 4$, $\|w_2\|^2 = 6$, and $\|w_3\|^2 = 50$, the following vectors form an orthonormal basis of U :

$$u_1 = \frac{1}{2}(1, 1, 1, 1), \quad u_2 = \frac{1}{\sqrt{6}}(-1, -1, 0, 2), \quad u_3 = \frac{1}{5\sqrt{2}}(1, 3, -6, 2)$$

7.22. Consider the vector space $\mathbf{P}(t)$ with inner product $\langle f, g \rangle = \int_0^1 f(t)g(t) dt$. Apply the Gram-Schmidt algorithm to the set $\{1, t, t^2\}$ to obtain an orthogonal set $\{f_0, f_1, f_2\}$ with integer coefficients.

First set $f_0 = 1$. Then find

$$t - \frac{\langle t, 1 \rangle}{\langle 1, 1 \rangle} \cdot 1 = t - \frac{\frac{1}{2}}{1} \cdot 1 = t - \frac{1}{2}$$

Clear fractions to obtain $f_1 = 2t - 1$. Then find

$$t^2 - \frac{\langle t^2, 1 \rangle}{\langle 1, 1 \rangle} (1) - \frac{\langle t^2, 2t - 1 \rangle}{\langle 2t - 1, 2t - 1 \rangle} (2t - 1) = t^2 - \frac{1}{3} (1) - \frac{1}{3} (2t - 1) = t^2 - t + \frac{1}{6}$$

Clear fractions to obtain $f_2 = 6t^2 - 6t + 1$. Thus, $\{1, 2t - 1, 6t^2 - 6t + 1\}$ is the required orthogonal set.

7.23. Suppose $v = (1, 3, 5, 7)$. Find the projection of v onto W or, in other words, find $w \in W$ that minimizes $\|v - w\|$, where W is the subspace of $\mathbf{R}^4$ spanned by

(a) $u_1 = (1, 1, 1, 1)$ and $u_2 = (1, -3, 4, -2)$,

(b) $v_1 = (1, 1, 1, 1)$ and $v_2 = (1, 2, 3, 2)$.

(a) Because u_1 and u_2 are orthogonal, we need only compute the Fourier coefficients:

$$c_1 = \frac{\langle v, u_1 \rangle}{\langle u_1, u_1 \rangle} = \frac{1 + 3 + 5 + 7}{1 + 1 + 1 + 1} = \frac{16}{4} = 4$$

$$c_2 = \frac{\langle v, u_2 \rangle}{\langle u_2, u_2 \rangle} = \frac{1 - 9 + 20 - 14}{1 + 9 + 16 + 4} = \frac{-2}{30} = -\frac{1}{15}$$

Then

$$w = \text{proj}(v, W) = c_1 u_1 + c_2 u_2 = 4(1, 1, 1, 1) - \frac{1}{15}(1, -3, 4, -2) = \left(\frac{59}{15}, \frac{63}{5}, \frac{56}{15}, \frac{62}{15}\right)$$

(b) Because v_1 and v_2 are not orthogonal, first apply the Gram-Schmidt algorithm to find an orthogonal basis for W . Set $w_1 = v_1 = (1, 1, 1, 1)$. Then find

$$v_2 - \frac{\langle v_2, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1 = (1, 2, 3, 2) - \frac{8}{4}(1, 1, 1, 1) = (-1, 0, 1, 0)$$

Set $w_2 = (-1, 0, 1, 0)$. Now compute

$$c_1 = \frac{\langle v, w_1 \rangle}{\langle w_1, w_1 \rangle} = \frac{1 + 3 + 5 + 7}{1 + 1 + 1 + 1} = \frac{16}{4} = 4$$

$$c_2 = \frac{\langle v, w_2 \rangle}{\langle w_2, w_2 \rangle} = \frac{-1 + 0 + 5 + 0}{1 + 0 + 1 + 0} = \frac{4}{2} = 2$$

Then $w = \text{proj}(v, W) = c_1 w_1 + c_2 w_2 = 4(1, 1, 1, 1) + 2(-1, 0, 1, 0) = (2, 4, 6, 4)$.

7.24. Suppose w_1 and w_2 are nonzero orthogonal vectors. Let v be any vector in V . Find c_1 and c_2 so that v' is orthogonal to w_1 and w_2 , where $v' = v - c_1 w_1 - c_2 w_2$.

If v' is orthogonal to w_1 , then

$$0 = \langle v - c_1 w_1 - c_2 w_2, w_1 \rangle = \langle v, w_1 \rangle - c_1 \langle w_1, w_1 \rangle - c_2 \langle w_2, w_1 \rangle$$

$$= \langle v, w_1 \rangle - c_1 \langle w_1, w_1 \rangle - c_2 0 = \langle v, w_1 \rangle - c_1 \langle w_1, w_1 \rangle$$

Thus, $c_1 = \langle v, w_1 \rangle / \langle w_1, w_1 \rangle$. (That is, c_1 is the component of v along w_1 .) Similarly, if v' is orthogonal to w_2 , then

$$0 = \langle v - c_1 w_1 - c_2 w_2, w_2 \rangle = \langle v, w_2 \rangle - c_2 \langle w_2, w_2 \rangle$$

Thus, $c_2 = \langle v, w_2 \rangle / \langle w_2, w_2 \rangle$. (That is, c_2 is the component of v along w_2 .)

7.25. Prove [Theorem 7.8](#): Suppose $w_1, w_2, \dots, w_r$ form an orthogonal set of nonzero vectors in V . Let $v \in V$. Define

$$v' = v - (c_1 w_1 + c_2 w_2 + \dots + c_r w_r), \quad \text{where} \quad c_i = \frac{\langle v, w_i \rangle}{\langle w_i, w_i \rangle}$$

Then v' is orthogonal to $w_1, w_2, \dots, w_r$.

For $i = 1, 2, \dots, r$ and using $\langle w_i, w_j \rangle = 0$ for $i \neq j$, we have

$$\begin{aligned}\langle v - c_1 w_1 - c_2 w_2 - \dots - c_r w_r, w_i \rangle &= \langle v, w_i \rangle - c_1 \langle w_1, w_i \rangle - \dots - c_i \langle w_i, w_i \rangle - \dots - c_r \langle w_r, w_i \rangle \\ &= \langle v, w_i \rangle - c_1 \cdot 0 - \dots - c_i \langle w_i, w_i \rangle - \dots - c_r \cdot 0 \\ &= \langle v, w_i \rangle - c_i \langle w_i, w_i \rangle = \langle v, w_i \rangle - \frac{\langle v, w_i \rangle}{\langle w_i, w_i \rangle} \langle w_i, w_i \rangle = 0\end{aligned}$$

The theorem is proved.

7.26. Prove [Theorem 7.9](#): Let $\{v_1, v_2, \dots, v_n\}$ be any basis of an inner product space V . Then there exists an orthonormal basis $\{u_1, u_2, \dots, u_n\}$ of V such that the change-of-basis matrix from $\{v_i\}$ to $\{u_i\}$ is *triangular*; that is, for $k = 1, 2, \dots, n$,

$$u_k = a_{k1} v_1 + a_{k2} v_2 + \dots + a_{kk} v_k$$

The proof uses the Gram-Schmidt algorithm and Remarks 1 and 3 of Section 7.7. That is, apply the algorithm to $\{v_i\}$ to obtain an orthogonal basis $\{w_1, \dots, w_n\}$, and then normalize $\{w_i\}$ to obtain an orthonormal basis $\{u_i\}$ of V . The specific algorithm guarantees that each w_k is a linear combination of $v_1, \dots, v_k$, and hence, each u_k is a linear combination of $v_1, \dots, v_k$.

7.27. Prove [Theorem 7.10](#): Suppose $S = \{w_1, w_2, \dots, w_r\}$, is an orthogonal basis for a subspace W of V . Then one may extend S to an orthogonal basis for V ; that is, one may find vectors $w_{r+1}, \dots, w_n$ such that $\{w_1, w_2, \dots, w_n\}$ is an orthogonal basis for V .

Extend S to a basis $S' = \{w_1, \dots, w_r, v_{r+1}, \dots, v_n\}$ for V . Applying the Gram-Schmidt algorithm to S' , we

first obtain $w_1, w_2, \dots, w_r$ because S is orthogonal, and then we obtain vectors $w_{r+1}, \dots, w_n$, where $\{w_1, w_2, \dots, w_n\}$ is an orthogonal basis for V . Thus, the theorem is proved.

7.28. Prove [Theorem 7.4](#): Let W be a subspace of V . Then $V = W \oplus W^\perp$.

By [Theorem 7.9](#), there exists an orthogonal basis $\{u_1, \dots, u_r\}$ of W , and by [Theorem 7.10](#) we can extend it to an orthogonal basis $\{u_1, u_2, \dots, u_n\}$ of V . Hence, $u_{r+1}, \dots, u_n \in W^\perp$. If $v \in V$, then

$$v = a_1 u_1 + \dots + a_n u_n, \text{ where } a_1 u_1 + \dots + a_r u_r \in W \text{ and } a_{r+1} u_{r+1} + \dots + a_n u_n \in W^\perp$$

Accordingly, $V = W + W^\perp$.

On the other hand, if $w \in W \cap W^\perp$, then $\langle w, w \rangle = 0$. This yields $w = 0$. Hence, $W \cap W^\perp = \{0\}$.

The two conditions $V = W + W^\perp$ and $W \cap W^\perp = \{0\}$ give the desired result $V = W \oplus W^\perp$.

Remark: Note that we have proved the theorem for the case that V has finite dimension. We remark that the theorem also holds for spaces of arbitrary dimension.

7.29. Suppose W is a subspace of a finite-dimensional space V . Prove that $W = W^{\perp\perp}$.

By [Theorem 7.4](#), $V = W \oplus W^\perp$, and also $V = W^\perp \oplus W^{\perp\perp}$. Hence,

$$\dim W = \dim V - \dim W^\perp \quad \text{and} \quad \dim W^{\perp\perp} = \dim V - \dim W^\perp$$

This yields $\dim W = \dim W^{\perp\perp}$. But $W \subseteq W^{\perp\perp}$ (see [Problem 7.14](#)). Hence, $W = W^{\perp\perp}$, as required.

7.30. Prove the following: Suppose $w_1, w_2, \dots, w_r$ form an orthogonal set of nonzero vectors in V . Let v be any vector in V and let c_i be the component of v along w_i . Then, for any scalars $a_1, \dots, a_r$ we have

$$\left\| v - \sum_{k=1}^r c_k w_k \right\| \leq \left\| v - \sum_{k=1}^r a_k w_k \right\|$$

That is, $\sum c_i w_i$ is the closest approximation to v as a linear combination of $w_1, \dots, w_r$.

By [Theorem 7.8](#), $v - \sum c_k w_k$ is orthogonal to every w_i and hence orthogonal to any linear combination of $w_1, w_2, \dots, w_r$. Therefore, using the Pythagorean theorem and summing from $k = 1$ to r ,

$$\begin{aligned} \|v - \sum a_k w_k\|^2 &= \|v - \sum c_k w_k + \sum (c_k - a_k) w_k\|^2 = \|v - \sum c_k w_k\|^2 + \|\sum (c_k - a_k) w_k\|^2 \\ &\geq \|v - \sum c_k w_k\|^2 \end{aligned}$$

The square root of both sides gives our theorem.

7.31. Suppose $\{e_1, e_2, \dots, e_r\}$ is an orthonormal set of vectors in V . Let v be any vector in V and let c_i be the Fourier coefficient of v with respect to e_i . Prove Bessel's inequality:

$$\sum_{k=1}^r c_k^2 \leq \|v\|^2$$

Note that $c_i = \langle v, e_i \rangle$, because $\|e_i\| = 1$. Then, using $\langle e_i, e_j \rangle = 0$ for $i \neq j$ and summing from $k = 1$ to r , we get

$$0 \leq \langle v - \sum c_k e_k, v - \sum c_k e_k \rangle = \langle v, v \rangle - 2\langle v, \sum c_k e_k \rangle + \sum c_k^2 = \langle v, v \rangle - \sum 2c_k \langle v, e_k \rangle + \sum c_k^2 \\ = \langle v, v \rangle - \sum 2c_k^2 + \sum c_k^2 = \langle v, v \rangle - \sum c_k^2$$

This gives us our inequality.

Orthogonal Matrices

7.32. Find an orthogonal matrix P whose first row is

$$u_1 = \left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right).$$

First find a nonzero vector $w_2 = (x, y, z)$ that is orthogonal to u_1 —that is, for which

$$0 = \langle u_1, w_2 \rangle = \frac{x}{3} + \frac{2y}{3} + \frac{2z}{3} = 0 \quad \text{or} \quad x + 2y + 2z = 0$$

One such solution is $w_2 = (0, 1, -1)$. Normalize w_2 to obtain the second row of P :

$$u_2 = (0, 1/\sqrt{2}, -1/\sqrt{2})$$

Next find a nonzero vector $w_3 = (x, y, z)$ that is orthogonal to both u_1 and u_2 —that is, for which

$$0 = \langle u_1, w_3 \rangle = \frac{x}{3} + \frac{2y}{3} + \frac{2z}{3} = 0 \quad \text{or} \quad x + 2y + 2z = 0$$

$$0 = \langle u_2, w_3 \rangle = \frac{y}{\sqrt{2}} - \frac{z}{\sqrt{2}} = 0 \quad \text{or} \quad y - z = 0$$

Set $z = -1$ and find the solution $w_3 = (4, -1, -1)$. Normalize w_3 and obtain the third row of P ; that is,

$$u_3 = (4/\sqrt{18}, -1/\sqrt{18}, -1/\sqrt{18})$$

Thus,

$$P = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ 0 & 1/\sqrt{2} & -1/\sqrt{2} \\ 4/3\sqrt{2} & -1/3\sqrt{2} & -1/3\sqrt{2} \end{bmatrix}$$

We emphasize that the above matrix P is not unique.

7.33. Let $A = \begin{bmatrix} 1 & 1 & -1 \\ 1 & 3 & 4 \\ 7 & -5 & 2 \end{bmatrix}$. Determine whether or not:

(a) the rows of A are orthogonal;

(b) A is an orthogonal matrix; (c) the columns of A are orthogonal.

(a) Yes, because $(1, 1, -1) \cdot (1, 3, 4) = 1 + 3 - 4 = 0$, $(1, 1, -1) \cdot (7, -5, 2) = 7 - 5 - 2 = 0$, and $(1, 3, 4) \cdot (7, -5, 2) = 7 - 15 + 8 = 0$.

(b) No, because the rows of A are not unit vectors, for example, $(1, 1, -1)^2 = 1 + 1 + 1 = 3$.

(c) No; for example, $(1, 1, 7) \cdot (1, 3, -5) = 1 + 3 - 35 = -31 \neq 0$.

7.34. Let B be the matrix obtained by normalizing each row of A in [Problem 7.33](#).

(a) Find B .

(b) Is B an orthogonal matrix?

(c) Are the columns of B orthogonal?

(a) We have

$$\begin{aligned} \|(1, 1, -1)\|^2 &= 1 + 1 + 1 = 3, & \|(1, 3, 4)\|^2 &= 1 + 9 + 16 = 26 \\ \|(7, -5, 2)\|^2 &= 49 + 25 + 4 = 78 \end{aligned}$$

Thus,

$$B = \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{3} & -1/\sqrt{3} \\ 1/\sqrt{26} & 3/\sqrt{26} & 4/\sqrt{26} \\ 7/\sqrt{78} & -5/\sqrt{78} & 2/\sqrt{78} \end{bmatrix}$$

(b) Yes, because the rows of B are still orthogonal and are now unit vectors.

(c) Yes, because the rows of B form an orthonormal set of vectors. Then, by [Theorem 7.11](#), the columns of B must automatically form an orthonormal set.

7.35. Prove each of the following:

(a) P is orthogonal if and only if P^T is orthogonal.

(b) If P is orthogonal, then P^{-1} is orthogonal.

(c) If P and Q are orthogonal, then PQ is orthogonal.

(a) We have $(P^T)^T = P$. Thus, P is orthogonal if and only if $PP^T = I$ if and only if $P^{TT} P^T = I$ if and only if P^T is orthogonal.

(b) We have $P^T = P^{-1}$, because P is orthogonal. Thus, by part (a), P^{-1} is orthogonal.

(c) We have $P^T = P^{-1}$ and $Q^T = Q^{-1}$. Thus, $(PQ)(PQ)^T = PQQ^T P^T = PQQ^{-1} P^{-1} = I$. Therefore, $(PQ)^T = (PQ)^{-1}$, and so PQ is orthogonal.

7.36. Suppose P is an orthogonal matrix. Show that

(a) $\langle Pu, Pv \rangle = \langle u, v \rangle$ for any $u, v \in V$;

(b) $\|Pu\| = \|u\|$ for every $u \in V$.

Use $P^T P = I$ and $\langle u, v \rangle = u^T v$.

(a) $\langle Pu, Pv \rangle = (Pu)^T (Pv) = u^T P^T P v = u^T v = \langle u, v \rangle$.

(b) We have

$$\|Pu\|^2 = \langle Pu, Pu \rangle = u^T P^T P u = u^T u = \langle u, u \rangle = \|u\|^2$$

Taking the square root of both sides gives our result.

7.37. Prove [Theorem 7.12](#): Suppose $E = \{e_i\}$ and $E' = \{e'_i\}$ are orthonormal bases of V . Let P be the change-of-basis matrix from E to E' . Then P is orthogonal.

Suppose

$$e'_i = b_{i1}e_1 + b_{i2}e_2 + \cdots + b_{in}e_n, \quad i = 1, \dots, n \quad (1)$$

Using [Problem 7.18](#)(b) and the fact that E' is orthonormal, we get

$$\delta_{ij} = \langle e'_i, e'_j \rangle = b_{i1}b_{j1} + b_{i2}b_{j2} + \cdots + b_{in}b_{jn} \quad (2)$$

Let $B = [b_{ij}]$ be the matrix of the coefficients in (1). (Then $P = B^T$.) Suppose $BB^T = [c_{ij}]$. Then

$$c_{ij} = b_{i1}b_{j1} + b_{i2}b_{j2} + \cdots + b_{in}b_{jn} \quad (3)$$

By (2) and (3), we have $c_{ij} = \delta_{ij}$. Thus, $BB^T = I$. Accordingly, B is orthogonal, and hence, $P = B^T$ is orthogonal.

7.38. Prove [Theorem 7.13](#): Let $\{e_1, \dots, e_n\}$ be an orthonormal basis of an inner product space V . Let $P = [a_{ij}]$ be an orthogonal matrix. Then the following n vectors form an orthonormal basis for V :

$$e'_i = a_{1i}e_1 + a_{2i}e_2 + \cdots + a_{ni}e_n, \quad i = 1, 2, \dots, n$$

Because $\{e_j\}$ is orthonormal, we get, by [Problem 7.18\(b\)](#),

$$\langle e'_i, e'_j \rangle = a_{1i}a_{1j} + a_{2i}a_{2j} + \cdots + a_{ni}a_{nj} = \langle C_i, C_j \rangle$$

where C_i denotes the i th column of the orthogonal matrix $P = [a_{ij}]$: Because P is orthogonal, its columns form an orthonormal set. This implies $\langle e'_i, e'_j \rangle = \langle C_i, C_j \rangle = \delta_{ij}$. Thus, $\{e'_i\}$ is an orthonormal basis.

Inner Products And Positive Definite Matrices

7.39. Which of the following symmetric matrices are positive definite?

$$(a) \ A = \begin{bmatrix} 3 & 4 \\ 4 & 5 \end{bmatrix}, (b) \ B = \begin{bmatrix} 8 & -3 \\ -3 & 2 \end{bmatrix}, (c) \ C = \begin{bmatrix} 2 & 1 \\ 1 & -3 \end{bmatrix}, (d) \ D = \begin{bmatrix} 3 & 5 \\ 5 & 9 \end{bmatrix}$$

Use [Theorem 7.14](#) that a 2×2 real symmetric matrix is positive definite if and only if its diagonal entries are positive and if its determinant is positive.

- (a) No, because $|A| = 15 - 16 = -1$ is negative.
- (b) Yes.
- (c) No, because the diagonal entry -3 is negative.
- (d) Yes.

7.40. Find the values of k that make each of the following matrices positive definite:

$$(a) \ A = \begin{bmatrix} 2 & -4 \\ -4 & k \end{bmatrix}, (b) \ B = \begin{bmatrix} 4 & k \\ k & 9 \end{bmatrix}, (c) \ C = \begin{bmatrix} k & 5 \\ 5 & -2 \end{bmatrix}$$

- (a) First, k must be positive. Also, $|A| = 2k - 16$ must be positive; that is, $2k - 16 > 0$. Hence, $k > 8$.

(b) We need $|B| = 36 - k^2$ positive; that is, $36 - k^2 > 0$. Hence, $k^2 < 36$ or $-6 < k < 6$.

(c) C can never be positive definite, because C has a negative diagonal entry -2 .

7.41. Find the matrix A that represents the usual inner product on $\mathbf{R}^2$ relative to each of the following bases of $\mathbf{R}^2$: (a) $\{\nu_1 = (1, 3), \nu_2 = (2, 5)\}$; (b) $\{w_1 = (1, 2), w_2 = (4, 2)\}$:

(a) Compute $\langle \nu_1, \nu_1 \rangle = 1 + 9 = 10$, $\langle \nu_1, \nu_2 \rangle = 2 + 15 = 17$, $\langle \nu_2, \nu_2 \rangle = 4 + 25 = 29$. Thus, $A = \begin{bmatrix} 10 & 17 \\ 17 & 29 \end{bmatrix}$.

(b) Compute $\langle w_1, w_1 \rangle = 1 + 4 = 5$, $\langle w_1, w_2 \rangle = 4 - 4 = 0$, $\langle w_2, w_2 \rangle = 16 + 4 = 20$. Thus, $A = \begin{bmatrix} 5 & 0 \\ 0 & 20 \end{bmatrix}$.

(Because the basis vectors are orthogonal, the matrix A is diagonal.)

7.42. Consider the vector space $\mathbf{P}_2(t)$ with inner product $\langle f, g \rangle = \int_{-1}^1 f(t)g(t) dt$.

(a) Find $\langle f, g \rangle$, where $f(t) = t + 2$ and $g(t) = t^2 - 3t + 4$.

(b) Find the matrix A of the inner product with respect to the basis $\{1, t, t^2\}$ of V .

(c) Verify [Theorem 7.16](#) by showing that $\langle f, g \rangle = [f]^T A [g]$ with respect to the basis $\{1, t, t^2\}$.

$$(a) \quad \langle f, g \rangle = \int_{-1}^1 (t+2)(t^2-3t+4) dt = \int_{-1}^1 (t^3 - t^2 - 2t + 8) dt = \left(\frac{t^4}{4} - \frac{t^3}{3} - t^2 + 8t \right) \Big|_{-1}^1 = \frac{46}{3}$$

(b) Here we use the fact that if $r + s = n$,

$$\langle t^r, t^r \rangle = \int_{-1}^1 t^n dt = \frac{t^{n+1}}{n+1} \Big|_{-1}^1 = \begin{cases} 2/(n+1) & \text{if } n \text{ is even,} \\ 0 & \text{if } n \text{ is odd.} \end{cases}$$

Then

$$\langle 1, 1 \rangle = 2, \langle 1, t \rangle = 0, \langle 1, t^2 \rangle = \frac{2}{3}, \langle t, t \rangle = \frac{2}{3}, \langle t, t^2 \rangle = 0, \langle t^2, t^2 \rangle = \frac{2}{5}.$$

Thus,

$$A = \begin{bmatrix} 2 & 0 & \frac{2}{3} \\ 0 & \frac{2}{3} & 0 \\ \frac{2}{3} & 0 & \frac{2}{5} \end{bmatrix}$$

(c) We have $[f]^T = (2, 1, 0)$ and $[g]^T = (4, -3, 1)$ relative to the given basis. Then

$$[f]^T A [g] = (2, 1, 0) \begin{bmatrix} 2 & 0 & \frac{2}{3} \\ 0 & \frac{2}{3} & 0 \\ \frac{2}{3} & 0 & \frac{2}{5} \end{bmatrix} \begin{bmatrix} 4 \\ -3 \\ 1 \end{bmatrix} = (4, \frac{2}{3}, \frac{4}{3}) \begin{bmatrix} 4 \\ -3 \\ 1 \end{bmatrix} = \frac{46}{3} = \langle f, g \rangle$$

7.43. Prove [Theorem 7.14](#): $A = \begin{bmatrix} a & b \\ b & d \end{bmatrix}$ is positive definite

if and only if a and d are positive and $|A| = ad - b^2$ is positive.

Let $u = [x, y]^T$. Then

$$f(u) = u^T A u = [x, y] \begin{bmatrix} a & b \\ b & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = ax^2 + 2bxy + dy^2$$

Suppose $f(u) > 0$ for every $u \neq 0$. Then $f(1, 0) = a > 0$ and $f(0, 1) = d > 0$. Also, we have $f(b, -a) = a(ad - b^2) > 0$. Because $a > 0$, we get $ad - b^2 > 0$.

Conversely, suppose $a > 0$, $d > 0$, $ad - b^2 > 0$. Completing the square gives us

$$f(u) = a\left(x^2 + \frac{2b}{a}xy + \frac{b^2}{a^2}y^2\right) + dy^2 - \frac{b^2}{a}y^2 = a\left(x + \frac{by}{a}\right)^2 + \frac{ad - b^2}{a}y^2$$

Accordingly, $f(u) > 0$ for every $u \neq 0$.

7.44. Prove [Theorem 7.15](#): Let A be a real positive definite matrix. Then the function $\langle u, v \rangle = u^T A v$ is an inner product on $\mathbf{R}^n$.

For any vectors u_1, u_2 , and v ,

$$\langle u_1 + u_2, v \rangle = (u_1 + u_2)^T A v = (u_1^T + u_2^T) A v = u_1^T A v + u_2^T A v = \langle u_1, v \rangle + \langle u_2, v \rangle$$

and, for any scalar k and vectors u, v ,

$$\langle ku, v \rangle = (ku)^T A v = ku^T A v = k \langle u, v \rangle$$

Thus $[I_1]$ is satisfied.

Because $u^T A v$ is a scalar, $(u^T A v)^T = u^T A v$. Also, $A^T = A$ because A is symmetric. Therefore,

$$\langle u, v \rangle = u^T A v = (u^T A v)^T = v^T A^T u^{TT} = v^T A u = \langle v, u \rangle$$

Thus, $[I_2]$ is satisfied.

Last, because A is positive definite, $X^T A X > 0$ for any nonzero $X \in \mathbf{R}^n$. Thus, for any nonzero vector v , $\langle v, v \rangle = v^T A v > 0$. Also, $\langle 0, 0 \rangle = 0^T A 0 = 0$. Thus, $[I_3]$ is satisfied. Accordingly, the function $\langle u, v \rangle = u^T A v$ is an inner product.

7.45. Prove [Theorem 7.16](#): Let A be the matrix representation of an inner product relative to a basis S of V . Then, for any vectors $u, v \in V$, we have

$$\langle u, v \rangle = [u]^T A [v]$$

Suppose $S = \{w_1, w_2, \dots, w_n\}$ and $A = [k_{ij}]$. Hence, $k_{ij} = \langle w_i, w_j \rangle$. Suppose

$$u = a_1 w_1 + a_2 w_2 + \dots + a_n w_n \quad \text{and} \quad v = b_1 w_1 + b_2 w_2 + \dots + b_n w_n$$

Then

$$\langle u, v \rangle = \sum_{i=1}^n \sum_{j=1}^n a_i b_j \langle w_i, w_j \rangle \quad (1)$$

On the other hand,

$$\begin{aligned} [u]^T A [v] &= (a_1, a_2, \dots, a_n) \begin{bmatrix} k_{11} & k_{12} & \dots & k_{1n} \\ k_{21} & k_{22} & \dots & k_{2n} \\ \dots & \dots & \dots & \dots \\ k_{n1} & k_{n2} & \dots & k_{nn} \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \\ &= \left(\sum_{i=1}^n a_i k_{i1}, \sum_{i=1}^n a_i k_{i2}, \dots, \sum_{i=1}^n a_i k_{in} \right) \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} = \sum_{j=1}^n \sum_{i=1}^n a_i b_j k_{ij} \end{aligned} \quad (2)$$

Equations (1) and (2) give us our result.

7.46. Prove [Theorem 7.17](#): Let A be the matrix representation of any inner product on V . Then A is a positive definite matrix.

Because $\langle w_i, w_j \rangle = \langle w_j, w_i \rangle$ for any basis vectors w_i and w_j , the matrix A is symmetric. Let X be any nonzero vector in $\mathbf{R}^n$. Then $[u] = X$ for some nonzero

vector $u \in V$. [Theorem 7.16](#) tells us that $X^TAX = [u]^T A[u] = \langle u, u \rangle > 0$. Thus, A is positive definite.

Complex Inner Product Spaces

7.47. Let V be a complex inner product space. Verify the relation

$$\langle u, av_1 + bv_2 \rangle = \bar{a}\langle u, v_1 \rangle + \bar{b}\langle u, v_2 \rangle$$

Using $[I_2^*]$, $[I_1^*]$, and then $[I_2^*]$, we find

$$\langle u, av_1 + bv_2 \rangle = \overline{\langle av_1 + bv_2, u \rangle} = \overline{a\langle v_1, u \rangle + b\langle v_2, u \rangle} = \bar{a}\overline{\langle v_1, u \rangle} + \bar{b}\overline{\langle v_2, u \rangle} = \bar{a}\langle u, v_1 \rangle + \bar{b}\langle u, v_2 \rangle$$

7.48. Suppose $\langle u, v \rangle = 3 + 2i$ in a complex inner product space V . Find

(a) $\langle (2 - 4i)u, v \rangle$; (b) $\langle u, (4 + 3i)v \rangle$; (c) $\langle (3 - 6i)u, (5 - 2i)v \rangle$.

(a) $\langle (2 - 4i)u, v \rangle = (2 - 4i)\langle u, v \rangle = (2 - 4i)(3 + 2i) = 14 - 8i$

(b) $\langle u, (4 + 3i)v \rangle = \overline{(4 + 3i)}\langle u, v \rangle = (4 - 3i)(3 + 2i) = 18 - i$

(c) $\langle (3 - 6i)u, (5 - 2i)v \rangle = (3 - 6i)\overline{(5 - 2i)}\langle u, v \rangle = (3 - 6i)(5 + 2i)(3 + 2i) = 129 - 18i$

7.49. Find the Fourier coefficient (component) c and the projection cw of $v = (3 + 4i, 2 - 3i)$ along $w = (5 + i, 2i)$ in $\mathbf{C}^2$.

Recall that $c = \langle v, w \rangle / \langle w, w \rangle$. Compute

$$\begin{aligned} \langle v, w \rangle &= (3 + 4i)\overline{(5 + i)} + (2 - 3i)\overline{(2i)} = (3 + 4i)(5 - i) + (2 - 3i)(-2i) \\ &= 19 + 17i - 6 - 4i = 13 + 13i \end{aligned}$$

$$\langle w, w \rangle = 25 + 1 + 4 = 30$$

Thus, $c = (13 + 13i)/30 = \frac{13}{30} + \frac{13}{30}i$. Accordingly,
 $\text{proj}(v, w) = cw = (\frac{26}{15} + \frac{39}{15}i, -\frac{13}{15} + \frac{1}{15}i)$

7.50. Prove [Theorem 7.18](#) (Cauchy-Schwarz): Let V be a complex inner product space. Then $|\langle u, v \rangle| \leq \|u\| \|v\|$

$v\|$.

If $v = 0$, the inequality reduces to $0 \leq 0$ and hence is valid. Now suppose $v \neq 0$. Using $z\bar{z} = |z|^2$ (for any complex number z) and $\langle v, u \rangle = \overline{\langle u, v \rangle}$, we expand $\|u - \langle u, v \rangle v\|^2 \leq 0$, where t is any real value:

$$\begin{aligned} 0 &\leq \|u - \langle u, v \rangle v\|^2 = \langle u - \langle u, v \rangle v, u - \langle u, v \rangle v \rangle \\ &= \langle u, u \rangle - \overline{\langle u, v \rangle} t \langle u, v \rangle - \langle u, v \rangle t \langle v, u \rangle + \langle u, v \rangle \overline{\langle u, v \rangle} t^2 \langle v, v \rangle \\ &= \|u\|^2 - 2t |\langle u, v \rangle|^2 + |\langle u, v \rangle|^2 t^2 \|v\|^2 \end{aligned}$$

Set $t = 1/\|v\|^2$ to find $0 \leq \|u\|^2 - \frac{|\langle u, v \rangle|^2}{\|v\|^2}$, from which $|\langle u, v \rangle|^2 \leq \|u\|^2 \|v\|^2$. Taking the square root of both sides, we obtain the required inequality.

7.51. Find an orthogonal basis for $u^\perp$ in C^3 where $u = (1, i, 1 + i)$.

Here $u^\perp$ consists of all vectors $s = (x, y, z)$ such that

$$\langle w, u \rangle = x - iy + (1 - i)z = 0$$

Find one solution, say $w_1 = (0, 1 - i, i)$. Then find a solution of the system

$$x - iy + (1 - i)z = 0, \quad (1 + i)y - iz = 0$$

Here z is a free variable. Set $z = 1$ to obtain $y = i/(1 + i) = (1 + i)/2$ and $x = (3i - 3)/2$. Multiplying by 2 yields the solution $w_2 = (3i - 3, 1 + i, 2)$. The vectors w_1 and w_2 form an orthogonal basis for $u^\perp$.

7.52. Find an orthonormal basis of the subspace W of $\mathbf{C}^3$ spanned by

$$v_1 = (1, i, 0) \quad \text{and} \quad v_2 = (1, 2, 1 - i).$$

Apply the Gram-Schmidt algorithm. Set $w_1 = v_1 = (1, i, 0)$. Compute

$$v_2 - \frac{\langle v_2, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1 = (1, 2, 1 - i) - \frac{1 - 2i}{2} (1, i, 0) = \left(\frac{1}{2} + i, 1 - \frac{1}{2}i, 1 - i\right)$$

Multiply by 2 to clear fractions, obtaining $w_2 = (1 + 2i, 2 - i, 2 - 2i)$. Next find $\|w_1\| = \sqrt{2}$ and then $\|w_2\| = \sqrt{18}$. Normalizing $\{w_1, w_2\}$, we obtain the following orthonormal basis of W :

$$\left\{ u_1 = \left(\frac{1}{\sqrt{2}}, \frac{i}{\sqrt{2}}, 0 \right), u_2 = \left(\frac{1 + 2i}{\sqrt{18}}, \frac{2 - i}{\sqrt{18}}, \frac{2 - 2i}{\sqrt{18}} \right) \right\}$$

7.53. Find the matrix P that represents the usual inner product on $\mathbf{C}^3$ relative to the basis $\{1, i, 1 - i\}$.

Compute the following six inner products:

$$\begin{aligned} \langle 1, 1 \rangle &= 1, & \langle 1, i \rangle &= \bar{i} = -i, & \langle 1, 1 - i \rangle &= \overline{1 - i} = 1 + i \\ \langle i, i \rangle &= \bar{i}i = 1, & \langle i, 1 - i \rangle &= i\overline{(1 - i)} = -1 + i, & \langle 1 - i, 1 - i \rangle &= 2 \end{aligned}$$

Then, using $(u, v) = \overline{\langle v, u \rangle}$, we obtain

$$P = \begin{bmatrix} 1 & -i & 1 + i \\ i & 1 & -1 + i \\ 1 - i & -1 - i & 2 \end{bmatrix}$$

(As expected, P is Hermitian; that is, $P^H = P$.)

Normed Vector Spaces

7.54. Consider vectors $u = (1, 3, -6, 4)$ and $v = (3, -5, 1, -2)$ in $\mathbf{R}^4$. Find

(a) $\|u\|_\infty$ and $\|v\|_\infty$, (b) $\|u\|_1$ and $\|v\|_1$, (c) $\|u\|_2$ and $\|v\|_2$,

(d) $d_\infty(u, v)$, $d_1(u, v)$, $d_2(u, v)$.

(a) The infinity norm chooses the maximum of the absolute values of the components. Hence,

$$\|u\|_\infty = 6 \quad \text{and} \quad \|v\|_\infty = 5$$

(b) The one-norm adds the absolute values of the components. Thus,

$$\|u\|_1 = 1 + 3 + 6 + 4 = 14 \quad \text{and} \quad \|v\|_1 = 3 + 5 + 1 + 2 = 11$$

(c) The two-norm is equal to the square root of the sum of the squares of the components (i.e., the norm induced by the usual inner product on $\mathbf{R}^3$). Thus,

$$\|u\|_2 = \sqrt{1 + 9 + 36 + 16} = \sqrt{62} \quad \text{and} \quad \|v\|_2 = \sqrt{9 + 25 + 1 + 4} = \sqrt{39}$$

(d) First find $u - v = (-2, 8, -7, 6)$. Then

$$d_\infty(u, v) = \|u - v\|_\infty = 8$$

$$d_1(u, v) = \|u - v\|_1 = 2 + 8 + 7 + 6 = 23$$

$$d_2(u, v) = \|u - v\|_2 = \sqrt{4 + 64 + 49 + 36} = \sqrt{153}$$

7.55. Consider the function $f(t) = t^2 - 4t$ in $C[0, 3]$.

(a) Find $\|f\|_\infty$, (b) Plot $f(t)$ in the plane $\mathbf{R}^2$, (c) Find $\|f\|_1$, (d) Find $\|f\|_2$.

(a) We seek $\|f\|_{\infty} = \max(|f(t)|)$. Because $f(t)$ is differentiable on $[0, 3]$, $|f(t)|$ has a maximum at a critical point of $f(t)$ (i.e., when the derivative $f'(t) = 0$), or at an endpoint of $[0, 3]$. Because $f'(t) = 2t - 4$, we set $2t - 4 = 0$ and obtain $t = 2$ as a critical point. Compute

$$f(2) = 4 - 8 = -4, \quad f(0) = 0 - 0 = 0, \quad f(3) = 9 - 12 = -3$$

Thus, $\|f\|_{\infty} = |f(2)| = |-4| = 4$.

(b) Compute $f(t)$ for various values of t in $[0, 3]$, for example,

t	0	1	2	3
$f(t)$	0	-3	-4	-3

Plot the points in $\mathbf{R}^2$ and then draw a continuous curve through the points, as shown in Fig. 7-8.

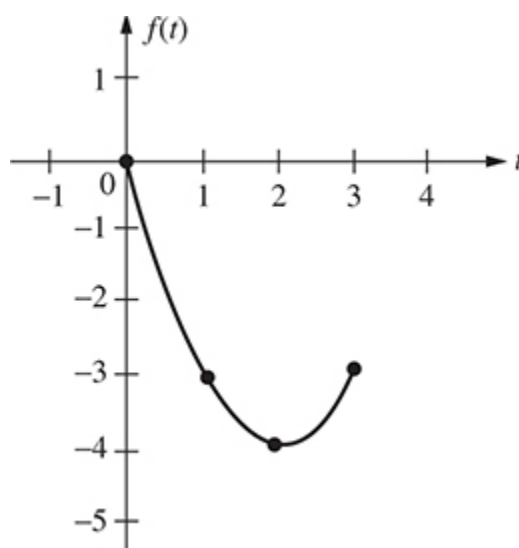


Figure 7-8

(c) We seek $\|f\|_1 = \int_0^3 |f(t)| dt$. As indicated in Fig. 7-3, $f(t)$ is negative in $[0, 3]$; hence, $|f(t)| = -(t^2 - 4t) =$

$$4t - t^2$$

Thus,

$$\|f\|_1 = \int_0^3 (4t - t^2) dt = \left(2t^2 - \frac{t^3}{3}\right) \Big|_0^3 = 18 - 9 = 9$$

(d)

$$\|f\|_2^2 = \int_0^3 f(t)^2 dt = \int_0^3 (t^4 - 8t^3 + 16t^2) dt = \left(\frac{t^5}{5} - 2t^4 + \frac{16t^3}{3}\right) \Big|_0^3 = \frac{153}{5}$$

Thus,

$$\|f\|_2 = \sqrt{\frac{153}{5}} .$$

7.56. Prove [Theorem 7.24](#): Let V be a normed vector space. Then the function $d(u, v) = \|u - v\|$ satisfies the following three axioms of a metric space:

[M₁] $d(u, v) \geq 0$; and $d(u, v) = 0$ iff $u = v$.

[M₂] $d(u, v) = d(v, u)$.

[M₃] $d(u, v) \leq d(u, w) + d(w, v)$.

If $u \neq v$, then $u - v \neq 0$, and hence, $d(u, v) = \|u - v\| > 0$. Also, $d(u, u) = \|u - u\| = \|0\| = 0$. Thus,

[M₁] is satisfied. We also have

$$d(u, v) = \|u - v\| = \| -1(v - u) \| = | -1 | \|v - u\| = \|v - u\| = d(v, u)$$

and

$$d(u, v) = \|u - v\| = \|(u - w) + (w - v)\| \leq \|u - w\| + \|w - v\| = d(u, w) + d(w, v)$$

Thus, $[M_2]$ and $[M_3]$ are satisfied.

SUPPLEMENTARY PROBLEMS

Inner Products

- 7.57.** Verify that the following is an inner product on $\mathbf{R}^2$, where $u = (x_1, x_2)$ and $v = (y_1, y_2)$:

$$f(u, v) = x_1 y_1 - 2x_1 y_2 - 2x_2 y_1 + 5x_2 y_2$$

- 7.58.** Find the values of k so that the following is an inner product on $\mathbf{R}^2$, where $u = (x_1, x_2)$ and $v = (y_1, y_2)$:

$$f(u, v) = x_1 y_1 - 3x_1 y_2 - 3x_2 y_1 + kx_2 y_2$$

- 7.59.** Consider the vectors $u = (1, -3)$ and $v = (2, 5)$ in $\mathbf{R}^2$. Find

(a) $\langle u, v \rangle$ with respect to the usual inner product in $\mathbf{R}^2$.

(b) $\langle u, v \rangle$ with respect to the inner product in $\mathbf{R}^2$ in [Problem 7.57](#).

(c) $\|v\|$ using the usual inner product in $\mathbf{R}^2$.

(d) $\|v\|$ using the inner product in $\mathbf{R}^2$ in [Problem 7.57](#).

- 7.60.** Show that each of the following is not an inner product on $\mathbf{R}^3$, where $u = (x_1, x_2, x_3)$ and $v = (y_1, y_2, y_3)$:

(a) $\langle u, v \rangle = x_1 y_1 + x_2 y_2$, (b) $\langle u, v \rangle = x_1 y_2 x_3 + y_1 x_2 y_3$.

- 7.61.** Let V be the vector space of $m \times n$ matrices over $\mathbf{R}$. Show that $\langle A, B \rangle = \text{tr}(B^T A)$ defines an inner product in V .
- 7.62.** Suppose $|\langle u, v \rangle| = \|u\| \|v\|$. (That is, the Cauchy-Schwarz inequality reduces to an equality.) Show that u and v are linearly dependent.
- 7.63.** Suppose $f(u, v)$ and $g(u, v)$ are inner products on a vector space V over $\mathbf{R}$. Prove
- (a) The sum $f + g$ is an inner product on V , where $(f + g)(u, v) = f(u, v) + g(u, v)$.
 - (b) The scalar product kf , for $k > 0$, is an inner product on V , where $(kf)(u, v) = kf(u, v)$.

Orthogonality, Orthogonal Complements, Orthogonal Sets

- 7.64.** Let V be the vector space of polynomials over $\mathbf{R}$ of degree ≤ 2 with inner product defined by $\langle f, g \rangle = \int_0^1 f(t)g(t)$. Find a basis of the subspace W orthogonal to $h(t) = 2t + 1$.
- 7.65.** Find a basis of the subspace W of $\mathbf{R}^4$ orthogonal to $u_1 = (1, -2, 3, 4)$ and $u_2 = (3, -5, 7, 8)$.
- 7.66.** Find a basis for the subspace W of $\mathbf{R}^5$ orthogonal to the vectors $u_1 = (1, 1, 3, 4, 1)$ and $u_2 = (1, 2, 1, 2, 1)$.
- 7.67.** Let $w = (1, -2, -1, 3)$ be a vector in $\mathbf{R}^4$. Find
- (a) an orthogonal basis for $w^\perp$, (b) an orthonormal basis for $w^\perp$.

7.68. Let W be the subspace of $\mathbf{R}^4$ orthogonal to $u_1 = (1, 1, 2, 2)$ and $u_2 = (0, 1, 2, -1)$. Find

(a) an orthogonal basis for W , (b) an orthonormal basis for W . (Compare with [Problem 7.65](#).)

7.69. Let S consist of the following vectors in $\mathbf{R}^4$:

$$u_1 = (1, 1, 1, 1), \quad u_2 = (1, 1, -1, -1), \quad u_3 = (1, -1, 1, -1), \quad u_4 = (1, -1, -1, 1)$$

(a) Show that S is orthogonal and a basis of $\mathbf{R}^4$.

(b) Write $v = (1, 3, -5, 6)$ as a linear combination of u_1, u_2, u_3, u_4 .

(c) Find the coordinates of an arbitrary vector $v = (a, b, c, d)$ in $\mathbf{R}^4$ relative to the basis S .

(d) Normalize S to obtain an orthonormal basis of $\mathbf{R}^4$.

7.70. Let $\mathbf{M} = \mathbf{M}_{2,2}$ with inner product $\langle A, B \rangle = \text{tr}(B^T A)$. Show that the following is an orthonormal basis for $\mathbf{M}$:

$$\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$$

7.71. Let $\mathbf{M} = \mathbf{M}_{2,2}$ with inner product $\langle A, B \rangle = \text{tr}(B^T A)$. Find an orthogonal basis for the orthogonal complement of (a) diagonal matrices, (b) symmetric matrices.

7.72. Suppose $\{u_1, u_2, \dots, u_r\}$ is an orthogonal set of vectors. Show that $\{k_1 u_1, k_2 u_2, \dots, k_r u_r\}$ is an orthogonal set for any scalars $k_1, k_2, \dots, k_r$.

7.73. Let U and W be subspaces of a finite-dimensional inner product space V . Show that

$$(a) \quad (U + W)^\perp = U^\perp \cap W^\perp, \quad (b) \quad (U \cap W)^\perp = U^\perp + W^\perp.$$

Projections, Gram-Schmidt Algorithm, Applications

7.74. Find the Fourier coefficient c and projection cw of v along w , where

$$(a) \quad v = (2, 3, -5) \text{ and } w = (1, -5, 2) \text{ in } \mathbf{R}^3.$$

$$(b) \quad v = (1, 3, 1, 2) \text{ and } w = (1, -2, 7, 4) \text{ in } \mathbf{R}^4.$$

$$(c) \quad v = t^2 \text{ and } w = t + 3 \text{ in } \mathbf{P}(t), \text{ with inner product } \langle f, g \rangle = \int_0^1 f(t)g(t) dt$$

$$(d) \quad v = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \text{ and } w = \begin{bmatrix} 1 & 1 \\ 5 & 5 \end{bmatrix} \text{ in } \mathbf{M} = \mathbf{M}_{2,2}, \text{ with inner product } \langle A, B \rangle = \text{tr}(B^T A).$$

7.75. Let U be the subspace of $\mathbf{R}^4$ spanned by

$$v_1 = (1, 1, 1, 1), \quad v_2 = (1, -1, 2, 2), \quad v_3 = (1, 2, -3, -4)$$

(a) Apply the Gram-Schmidt algorithm to find an orthogonal and an orthonormal basis for U .

(b) Find the projection of $v = (1, 2, -3, 4)$ onto U .

7.76. Suppose $v = (1, 2, 3, 4, 6)$. Find the projection of v onto W , or, in other words, find $w \in W$ that minimizes $\|v - w\|$, where W is the subspace of $\mathbf{R}^5$ spanned by

$$(a) \quad u_1 = (1, 2, 1, 2, 1) \text{ and } u_2 = (1, -1, 2, -1, 1), \quad (b) \quad v_1 = (1, 2, 1, 2, 1) \text{ and } v_2 = (1, 0, 1, 5, -1)$$

- 7.77.** Consider the subspace $W = \mathbf{P}_2(t)$ of $\mathbf{P}(t)$ with inner product $\langle f, g \rangle = \int_0^1 f(t)g(t) dt$. Find the projection of $f(t) = t^3$ onto W . (*Hint:* Use the orthogonal polynomials $1, 2t - 1, 6t^2 - 6t + 1$ obtained in [Problem 7.22](#).)
- 7.78.** Consider $\mathbf{P}(t)$ with inner product $\langle f, g \rangle = \int_{-1}^1 f(t)g(t) dt$ and the subspace $W = \mathbf{P}_3(t)$.
- (a) Find an orthogonal basis for W by applying the Gram-Schmidt algorithm to $\{1, t, t^2, t^3\}$.
- (b) Find the projection of $f(t) = t^5$ onto W .

Orthogonal Matrices

- 7.79.** Find the number and exhibit all 2×2 orthogonal matrices of the form $\begin{bmatrix} \frac{1}{3} & x \\ y & z \end{bmatrix}$.
- 7.80.** Find a 3×3 orthogonal matrix P whose first two rows are multiples of $u = (1, 1, 1)$ and $v = (1, -3, 2)$, respectively.
- 7.81.** Find a symmetric orthogonal matrix P whose first row is $(\frac{1}{3}, \frac{2}{3}, \frac{2}{3})$. (Compare with [Problem 7.32](#).)
- 7.82.** Real matrices A and B are said to be *orthogonally equivalent* if there exists an orthogonal matrix P such that $B = P^T A P$. Show that this relation is an equivalence relation.

Positive Definite Matrices and Inner Products

- 7.83.** Find the matrix A that represents the usual inner product on $\mathbf{R}^2$ relative to each of the following bases:
- (a) $\{v_1 = (1, 4), v_2 = (2, -3)\}$, (b) $\{w_1 = (1, -3), w_2 = (6, 2)\}$

7.84. Consider the following inner product on $\mathbf{R}^2$:

$$f(u, v) = x_1y_1 - 2x_1y_2 - 2x_2y_1 + 5x_2y_2, \quad \text{where} \quad u = (x_1, x_2) \quad v = (y_1, y_2)$$

Find the matrix B that represents this inner product on $\mathbf{R}^2$ relative to each basis in [Problem 7.83](#).

7.85. Find the matrix C that represents the usual basis on $\mathbf{R}^3$ relative to the basis S of $\mathbf{R}^3$ consisting of the vectors $u_1 = (1, 1, 1)$, $u_2 = (1, 2, 1)$, $u_3 = (1, -1, 3)$.

7.86. Let $V = \mathbf{P}_2(t)$ with inner product $\langle f, g \rangle = \int_0^1 f(t)g(t) dt$.

(a) Find $\langle f, g \rangle$, where $f(t) = t + 2$ and $g(t) = t^2 - 3t + 4$.

(b) Find the matrix A of the inner product with respect to the basis $\{1, t, t^2\}$ of V .

(c) Verify [Theorem 7.16](#) that $\langle f, g \rangle = [f]^T A [g]$ with respect to the basis $\{1, t, t^2\}$.

7.87. Determine which of the following matrices are positive definite:

$$(a) \begin{bmatrix} 1 & 3 \\ 3 & 5 \end{bmatrix}, (b) \begin{bmatrix} 3 & 4 \\ 4 & 7 \end{bmatrix}, (c) \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}, (d) \begin{bmatrix} 6 & -7 \\ -7 & 9 \end{bmatrix}$$

7.88. Suppose A and B are positive definite matrices. Show that:

(a) $A + B$ is positive definite and

(b) kA is positive definite for $k > 0$.

7.89. Suppose B is a real nonsingular matrix. Show that:

(a) $B^T B$ is symmetric and (b) $B^T B$ is positive definite.

Complex Inner Product Spaces

7.90. Verify that

$$\langle a_1 u_1 + a_2 u_2, b_1 v_1 + b_2 v_2 \rangle = a_1 \bar{b}_1 \langle u_1, v_1 \rangle + a_1 \bar{b}_2 \langle u_1, v_2 \rangle + a_2 \bar{b}_1 \langle u_2, v_1 \rangle + a_2 \bar{b}_2 \langle u_2, v_2 \rangle$$

More generally, prove that

$$\langle \sum_{i=1}^m a_i u_i, \sum_{j=1}^n b_j v_j \rangle = \sum_{i,j} a_i \bar{b}_j \langle u_i, v_j \rangle.$$

7.91. Consider $u = (1 + i, 3, 4 - i)$ and $v = (3 - 4i, 1 + i, 2i)$ in $\mathbf{C}^3$. Find

$$(a) \langle u, v \rangle, \quad (b) \langle v, u \rangle, \quad (c) \|u\|, \quad (d) \|v\|, \quad (e) d(u, v)$$

7.92. Find the Fourier coefficient c and the projection cw of
of

$$(a) \quad u = (3 + i, 5 - 2i) \text{ along } w = (5 + i, 1 + i) \text{ in } \mathbf{C}^2,$$

$$(b) \quad u = (1 - i, 3i, 1 + i) \text{ along } w = (1, 2 - i, 3 + 2i) \text{ in } \mathbf{C}^3$$

7.93. Let $u = (z_1, z_2)$ and $v = (w_1, w_2)$ belong to $\mathbf{C}^2$. Verify that the following is an inner product of $\mathbf{C}^2$:

$$f(u, v) = z_1 \bar{w}_1 + (1 + i)z_1 \bar{w}_2 + (1 - i)z_2 \bar{w}_1 + 3z_2 \bar{w}_2$$

7.94. Find an orthogonal basis and an orthonormal basis for the subspace W of $\mathbf{C}^3$ spanned by $u_1 = (1, i, 1)$ and $u_2 = (1 + i, 0, 2)$.

7.95. Let $u = (z_1, z_2)$ and $v = (w_1, w_2)$ belong to $\mathbf{C}^2$. For what values of $a, b, c, d \in \mathbf{C}$ is the following an inner product on $\mathbf{C}^2$?

$$f(u, v) = az_1 \bar{w}_1 + bz_1 \bar{w}_2 + cz_2 \bar{w}_1 + dz_2 \bar{w}_2$$

7.96. Prove the following form for an inner product in a complex space V :

$$\langle u, v \rangle = \frac{1}{4} \|u + v\|^2 - \frac{1}{4} \|u - v\|^2 + \frac{1}{4} \|u + iv\|^2 - \frac{1}{4} \|u - iv\|^2$$

[Compare with [Problem 7.7\(b\)](#).]

7.97. Let V be a real inner product space. Show that

- (i) $\|u\| = \|\nu\|$ if and only if $\langle u + \nu, u - \nu \rangle = 0$;
- (ii) $\|u + \nu\|^2 = \|u\|^2 + \|\nu\|^2$ if and only if $\langle u, \nu \rangle = 0$.

Show by counterexamples that the above statements are not true for, say, $\mathbf{C}^2$.

7.98. Find the matrix P that represents the usual inner product on $\mathbf{C}^3$ relative to the basis $\{1, 1 + i, 1 - 2i\}$.

7.99. A complex matrix A is *unitary* if it is invertible and $A^{-1} = A^H$. Alternatively, A is unitary if its rows (columns) form an orthonormal set of vectors (relative to the usual inner product of $\mathbf{C}^n$). Find a unitary matrix whose first row is:

- (a) a multiple of $(1, 1 - i)$;
- (b) a multiple of $(\frac{1}{2}, \frac{1}{2}i, \frac{1}{2} - \frac{1}{2}i)$.

Normed Vector Spaces

7.100. Consider vectors $u = (1, -3, 4, 1, -2)$ and $\nu = (3, 1, -2, -3, 1)$ in $\mathbf{R}^5$. Find

- (a) $\|u\|_\infty$ and $\|\nu\|_\infty$,
- (b) $\|u\|_1$ and $\|\nu\|_1$,
- (c) $\|u\|_2$ and $\|\nu\|_2$,
- (d) $d_\infty(u, \nu), d_1(u, \nu), d_2(u, \nu)$

7.101. Repeat [Problem 7.100](#) for $u = (1 + i, 2 - 4i)$ and $\nu = (1 - i, 2 + 3i)$ in $\mathbf{C}^2$.

7.102. Consider the functions $f(t) = 5t - t^2$ and $g(t) = 3t - t^2$ in $C[0, 4]$. Find

- (a) $d_\infty(f, g)$,
- (b) $d_1(f, g)$,
- (c) $d_2(f, g)$

7.103. Prove (a) $\|\cdot\|_1$ is a norm on $\mathbf{R}^n$. (b) $\|\cdot\|_\infty$ is a norm on $\mathbf{R}^n$.

7.104. Prove (a) $\|\cdot\|_1$ is a norm on $C[a, b]$. (b) $\|\cdot\|_\infty$ is a norm on $C[a, b]$.

ANSWERS TO SUPPLEMENTARY PROBLEMS

Notation: $M = [R_1; R_2; \dots]$ denotes a matrix M with rows $R_1, R_2, \dots$. Also, basis need not be unique.

7.58. $k > 9$

7.59. (a) -13 , (b) -71 , (c) $\sqrt{29}$, (d) $\sqrt{89}$

7.60. Let $u = (0, 0, 1)$; then $\langle u, u \rangle = 0$ in both cases

7.64. $\{7t^2 - 5t, 12t^2 - 5\}$

7.65. $\{(1, 2, 1, 0), (4, 4, 0, 1)\}$

7.66. $(-1, 0, 0, 0, 1), (-6, 2, 0, 1, 0), (-5, 2, 1, 0, 0)$

(a) $u_1 = (0, 0, 3, 1), u_2 = (0, 5, -1, 3), u_3 = (-14, -2, -1, 3)$,

7.67. (b) $u_1/\sqrt{10}, u_2/\sqrt{35}, u_3/\sqrt{210}$

7.68.

(a) $(0, 2, -1, 0), (-15, 1, 2, 5)$, (b) $(0, 2, -1, 0)/\sqrt{5}, (-15, 1, 2, 5)/\sqrt{255}$

7.69.

(b) $v = \frac{1}{4}(5u_1 + 3u_2 - 13u_3 + 9u_4)$,

(c) $[v] = \frac{1}{4}[a + b + c + d, a + b - c - d, a - b + c - d, a - b - c + d]$

7.71. (a) $[0, 1; 0, 0], [0, 0; 1, 0]$, (b) $[0, -1; 1, 0]$

7.74. (a) $c = -\frac{23}{30}$, (b) $c = \frac{1}{7}$, (c) $c = \frac{15}{148}$, (d) $c = \frac{19}{26}$

7.75. (a) $w_1 = (1, 1, 1, 1), w_2 = (0, -2, 1, 1), w_3 = (12, -4, -1, -7),$
 (b) $\text{proj}(v, U) = \frac{1}{5}(-1, 12, 3, 6)$

7.76.

(a) $\text{proj}(v, W) = \frac{1}{8}(23, 25, 30, 25, 23),$ (b) First find an orthogonal basis for $W;$
 say, $w_1 = (1, 2, 1, 2, 1)$ and $w_2 = (0, 2, 0, -3, 2).$ Then $\text{proj}(v, W) = \frac{1}{17}(34, 76, 34, 56, 42)$

7.77. $\text{proj}(f, W) = \frac{3}{2}t^2 - \frac{3}{5}t + \frac{1}{20}$

7.78. (a) $\{1, t, 3t^2 - 1, 5t^3 - 3t\},$ $\text{proj}(f, W) = \frac{10}{9}t^3 - \frac{5}{21}t$

7.79.

Four: $[a, b; \quad b, -a], [a, b; \quad -b, -a], [a, -b; \quad b, a], [a, -b; \quad -b, -a],$ where $a = \frac{1}{3}$ and $b = \frac{1}{3}\sqrt{8}$

7.80.

$P = [1/a, 1/a, 1/a; \quad 1/b, -3/b, 2/b; \quad 5/c, -1/c, -4/c],$ where $a = \sqrt{3}, b = \sqrt{14}, c = \sqrt{42}$

7.81. $\frac{1}{3}[1, 2, 2; \quad 2, -2, 1; \quad 2, 1, -2]$

7.83. (a) $[17, -10; \quad -10, 13],$ (b) $[10, 0; \quad 0, 40]$

7.84. (a) $[65, -68; \quad -68, 73],$ (b) $[58, 8; \quad 8, 8]$

7.85. $[3, 4, 3; \quad 4, 6, 2; \quad 3, 2, 11]$

7.86.

(a) $\frac{83}{12},$ (b) $[1, a, b; \quad a, b, c; \quad b, c, d],$ where $a = \frac{1}{2}, b = \frac{1}{3}, c = \frac{1}{4}, d = \frac{1}{5}$

7.87. (a) No, (b) Yes, (c) No, (d) Yes

7.91.

(a) $-4i,$ (b) $4i,$ (c) $\sqrt{28},$ (d) $\sqrt{31},$ (e) $\sqrt{59}$

7.92. (a) $c = \frac{1}{28}(19 - 5i),$ (b) $c = \frac{1}{19}(3 + 6i)$

7.94. $\{v_1 = (1, i, 1)/\sqrt{3}, \quad v_2 = (2i, 1 - 3i, 3 - i)/\sqrt{24}\}$

7.95. a and d real and positive, $c = \bar{b}$ and $ad - bc$ positive.

7.97. $u = (1, 2), v = (i, 2i)$

7.98. $P = [1, 1 - i, 1 + 2i; 1 + i, 2, -1 + 3i; 1 - 2i, -1 - 3i, 5]$

7.99.

(a) $(1/\sqrt{3})[1, 1 - i; 1 + i, -1],$

(b) $[a, ai, a - ai; bi, b, 0; a, ai, -a - ai],$ where $a = \frac{1}{2}$ and $b = 1/\sqrt{2}.$

7.100.

(a) 4 and 3, (b) 11 and 10, (c) $\sqrt{31}$ and $\sqrt{24},$ (d) 6, 19, 9

7.101.

(a) $\sqrt{20}$ and $\sqrt{13},$ (b) $\sqrt{2} + \sqrt{20}$ and $\sqrt{2} + \sqrt{13},$ (c) $\sqrt{22}$ and $\sqrt{15},$ (d) 7, 9, $\sqrt{53}$

7.102. (a) 8, (b) 16, (c) $16/\sqrt{3}$

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